US 20250283030A1

(19) **United States**

(12) **Patent Application Publication**

Merré et al.

(10) **Pub. No.: US 2025/0283030 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **BIOSYNTHESIS OF PHENYLPROPANOID COMPOUNDS**

(71) Applicant: **BGENE GENETICS**, Grenoble (FR)

(72) Inventors: **William Merré**, Grenoble (FR); **Ricardo de Andrade**, Grenoble (FR); **Caroline Ranquet**, Grenoble (FR)

(73) Assignee: **BGENE GENETICS**, Grenoble (FR)

(21) Appl. No.: **18/289,895**

(22) PCT Filed: **May 9, 2022**

(86) PCT No.: **PCT/FR2022/050880**

§ 371 (c)(1),
(2) Date: **Nov. 8, 2023**

(30) **Foreign Application Priority Data**

May 12, 2021 (FR) 2105009

Publication Classification

(51) **Int. Cl.**

C12N 1/20	(2006.01)
C12N 9/00	(2006.01)
C12N 9/02	(2006.01)
C12N 9/10	(2006.01)
C12N 9/88	(2006.01)
C12N 15/78	(2006.01)
C12P 7/26	(2006.01)
C12P 7/42	(2006.01)
C12P 17/06	(2006.01)
C12P 19/32	(2006.01)
C12R 1/40	(2006.01)

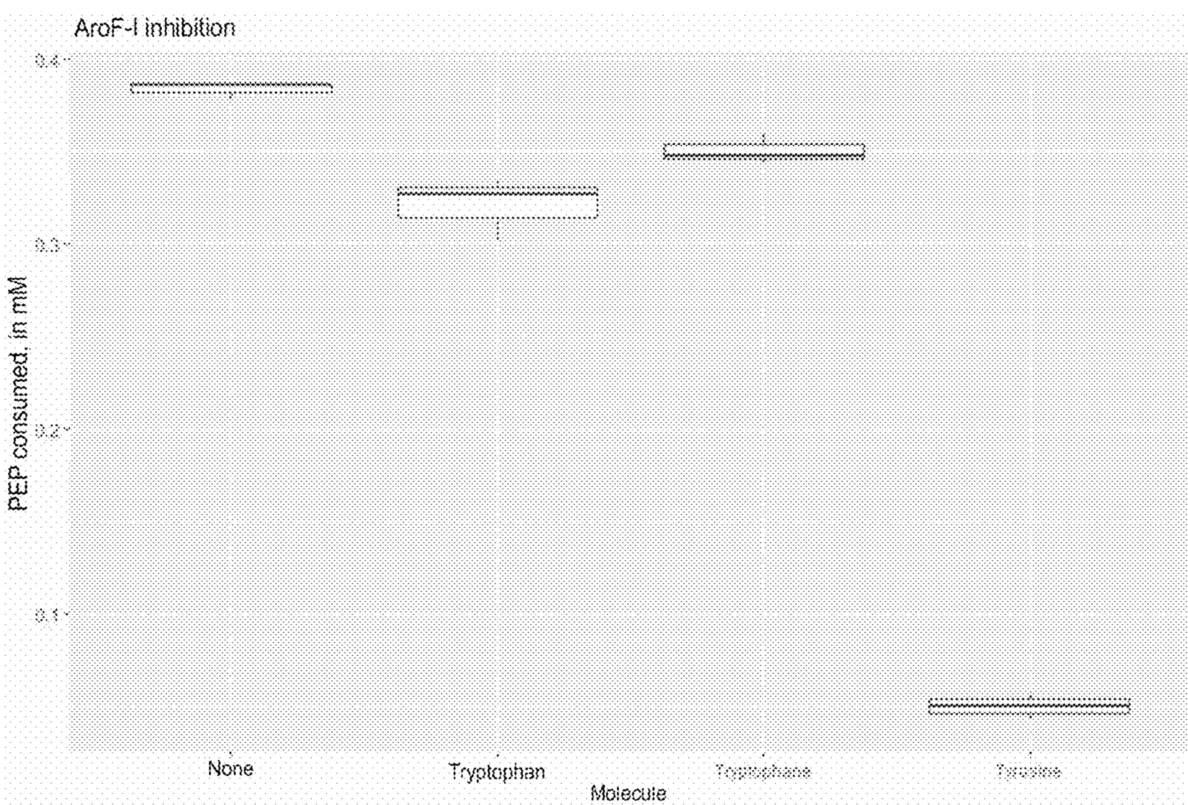
(52) **U.S. Cl.**

CPC ***C12N 1/20*** (2013.01); ***C12N 9/0071*** (2013.01); ***C12N 9/1029*** (2013.01); ***C12N 9/1085*** (2013.01); ***C12N 9/88*** (2013.01); ***C12N 9/93*** (2013.01); ***C12N 15/78*** (2013.01); ***C12P 7/26*** (2013.01); ***C12P 7/42*** (2013.01); ***C12P 17/06*** (2013.01); ***C12P 19/32*** (2013.01); *C12R 2001/40* (2021.05); *C12Y 114/16001* (2013.01); *C12Y 205/01054* (2013.01); *C12Y 402/01096* (2013.01); *C12Y 403/01023* (2013.01); *C12Y 602/01012* (2013.01)

(57) **ABSTRACT**

The present invention relates to the field of the production of phenylpropanoid compounds, especially that of genetically modified strains for the production of phenylpropanoid compounds. In particular, the invention relates to a genetically modified strain of *Pseudomonas putida* comprising a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP), and to the use thereof for the synthesis of phenylpropanoid compounds, in particular coumaric acid or frambinone.

Specification includes a Sequence Listing.



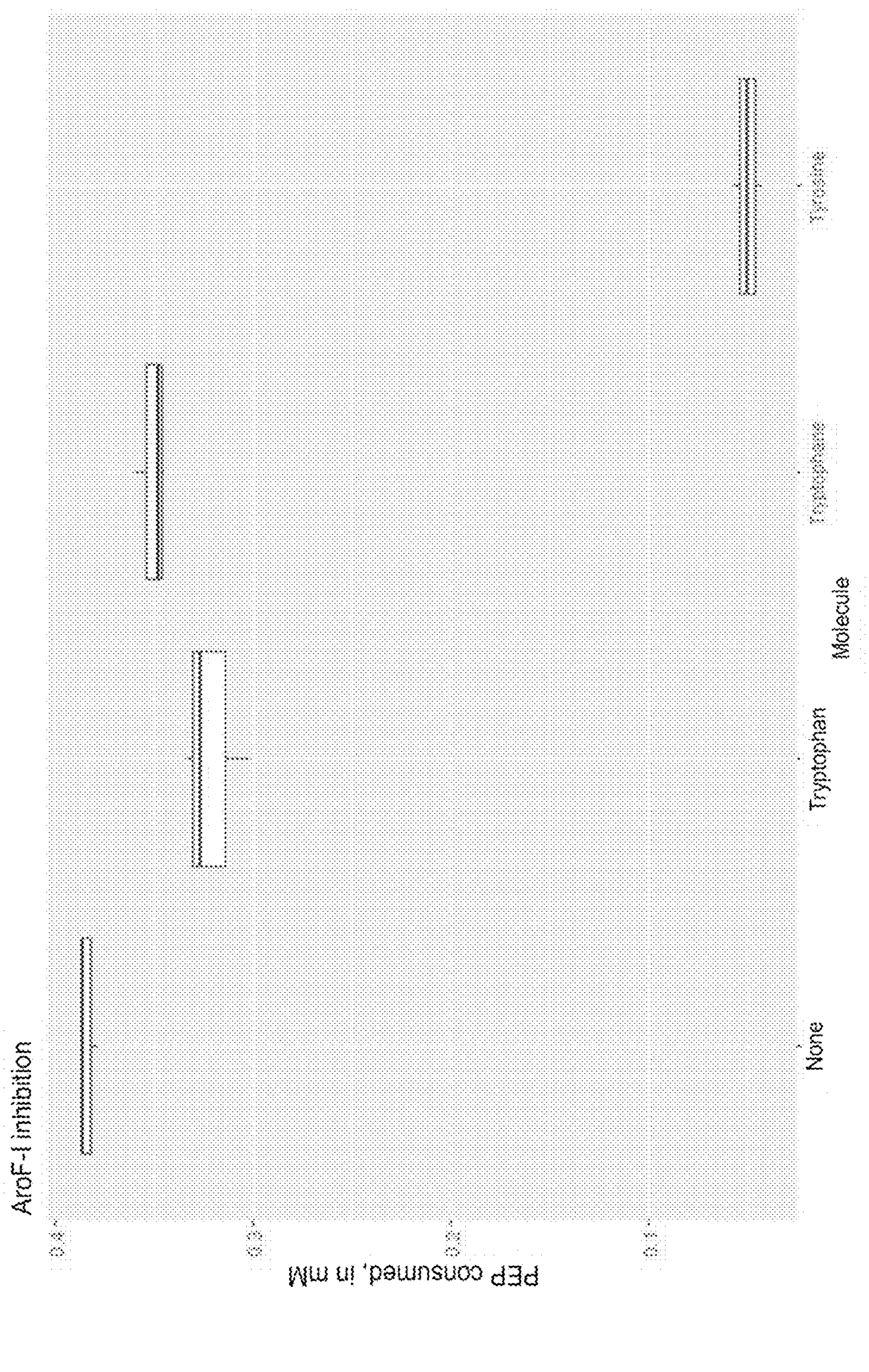

AroF-I inhibition
PEP consumed, in mM
None
Tryptophan
Tyrosine
Molecule

FIG. 1

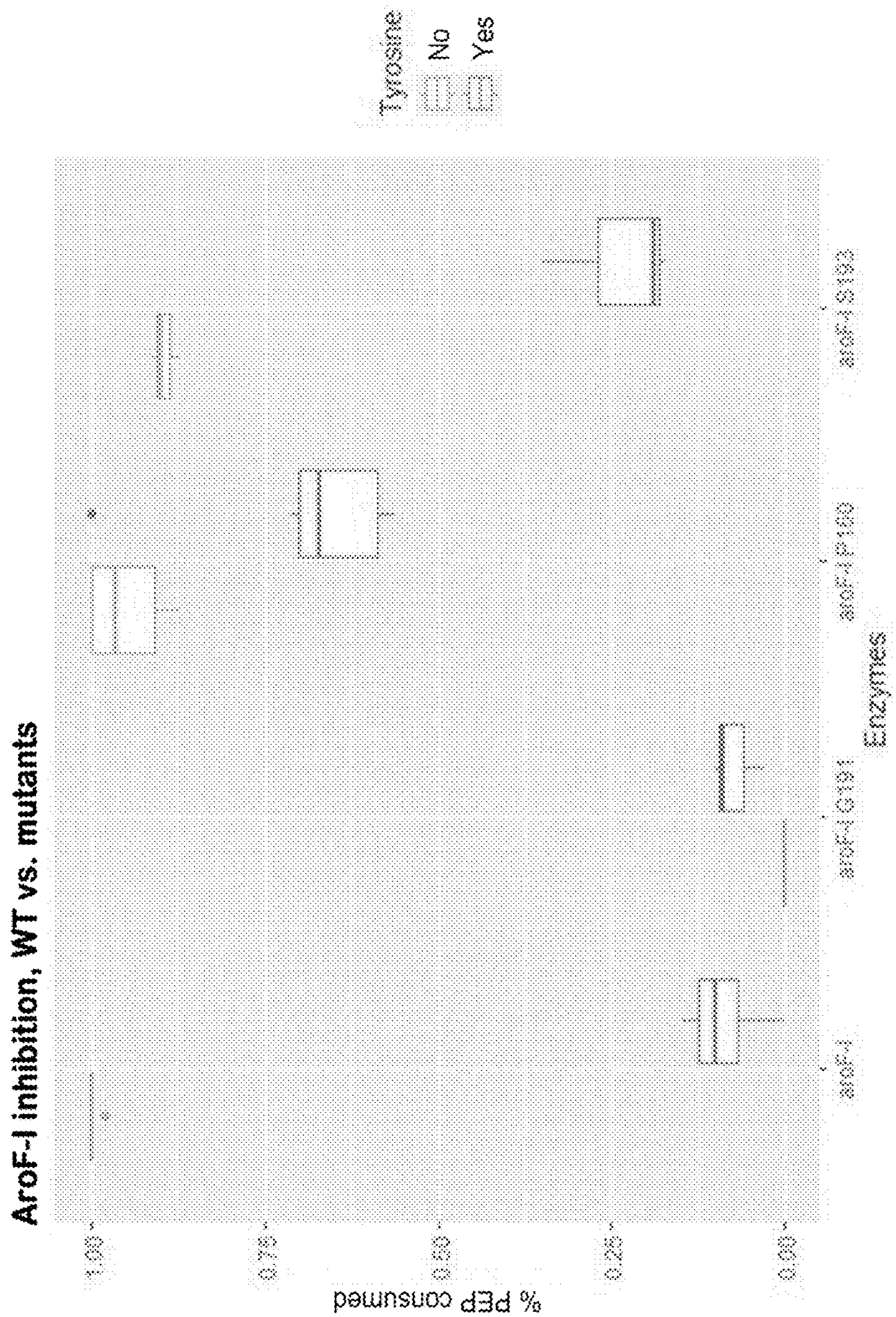

AroF-I inhibition, WT vs. mutants
% PEP consumed
1.00
0.75
0.50
0.25
0.00
aroF-I
Enzymes
Tyrosine
No
Yes

FIG. 2

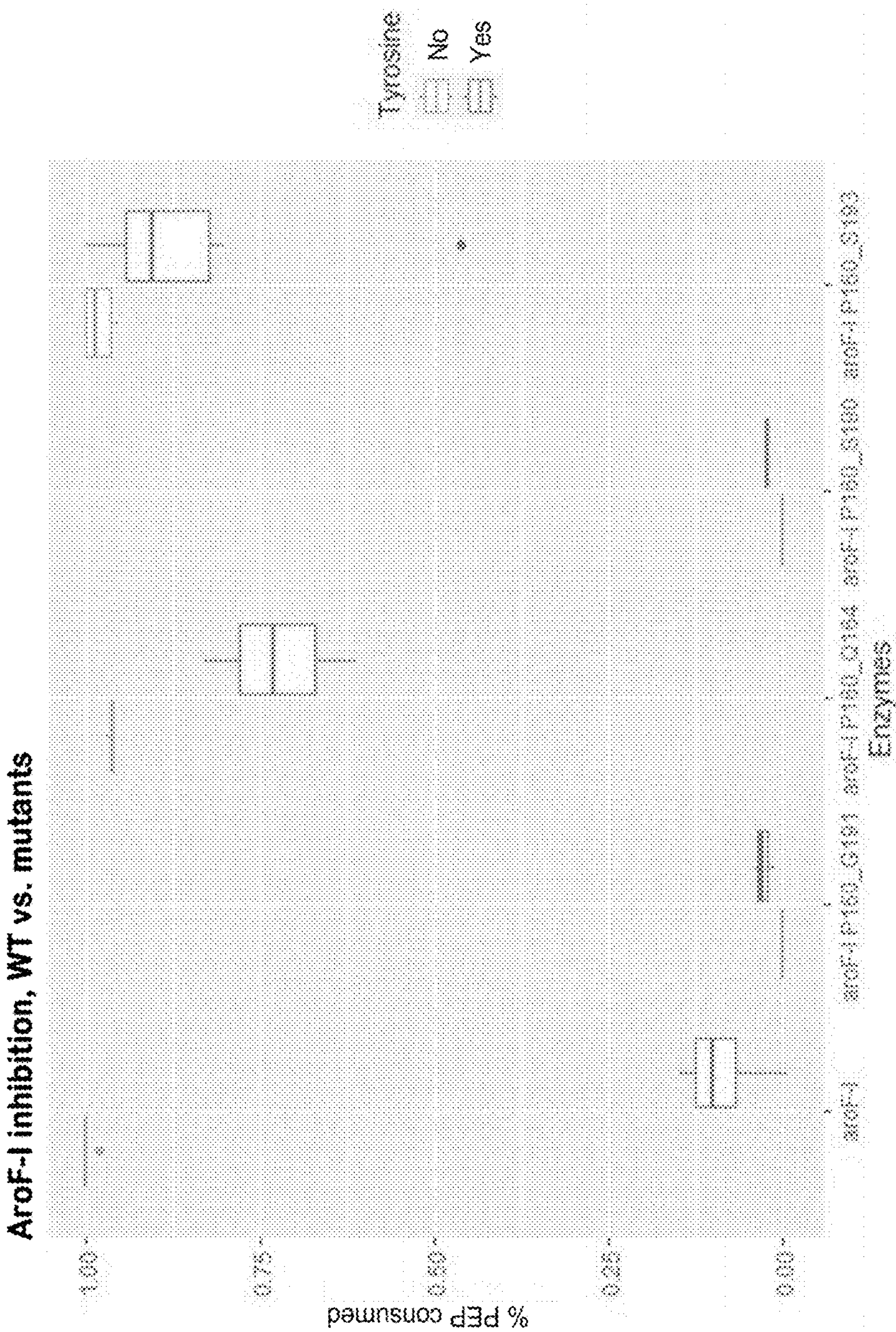

AroF-I inhibition, WT vs. mutants
% PEP consumed
1.00
0.75
0.50
0.25
0.00
aroF-I
Enzymes
Tyrosine
No
Yes

FIG. 3

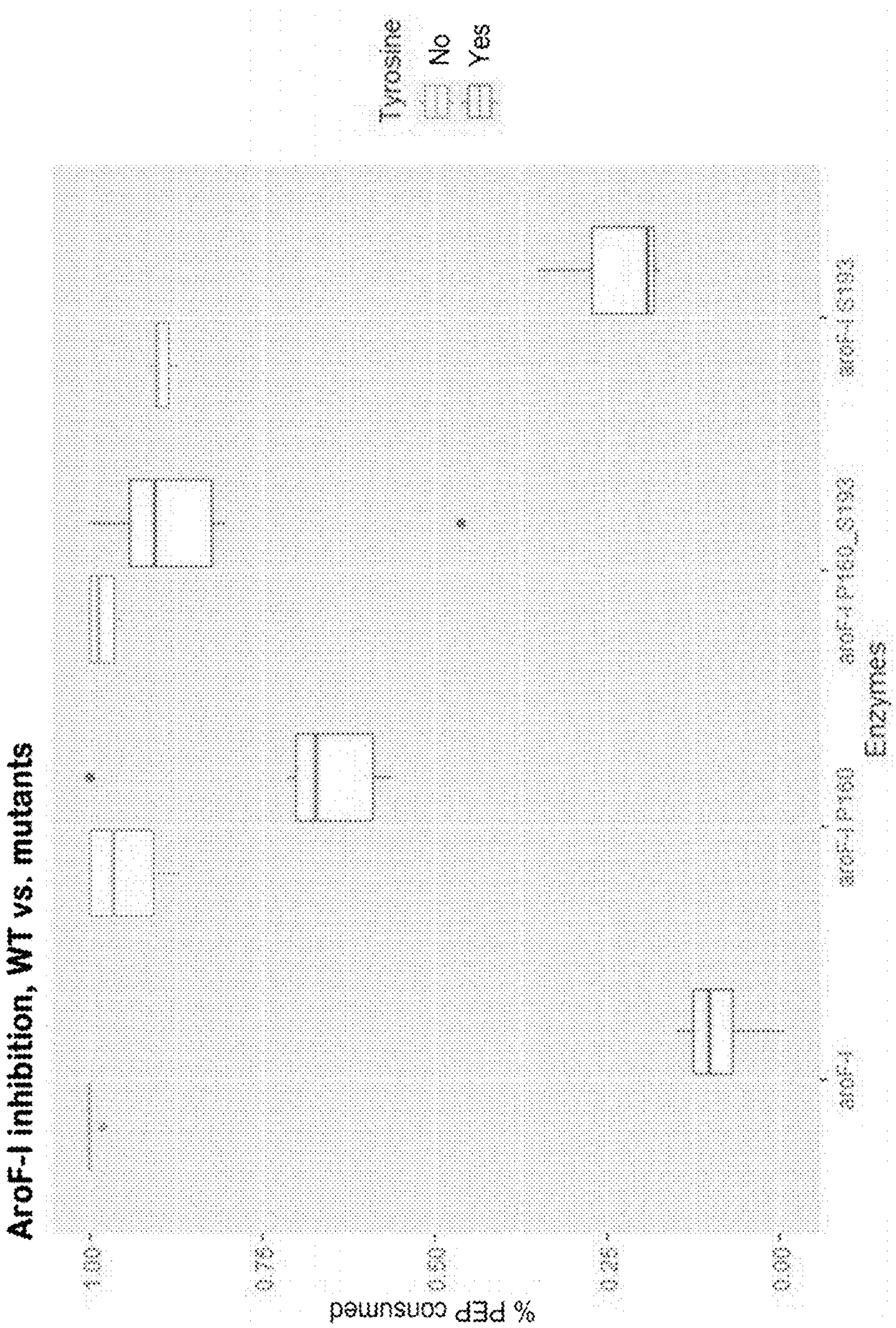

AroF-I inhibition, WT vs. mutants
% PEP consumed
1.00
0.75
0.50
0.25
0.00
aroF-I
aroF-I P160
aroF-I P160_S193
aroF-I S193
Enzymes
Tyrosine
No
Yes

FIG. 4

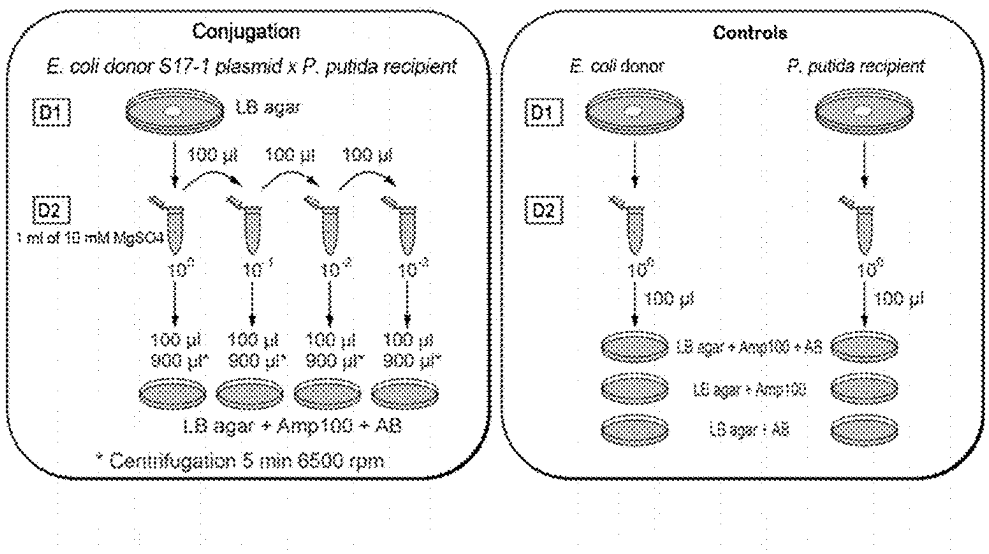

Conjugation
E. coli donor S17-1 plasmid x P. putida recipient
D1
LB agar
100 µl
100 µl
100 µl
D2
1 ml of 10 mM MgSO4
10^0
10^{-1}
10^{-2}
10^{-3}
100 µl
900 µl*
LB agar + Amp100 + AB
* Centrifugation 5 min 6500 rpm
Controls
E. coli donor
P. putida recipient
D1
D2
10^0
10^0
100 µl
100 µl
LB agar + Amp100 + AB
LB agar + Amp100
LB agar + AB

FIG.5

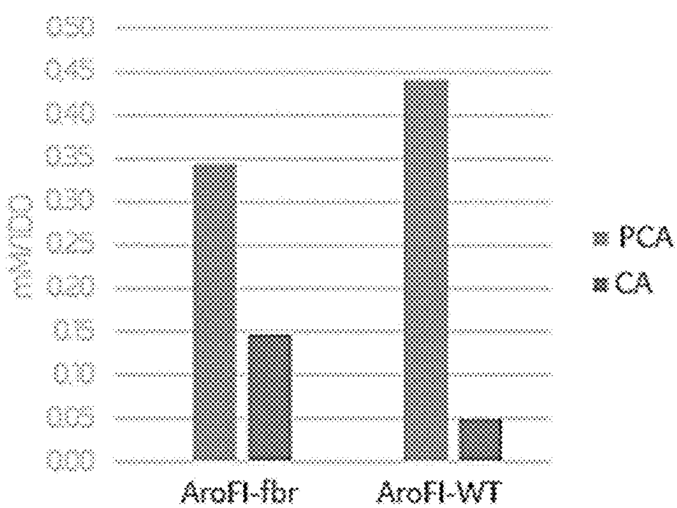
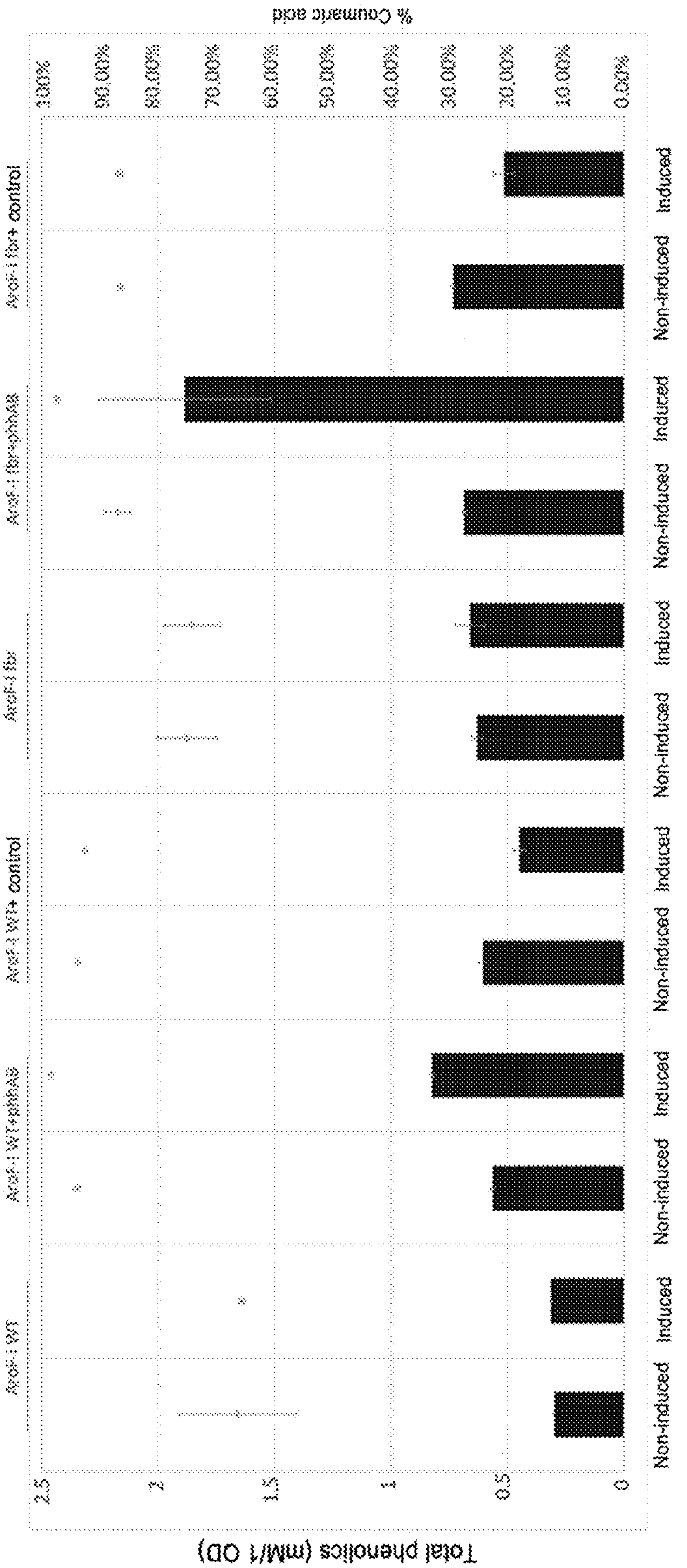

0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
mM/OD
PCA
CA
AroFI-fbr
AroFI-WT

FIG.6



% Coumaric acid
100%
90.00%
80.00%
70.00%
60.00%
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
Total phenolics (mM/1 OD)
2.5
2
1.5
1
0.5
0
Non-induced
Induced
Control

FIG.7

BIOSYNTHESIS OF PHENYLPROPANOID COMPOUNDS

TECHNICAL FIELD

[0001] The present invention relates to the field of the production of phenylpropanoid compounds, and particularly to genetically modified strains for the production of phenylpropanoid compounds, such as coumaric acid or frambinone.

PRIOR ART

[0002] For many years, the bioproduction of "natural" flavorings and fragrances has been an important area of research for industry, in order to meet the needs of consumers who seek to be increasingly environmentally responsible. The biology of synthesis, particularly using microorganisms, makes this natural production possible, but yields are not always sufficient for large-scale production.

[0003] The flavor of raspberries (*Rubus idaeus*) is associated with more than 200 compounds, but frambinone, a natural phenolic compounds, is the compound with the greatest impact, defining the characteristic raspberry flavor (Klesk et al., 2004, J. Agric. Food Chem. 52, 5155-61; Larsen et al., 1991, Acta Agric. Scand. 41, 447-54).

[0004] Since it is only present in small amounts in raspberries (1-4 mg per kg of fruit), natural frambinone is extremely valuable (Larsen et al., 1991). However, because the natural availability thereof is limited, producing it by means of biotechnology is highly desirable.

[0005] In this context, the biosynthesis pathway of phenylpropanoid compounds, particularly frambinone, can be recreated within a microorganism by inserting heterologous genes encoding some of the key enzymes of this pathway.

[0006] Tyrosine is an important amino acid for the biosynthesis of phenylpropanoid compounds because it is in particular the precursor of coumaric acid or of frambinone.

[0007] Indeed, frambinone can be obtained from the aromatic amino acid L-tyrosine as initial substrate, via a 4-step biosynthesis pathway.

[0008] In the first step, the tyrosine is deaminated by a tyrosine ammonia lyase (TAL, EC 4.3.1.23) to form coumaric acid. Catalyzed by a 4-coumarate:CoA ligase (4CL, EC 6.2.1.12), a molecule of Coenzyme A (CoA) is grafted onto the coumaric acid. The coumaroyl-CoA is then converted by a benzalacetone synthase (BAS, EC 2.3.1.212) into 4-hydroxybenzalacetone. This reaction is a decarboxylative condensation, and uses a unit of malonyl-CoA as co-substrate. The final step is the reduction of the 4-hydroxybenzalacetone into frambinone by a benzalacetone reductase.

[0009] Thus, the development of strains that overproduce tyrosine as a platform for the production of phenylpropanoid compounds, particularly frambinone, is a crucial starting point for the biosynthesis of these compounds.

[0010] However, the production of tyrosine in microorganisms is as complex as it is closely regulated. This is because some enzymes in the tyrosine production pathway, also known as the shikimate pathway, are regulated in a negative feedback loop by this amino acid, which inhibits their activity by bonding to an inhibition site when the concentration in the cell becomes too high. This system of regulation, when too great a quantity of tyrosine is present in the microorganism, is referred to as feedback inhibition.

[0011] In spite of this, it should be possible, by mutagenesis, to obtain enzymes that are "resistant" to this inhibition and therefore to deregulate the tyrosine production pathway. These mutants, which are resistant to tyrosine feedback inhibition, are referred to as fbr mutants, for "feedback resistant".

[0012] Numerous articles describe the deregulation of this pathway in *E. coli*, and one enzyme has been particularly well studied: DAHP synthase (DAHP=3-deoxy-D-arabino-heptulosonate-7-phosphate). The first reaction in the shikimate pathway consists of the condensation of a phospho (enol)pyruvate (PEP) and an erthythrose-4-phosphate (E4P) to give DAHP.

[0013] This DAHP synthase activity is carried out by 3 isoenzymes in *E. coli*: AroG (feedback-inhibited by phenylalanine), AroF (feedback-inhibited by tyrosine) and AroH (feedback-inhibited by tryptophan). The structures of these proteins are known, and the substructures (and also the amino acids involved) associated with feedback inhibition are also thoroughly described in the literature.

[0014] Kikuchi et al., 1997 and Cui et al., 2019, identified, described and studied, in *E. coli*, the mutants AroGfbr and AroFfbr which are resistant to phenylalanine and tyrosine feedback inhibition, respectively. Fbr mutants of the protein TyrA (mutant tyrAfbr) have also been described in *E. coli* by Lutke-Eversloh and Stephanopoulos, (2005) (Lutke-Eversloh and Stephanopoulos, Appl. Environ Microbiol. 2005 November; 71(11): 7224-8).

[0015] In 2007, Lutke-Eversloh and Stephanopoulos (Lutke-Eversloh and Stephanopoulos, Appl. Environ Microbiol. 2007 November; 75(1): 103-10) also described *E. coli* strains that overproduce tyrosine, obtained by combining these various mutated enzymes. These strains were used to produce phenylpropanoids such as coumaric acid (Kang et al., 2012), or naringenin, particularly via mutation of the rpoA gene in addition to the aroGfbr and tyrAfbr mutations (Santos et al., 2011).

[0016] A review of the modifications carried out to obtain *E. coli* strains that overproduce tyrosine is described in the article "Modular engineering of L-tyrosine production in *E. coli*" (Juminaga et al., 2011).

[0017] Phenylpropanoid compounds have also been synthesized from *Saccharomyces cerevisiae* strains that overexpress tyrosine, as described by Rodriguez et al., 2015.

[0018] Document GB 2 416 769 describes the possibility of producing frambinone using microorganisms containing genes encoding the enzymes 4CL and BAS, at least one of which comes from a heterologous source. The preferred microorganism is *E. coli* (strain BL21) and may further comprise a sequence encoding BAR, C4H, PAL and/or CHS, with the sequence encoding BAR advantageously being endogenous.

[0019] However, it appears that *E. coli* or *S. cerevisiae* do not have good tolerance for the toxicity of phenylpropanoid compounds, and are therefore not the most well-suited microorganisms for the production thereof.

[0020] Bacteria of the genus *Pseudomonas* are more tolerant to these highly toxic molecules, particularly the bacterium *Pseudomonas putida* (Calero et al., 2017). However, the enzymes involved in the production of aromatic amino acids in *P. putida* have not been described in detail.

[0021] Consequently, the overproduction of shikimate pathway derivatives such as tyrosine in *P. putida* involves the use of AroGfbr and TyrAfbr mutants originating from *E. coli*, which is not particularly effective (Calero et al., 2016).

[0022] The overproduction of tyrosine and phenol has been described in the bacterium *Pseudomonas taiwanensis* VLB120 by implementing point mutations in 3 genes: trpEP290S, aroF-I P148L and pheAT3101 (Wynands et al., Metab Eng. 2018 May, 47:121-133).

[0023] However, there is still a need to develop novel enzymes and novel strains of *Pseudomonas putida* that overproduce tyrosine, enabling the production of phenylpropanoid compounds, particularly coumaric acid and frambinone.

Technical Problem

[0024] The strains that overproduce tyrosine that have been developed to date are essentially *E. coli* and *S. Cerevisiae* strains. However, these strains do not have good tolerance for the toxicity of phenylpropanoid compounds, and are therefore not the most well-suited microorganisms for the production thereof.

[0025] Thus, there is a particular need to develop novel strains that overproduce tyrosine, enabling the production of phenylpropanoid compounds.

[0026] The present disclosure will improve the situation.

SUMMARY

[0027] One of the aspects of the present invention relates to a genetically modified strain of *Pseudomonas putida*, comprising a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, having the sequence SEQ ID NO:1 and having at least one P160L mutation, a P160L/Q164A double mutation, or a P160L/S193A double mutation, preferably a P160L/S193A double mutation.

[0028] Another aspect of the invention relates to a method for synthesizing a phenylpropanoid compound or a phenylpropanoid derivative by employing a genetically modified strain of *Pseudomonas putida* according to the invention.

[0029] Finally, the invention also relates to the use of a genetically modified strain of *Pseudomonas putida* for the synthesis of phenylpropanoid compounds.

[0030] The characteristics set out in the following paragraphs can optionally be implemented. They may be implemented independently or in combination with one another:

BRIEF DESCRIPTION OF THE FIGURES

[0031] Other features, details and advantages will become apparent upon reading the following detailed description and upon examining the appended drawings, in which:

[0032] FIG. **1** Graph of PEP consumed, in mM, in the presence or absence of aromatic amino acid, by the wild-type (WT) AroF-I protein. The results presented are from three independent replicas.

[0033] FIG. **2** Graph showing the share of PEP consumed, in mM, in the presence or absence of aromatic amino acid, by the AroF-I WT protein compared to the single mutants AroF-I-G191, AroF-I-P160 and AroF-I-S193. The results presented are from three independent replicas.

[0034] FIG. **3** Graph showing the share of PEP consumed, in mM, in the presence or absence of aromatic amino acid, by the AroF-I WT protein compared to the double mutants AroF-I-P160_G191, AroF-I-P160L_Q164, AroF-I-P160_S190 and AroF-I-P160_S193. The results presented are from three independent replicas.

[0035] FIG. **4** Graph showing the share of PEP consumed, in mM, in the presence or absence of aromatic amino acid, by the AroF-I WT protein compared to the single mutants AroF-I-P160 and AroF-I-S193 and to the double mutant AroF-I-P160_S193. The results presented are from three independent replicas.

[0036] FIG. **5** presents the conjugation protocol that can be used to transform and genetically modify *P. putida*.

[0037] FIG. **6** Graph presenting the production of coumaric acid (PCA) and cinnamic acid (CA) by *Pseudomonas putida* strains expressing the enzymes AroF-I WT/TAL (aroF-I WT) and aroF-I fbr P160L/S193A/TAL (aroF-I fbr).

[0038] FIG. **7** Graph presenting the production of total phenolics and the proportion of coumaric acid in the total phenolics (% PCA) by *Pseudomonas putida* strains expressing the enzymes enzymes AroF-I WT/TAL (aroF-I WT), AroF-I WT/TAL+empty plasmid (aroF-I WT+control), AroF-I WT/TAL+phhA/B plasmid (aroF-I WT+phhA/B), aroF-I fbr P160L/S193A/TAL (aroF-I fbr), aroF-I fbr P160L/S193A/TAL+empty plasmid (aroF-I fbr+control), and aroF-I fbr P160L/S193A/TAL+phhA/B plasmid (aroF-I fbr+phhA/B). Expression of the phhA/B genes from the araC/pBAD promoter is induced or non-induced.

DESCRIPTION OF THE EMBODIMENTS

[0039] A first subject of the present invention therefore relates to a genetically modified strain of *Pseudomonas putida*, comprising a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, the sequence of which has at least 80% identity to the sequence SEQ ID NO:1, and having at least one P160L mutation, a P160L/Q164A double mutation, or a P160L/S193A double mutation, preferably a P160L/S193A double mutation.

[0040] For the purposes of the present description, the expressions "genetically modified strain of *Pseudomonas putida*", "modified strain of *Pseudomonas putida*", "genetically modified strain" and "modified strain" are considered to be synonymous.

[0041] In particular, "genetically modified strain" means a strain which comprises either (i) at least one recombinant nucleic acid, or transgene, integrated stably into the genome thereof and/or present on a vector, for example a plasmid vector, or (ii) one or more non-natural mutations by nucleotide insertion, substitution or deletion, said mutations being obtained by transformation techniques or gene editing techniques known to those skilled in the art.

[0042] Indeed, techniques for genetic modification by transformation, mutagenesis or gene editing are well known to those skilled in the art and are described for example in "Strategies used for genetically modifying bacterial genome: site-directed mutagenesis, gene inactivation", Journal of Zhejiang Univ-Sci B (Biomed & Biotechnol) 2016 17(2):83-99., and in Martinez-Garcia and de Lorenzo, "*Pseudomonas putida* in the quest of programmable chemistry", Current Opinion in Biotechnology, 59:111-121, 2019.

[0043] Use will preferably be made, for integrating the mutated or recombinant genes in *Pseudomonas putida*, of the mutagenesis technique described in example 2.

[0044] In particular, a genetically modified strain may comprise a nucleic acid that modifies the expression of one or more genes that are naturally expressed in *Pseudomonas putida.*

[0045] According to a particular embodiment, a genetically modified strain may comprise a nucleic acid encoding one or more enzymes that are not naturally expressed in *Pseudomonas putida.*

[0046] The wild-type strains of *P. putida* KT2440 are available for example in the NBRC strain collection (National Institute of Technology and Evaluation Biological Resource center https://www.nite.go.jp/en/nbrc/, NBRC100650).

[0047] Furthermore, strains of *Pseudomonas putida* or *Pseudomonas taiwanensis* optimized for tyrosine production are known to those skilled in the art, who may use them as a founder strain to obtain the genetically modified strains according to the invention (Calero et al., 2016; Wierckx et al., 2005, Appl Environ Microbiol. 71(12):8221-7; Wynands et al., 2018; Otto et al. 2019, Front Bioeng Biotechnol November 20; 7: 312).

[0048] For the purposes of the present invention, the percentage identity refers to the percentage of identical residues in a nucleotide or amino acid sequence over a given fragment following alignment and comparison with a reference sequence. An alignment algorithm is used for the comparison and the sequences to be compared are input with the corresponding parameters of the algorithm. The default algorithm parameters can be used.

[0049] Preferably, for a nucleic acid sequence comparison and to determine a percentage identity, the BLAST algorithm, as described at https://blast.ncbi.nlm.nih.gov/Blast.cgi, is used with the default parameters.

[0050] For the purposes of the present invention, "mutated AroF1 gene" means a nucleic acid comprising at least a portion encoding a mutated version of 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase under the control of a promoter enabling the expression thereof in the genetically modified strain.

[0051] For the purposes of the present invention, 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase means the enzymes (EC 2.5.1.54) which are able, in bacteria, to carry out the first reaction in the shikimate pathway, which consists of the condensation of a phospho(enol) pyruvate (PEP) and an erthythrose-4-phosphate (E4P) to give DAHP.

[0052] According to a particular embodiment, the genetically modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase, the amino acid sequence of which has at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO:1, and having at least one P160L mutation, a P160L/Q164A double mutation, or a P160L/S193A double mutation, preferably a P160L/S193A double mutation.

[0053] The endogenous *Pseudomonas putida* AroF-I gene encodes DAHP synthase of amino acid sequence SEQ ID NO:1. In a particular embodiment, therefore, the genetically modified strain according to the present invention may comprise, in addition to the endogenous AroF-I gene, at least one recombinant mutated AroF-I nucleic acid sequence encoding a mutated protein comprising the P160L mutation, the P160L/Q164A double mutation, or the P160L/S193A double mutation, preferably the P160L/S193A double mutation.

[0054] According to a particular embodiment, the genetically modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase comprising a P160L mutation as defined by amino acid sequence SEQ ID NO:2.

[0055] According to a variant of this embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase, the sequence of which has at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 2 and contains the P160L mutation.

[0056] According to another particular embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase and having at least one P160L/Q164A double mutation. According to this embodiment, the mutated AroF-I gene encodes DAHP synthase comprising the P160L/Q164A double mutation defined by amino acid sequence SEQ ID NO: 3.

[0057] According to a variant of this embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase, the sequence of which has at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 3 and contains the P160L/Q164A double mutation.

[0058] According to a preferred embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase, having at least one P160L/S193A double mutation. According to this embodiment, the mutated AroF-I gene encodes DAHP synthase comprising the P160L/S193A double mutation defined by amino acid sequence SEQ ID NO: 4.

[0059] According to a variant of this embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase, the sequence of which has at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 4 and contains the P160L/S193A double mutation.

[0060] In one embodiment, which may be combined with the preceding embodiments, the coding sequence of the mutated AroF-I gene is placed under the control of a heterologous promoter, in particular a constitutive or inducible promoter, for example selected from the promoters ptrc, xyls/pm or araC/pBAD, which makes it possible to overexpress the mutated AroF-I gene in the genetically modified strain according to the invention. In a preferred embodiment, the coding sequence of the mutated AroF-I gene in the genetically modified strain according to the invention is inserted so as to make the AroH gene non-functional, for example by disrupting the AroH gene or by deleting the AroH gene and in particular all or part of the coding sequence thereof.

[0061] The AroH gene encodes another DAHP synthase (isoenzyme of AroF). Thus, in one embodiment, the genetically modified strain according to the present invention comprises the deleted or disrupted AroH gene and at least one recombinant nucleic acid comprising the mutated AroF-I gene as described previously or a coding sequence of the mutated AroF-I gene.

[0062] In an entirely advantageous way, by implementing specific mutations the Applicant has developed a strain of *Pseudomonas putida* that is able to express a recombinant DAHP synthase which is resistant to feedback inhibition by tyrosine, thereby deregulating the tyrosine production pathway. In other words, the present invention makes it possible to produce strains that overproduce tyrosine as end product or as intermediate product in the synthesis of phenylpropanoid compounds. In particular, this overproduction is particularly advantageous for the production of phenylpropanoid compounds by the modified strains according to the invention.

[0063] Phenylpropanoid compounds are a class of plant-derived organic compounds that are biosynthesized from phenylalanine or tyrosine. As examples of phenylpropanoid compound, mention may particularly be made of coumaric acid, p-coumaroyl-CoA, 4-hydroxybenzalacetone, frambinone, zingerone, vanillin, flavonoids and stilbenoids.

[0064] For the purposes of the present invention, "strain that overproduces tyrosine" means a modified strain of *Pseudomonas putida* that is able to produce a greater quantity of tyrosine than a wild-type strain of *Pseudomonas putida* comprising the AroF-I gene encoding DAHP synthase of amino acid sequence SEQ ID NO:1 (non-mutated gene), either as end product or as intermediate product.

[0065] In order to be able to convert the tyrosine into phenylpropanoid compounds such as coumaric acid, the modified strain according to the invention may advantageously further comprise at least one additional recombinant gene.

[0066] For the purposes of the present invention, the expression "additional recombinant gene" means any recombinant gene present in the *Pseudomonas putida* strain in addition to the mutated AroF-I gene as defined previously. The additional recombinant gene may result from the insertion of a heterologous promoter, for example a strong promoter to overexpress an endogenous *Pseudomonas putida* gene, or a recombinant coding sequence encoding a protein that is not naturally expressed in *Pseudomonas putida*.

[0067] According to a particular embodiment, the genetically modified strain of *Pseudomonas putida* comprises at least an additional recombinant gene encoding a polypeptide having phenylalanine hydroxylase (phhA) activity and an additional recombinant gene encoding a polypeptide having tetrahydrobiopterin dehydratase (phhB) activity.

[0068] The Applicant observed that it was possible for these two enzymes, although present endogenously in Pseudomonasputida, to not be sufficiently active, and/or for their expression to be insufficient in the context of use of the strain for the production of phenylpropanoid compounds. According to this embodiment, activities have therefore been optimized by overexpressing these two enzymes, phhA and phhB. Overexpression may for example be obtained by placing the additional recombinant genes for these enzymes under the control of a heterologous promoter, in particular a constitutive or inducible promoter. Examples of such promoters are in particular the promoters ptrc, xyIS/pm, araC/pBAD.

[0069] The overexpression of the enzymes phhA and phhB is preferably obtained by placing the additional recombinant genes for these enzymes under the control of a strong inducible promoter araC/pBADopt (Prior et al., 2010).

[0070] The overexpression of a gene means greater expression of said gene in a genetically modified strain than in the same strain but in which the gene is expressed solely under the control of the natural promoter. Overexpression can be obtained by inserting one or more copies of the gene directly into the genome of the strain, preferably under the control of a strong promoter, or also by cloning in plasmids, in particular multicopy plasmids, preferably also under the control of a strong promoter.

[0071] In another embodiment, which can be combined with the preceding embodiment, the endogenous phhA and phhB coding sequences are placed under the control of a heterologous promoter as defined previously, for example in order to overexpress the corresponding endogenous coding sequences in the genetically modified strain according to the invention.

[0072] In particular, the modified strain may comprise an additional recombinant gene encoding a phenylalanine hydroxylase (phhA) (EC 1.14.16.1), the sequence of which is defined by the amino acid sequence SEQ ID NO: 5 or by a sequence having at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 5 and encoding an enzyme having phhA activity, and an additional recombinant gene encoding a tetrahydrobiopterin dehydratase (phhB) (EC 4.2.1.96), the sequence of which is defined by the amino acid sequence SEQ ID NO: 6 or by a sequence having at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 6 and encoding an enzyme having phhB activity, in particular under the control of a heterologous promoter enabling their overexpression.

[0073] According to this embodiment, the genetically modified strain according to the invention is able to overproduce tyrosine and also to convert phenylalanine into tyrosine via the phhA and phhB enzymes.

[0074] According to a particular embodiment, the additional recombinant genes encoding phenylalanine hydroxylase (phhA) and tetrahydrobiopterin dehydratase (phhB) comprise the corresponding coding sequences of *Pseudomonas fluorescens* (phhA VVN86558.1/phhB: AYF50180.1) or *Pseudomonas aeruginosa* (phhA AAA25936.1/phhB AAA25937.1).

[0075] According to a preferred embodiment, the modified strain of *Pseudomonas putida* comprises a mutated AroF-I gene encoding DAHP synthase comprising the P160L/S193A double mutation defined by the amino acid sequence SEQ ID NO: 4 or a sequence having at least, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 4, and the two additional recombinant genes below:

- **[0076]** a recombinant gene encoding a phenylalanine hydroxylase (phhA), preferably a phhA defined by the sequence SEQ ID NO: 5 or a sequence having at least, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 5, and
- **[0077]** a recombinant gene encoding a tetrahydrobiopterin dehydratase (phhB), preferably a phhB defined by the sequence SEQ ID NO: 6 or a sequence having at least, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 6;
- **[0078]** said genes phhA and phhB preferably being placed under the control of a heterologous promoter, enabling their overexpression.

[0079] According to another particular embodiment, which may preferably be combined with the preceding embodiment, the genetically modified strain of *Pseudomonas putida* comprises an additional recombinant gene encoding a polypeptide having tyrosine ammonia lyase (TAL) activity. A recombinant gene encoding TAL may originate from the microorganism *Rhodotorula glutinis* and optimized according to the reference Zhou et al., 2015 (three point mutations in this TAL enzyme makes it more effective: S9N; A11T; E518V). This TAL enzyme is referred to as TAL_rg_opt. In particular, the modified strain may comprise a recombinant gene encoding a tyrosine ammonia lyase (TAL) (EC 4.3.1.23), the sequence of which is defined by the amino acid sequence SEQ ID NO: 7 (TAL_rg_opt) or by a sequence having at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 7 and encoding an enzyme having TAL activity.

[0080] According to this particular embodiment, the genetically modified strain according to the invention is able to overproduce tyrosine and also to convert it into coumaric acid via the TAL enzyme.

[0081] According to another particular embodiment, which may be combined with the preceding embodiments, the genetically modified strain of *Pseudomonas putida* comprises an additional recombinant gene encoding 4-coumarate-CoA ligase (4-CL). In particular, the modified strain may comprise a recombinant gene encoding a 4-CL (EC 6.2.1.12), the sequence of which is defined by the amino acid sequence SEQ ID NO: 8 or by a sequence having at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 8 and encoding an enzyme having 4-CL activity.

[0082] According to this particular embodiment, the genetically modified strain is able to convert coumaric acid into p-coumaroyl-CoA via the 4-CL enzyme.

[0083] According to another particular embodiment, which may preferably be combined with the preceding embodiments, the genetically modified strain of *Pseudomonas putida* comprises an additional recombinant gene encoding a polypeptide having benzalacetone synthase (BAS) activity. In particular, the modified strain may comprise a recombinant gene encoding a BAS (EC 2.3.1.212), the sequence of which is defined by the amino acid sequence SEQ ID NO: 9 or by a sequence having at least 80%, 85%, 90%, 95% and most particularly at least 98% identity to the sequence SEQ ID NO: 9 and encoding an enzyme having BAS activity.

[0084] According to this particular embodiment, the genetically modified strain is able to convert p-coumaroyl-CoA into 4-hydroxybenzalcetone via the BAS enzyme.

[0085] According to another particular embodiment, the genetically modified strain of *Pseudomonas putida* comprises a plurality of additional recombinant genes, namely the five additional recombinant genes below:

- **[0086]** a recombinant gene encoding a phenylalanine hydroxylase (phhA), preferably a phhA defined by the sequence SEQ ID NO: 5, in particular under the control of a heterologous promoter enabling its overexpression,
- **[0087]** a recombinant gene encoding a tetrahydrobiopterin dehydratase (phhB), preferably a phhB defined by the sequence SEQ ID NO: 6, in particular under the control of a heterologous promoter enabling its overexpression,
- **[0088]** a recombinant gene encoding a tyrosine ammonia lyase (TAL_RG_OPT), preferably a TAL_RG_OPT defined by the sequence SEQ ID NO: 7,
- **[0089]** a recombinant gene encoding a 4-coumarate-CoA ligase (4-CL), preferably a 4-CL defined by the sequence SEQ ID NO:8,
- **[0090]** a recombinant gene encoding a benzalacetone synthase (BAS), preferably a BAS defined by the sequence SEQ ID NO:9.

[0091] The enzymes TAL, 4-CL and BAS are all enzymes involved in the synthesis of phenylpropanoid compounds.

[0092] According to this preferred embodiment, the modified strain is able to produce a multitude of phenylpropanoid compounds, namely in particular coumaric acid, p-coumaroyl-CoA and 4-hydroxybenzalacetone.

[0093] Another subject of the present invention relates to a genetically modified strain of *Pseudomonas putida* comprising an additional recombinant AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, the sequence of which has at least 80% identity to the sequence SEQ ID NO:1, and in which the genes encoding the enzymes phhA and phhB are overexpressed.

Method for Synthesizing a Phenylpropanoid Compound

[0094] Another subject of the invention relates to a method for synthesizing one or more phenylpropanoid compounds.

[0095] The phenylpropanoid compounds may be as defined previously.

[0096] The synthesis method according to the invention comprises the implementation of a step of growing a genetically modified strain of *Pseudomonas putida* as defined previously in a culture medium under conditions that enable the expression of the mutated gene and/or of the additional recombinant genes required for the synthesis of one or more phenylpropanoid compounds.

[0097] According to a particular embodiment, the synthesis method according to the invention makes it possible to produce large quantities of coumaric acid.

[0098] According to a variant of this embodiment, the method may comprise a step of growing a genetically modified strain of *Pseudomonas putida* comprising a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, for example of sequence SEQ ID NO:2, SEQ ID NO:3 or SEQ ID NO:4, and at least the additional recombinant genes below, encoding:

- **[0099]** a phenylalanine hydroxylase (phhA) of sequence SEQ ID NO: 5,
- **[0100]** a tetrahydrobiopterin dehydratase (phhB) of sequence SEQ ID NO: 6,
- **[0101]** and a tyrosine ammonia lyase (TAL_RG_OPT) of sequence SEQ ID NO: 7.

[0102] According to another particular embodiment, the synthesis method according to the invention makes it possible to produce frambinone, in particular industrial quantities thereof, and particularly with a yield at least equal to 20 g/l of frambinone in a fermenter of at least 500 l.

[0103] According to this embodiment, the method comprises a step of growing a genetically modified strain of *Pseudomonas putida* comprising a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase of sequence SEQ ID NO:2, SEQ ID NO:3 or SEQ ID NO:4, and comprising the additional recombinant genes below, encoding:

- **[0104]** a tyrosine ammonia lyase (TAL_RG_OPT), preferably a TAL_RG_OPT defined by the sequence SEQ ID NO: 7,
- **[0105]** a 4-coumarate-CoA ligase (4-CL), preferably a 4-CL defined by the sequence SEQ ID NO:8,

[0106] a benzalacetone synthase (BAS), preferably a BAS defined by the sequence SEQ ID NO:9.

[0107] The synthesis method according to the invention may also comprise a step of purifying and/or recovering the phenylpropanoid compound such as coumaric acid or frambinone.

[0108] According to a particular embodiment, the strain is a genetically modified strain of *Pseudomonas putida* comprising an additional recombinant AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, the sequence of which has at least 80% identity to the sequence SEQ ID NO:1, and in which the genes encoding the enzymes phhA and phhB are overexpressed.

Uses of the Strains According to the Invention

[0109] Another subject of the invention relates to the use of a strain of *Pseudomonas putida* as defined previously for the synthesis of phenylpropanoid compounds.

[0110] According to a preferred embodiment, the phenylpropanoid compound is selected from coumaric acid, p-coumaroyl-CoA, 4-hydroxybenzalacetone and frambinone, preferably from coumaric acid and/or frambinone.

[0111] The present invention will be better understood in light of the following nonlimiting examples, given solely by way of illustration and not intended to limit the scope of this invention as defined by the claims.

EXAMPLES

Example 1: Development of Mutants Enabling the Production of 3-Deoxy-D-Arabino-Heptulosonate-7-Phosphate (DAHP) Resistant to Tyrosine Feedback Inhibition

A. In Silico Study

[0112] The starting enzyme used is the AroF-I enzyme identified as 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) in *P. putida* (Uniprot identifier Q88KG6).

[0113] The tertiary structure (in PDB format) of this enzyme was reconstructed from the protein sequence thereof, using the methodology described in Waterhouse, A. et al., (SWISS-MODEL: homology modeling of protein structures and complexes. Nucleic Acids Res. 46(W1), W296-W303 (2018)). The basic structure selected by sequence homology was that of the 3-deoxy-D-arabino-heptulosonate-7-phosphate synthase of *Saccharomyces cerevisiae*.

[0114] The region characterizing tyrosine feedback inhibition has been identified in the literature via a homologous protein in *E. coli*. Using an alignment of the tertiary structures of these two enzymes, the region corresponding to this feedback inhibition in AroF-I was identified.

[0115] Next, the docking of a tyrosine molecule with the AroF-I enzyme was carried out using the methodology described by O. Trott et al., ("AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization and multithreading", Journal of Computational Chemistry, 2010).

[0116] This method made it possible to identify the amino acids involved in the formation of chemical bonds with the phenylalanine in the docking pocket; namely, inter alia, the amino acids at positions 160 (P160), 164 (Q164), 190 (S190), 191 (G191), 193 (S193) and 225 (1225).

[0117] By searching in the literature and also in the NCBI and Uniprot databases, we identified protein sequences similar to AroF-I but having a certain amount of amino acid diversity, particularly at the positions corresponding to those involved in docking.

[0118] This approach made it possible to identify amino acids that can be substituted for the amino acids naturally found in the docking pocket of the AroF-I protein without modifying the structure or activity thereof.

[0119] Thus, only binding to tyrosine is inhibited by the mutations shown. Table 1 below lists the positions and original amino acids as well as the substitutes which characterize the mutations that could potentially deactivate the AroF-I feedback inhibition function.

TABLE 1

Original amino acid (AroF-I)	Position in FASTA	Substitute amino acids
P	160	A, L, T, V, M
Q	164	A, R
S	190	A, F
G	191	K
S	193	A, G
I	225	P, V, T

[0120] Of the various mutations identified in table 1, a subset of mutations was chosen according to the properties of the substitute amino acids and also according to the impact of these mutations in evolutionarily-close proteins already described in the literature ((Cui, D et al., Molecular basis for feedback inhibition of tyrosine-regulated 3-deoxy-d-arabino-heptulosonate-7-phosphate synthase from *Escherichia coli,* 2019), (Ding, R. et al. Introduction of two mutations into AroG increases phenylalanine production in *Escherichia coli*. Biotechnol Lett 36, 2103-2108 (2014)), (Kikuchi Y et al. Mutational analysis of the feedback sites of phenylalanine-sensitive 3-deoxy-d-arabino-heptulosonate-7-phosphate synthase of *Escherichia coli*. Appl Environ Microbiol 63:761-762 (1997)).

[0121] The subset chosen is as follows: P160L, Q164A, S190A, G191K, S193A, and 1225P.

B. Experiments

Materials and Methods

Cloning of Genes Encoding the AroF-I Enzyme and the Derived Mutants

[0122] In order to express and purify the AroF-I enzyme and the mutants derived therefrom, the corresponding genes are cloned in a pET28_b(+) plasmid with a His tag grafted to the N-terminal part of the protein (upstream of the gene's ATG).

[0123] To produce these constructs, use is made of the Gibson Assembly method (Gibson et al., 2009). The reaction product is transformed in *E. coli* BL21(DE3) strains (Novagen). The transformed cells are plated out on agar medium with an antibiotic for selection. The colonies obtained are subsequently verified using PCR and sequenced in order to validate them.

Expression and Purification of the Enzymes

[0124] The *E. coli* BL21(DE3) strains containing the plasmids are cultured for 24 h in a ZYM autoinduction medium having the following composition:

[0125] a) Saline base:

[0126] yeast extract: 5 g/l

[0127] tryptone: 10 g/l

[0128] Na_2SO_4: 0.7 g/l

[0129] NH_4Cl: 2.7 g/l

[0130] KH_2PO_4: 3.4 g/l

[0131] Na_2HPO_4: 4.45 g/l

[0132] b) Solution of sugars (concentrated 50-fold):

[0133] Glycerol: 250 g/l

[0134] glucose: 25 g/l

[0135] lactose: 100 g/l

[0136] c) Solution of metals (concentrated 5000-fold):

TABLE 2

Solution of metals	Molar mass (g/mol)	Concentration (g/l)	Concentration (mM)	Final concentration (mM)	Per 100 ml(g)
$FeCl_3$	162.2	40.55	250	0.050	4.05
$CaCl_2$•$2H_2O$	147.01	14.70	100	0.100	1.47
$MnCl_2$	125.84	6.29	50	0.050	0.63
$ZnSO_4$•$7H_2O$	287.56	14.38	50	0.050	1.44
$CoCl_2$	129.84	1.30	10	0.010	0.13
$CuCl_2$	134.45	1.35	10	0.010	0.13
$NiCl_2$	129.60	1.30	10	0.010	0.13
Na_2MoO_4•$2H_2O$	241.95	2.42	10	0.010	0.24
Na_2SeO_3	172.94	1.73	10	0.010	0.17
$H3BO_3$	61.83	0.62	10	0.010	0.06

[0137] d) Preparation of complete medium:

TABLE 3

ZYM-5052	Per 50 ml
ZYM saline base (ml)	49.5
1M $MgSO_4$ (in µl)	100
Solution of metals, 5000-fold (in µl)	10
5052 solution, 50-fold (in µl)	1
Antibiotic 50 mg/ml (in µl)	50

[0138] The medium is seeded at OD=0.05 from an overnight preculture. The culture is carried out in a final volume of 50 ml in a 250 ml Erlenmeyer flask. The culture parameters are as follows: agitation at 140 rpm, 5 hours at 25° C. then 19 hours at 15° C.

[0139] All the chemical products used to prepare the medium come from Sigma-Aldrich, Roth or Euromedex.

[0140] After incubation, the 50 ml of culture are centrifuged for 10 minutes at 5000×g, 4° C. The supernatant is discarded and the pellet is taken up in 3 ml of LEW buffer (Macherey-Nagel, see below), 1 mg/ml of lyzosyme are added and the suspension is left over ice for 30 minutes.

[0141] The cells are subsequently lyzed using sonication (Omni Sonic Ruptor 400, Omni International, power 30%, pulse 40 for 8 minutes). After adding 0.22% of streptomycin, the samples are subsequently centrifuged for 30 minutes at 15 000×g, 4° C. The supernatant is recovered and the soluble proteins are purified following the supplier's recommendations, using the Protino® Ni-TED 2000 kit (Macherey-Nagel).

[0142] Finally, the purified proteins are concentrated ~10-fold using Amicon Ultra-4 30Kd centrifugal filters (Merck), and the elution buffer is replaced with 0.1 M Tris-HCl, pH 7.5+10% glycerol. The purified proteins are stored at −80° C.

Enzymatic Activity Assay

[0143] The activity of the enzymes is measured by monitoring the disappearance of the substrate using HPLC (High Pressure Liquid Chromatography) detection.

[0144] The reaction is performed in a final volume of 200 µl with 40 mM of phosphate buffer, pH 7, 300 µM of phospho(enol)pyruvate (PEP), 20 µg of purified enzyme, 1 mM of tyrosine (or another aromatic amino acid) or 3 µM of HCl, the volume being completed with ultrapure water.

[0145] After pre-incubation for 2 minutes at 30° C., the reaction is initiated by adding 300 µM of erythrose-4-phosphate (E4P) and the reaction mixture is incubated for 60 minutes at 30° C.

[0146] Finally, the mixture is heated for 5 minutes at 80° C. in order to stop the reaction and precipitate the proteins.

Measurement of the Consumption of PEP Using HPLC

[0147] The enzymatic activity assays are firstly centrifuged for 10 minutes at 15 000×g in order to precipitate the proteins. The supernatant obtained is filtered over a 0.22 µm membrane before HPLC analysis.

[0148] The method used is described below:

[0149] System: Agilent 1100 (Agilent)

[0150] Column: Luna OMEGA polar C18 (Phenomenex)

[0151] Mobile phase: 1 mM phosphoric acid

[0152] Flow rate: 0.3 ml/min

[0153] UV detection: 220 nm

[0154] Injection: 5 µl

[0155] Duration of analysis: 15 minutes

[0156] In order to quantify the amount of PEP present in the samples, a series of standards is analyzed, with PEP concentrations from 0.1 mM to 1 mM.

Measurement of the Concentration of PCA and CA Using HPLC

[0157] The column used is as follows: Kinetex® 5 µm F5 100A 150×4.6 mm

[0158] Mobile phases: 0.1% formic acid and acetonitrile

[0159] Gradient:

[0160] 5 min: 100% formic acid

[0161] 5 min to 25 min: 75% formic acid-25% acetonitrile

[0162] 25 min to 30 min: 62% formic acid-38% acetonitrile

[0163] 30 min to 35 min: 100% formic acid

[0164] Flow rate: 1 ml/min

[0165] Oven temperature: 40° C.

[0166] Sample injection: 10 µl

[0167] Coumaric acid detection: 315 nm

[0168] Cinnamic acid detection: 280 nm

Cloning the phhaA/B Genes in an Arabinose-Dependent Expression Plasmid

[0169] The phhA/B genes are amplified by PCR directly from gDNA of the *Pseudomonas putida* strain KT2440 and are cloned in a pBBR1-MCS2 plasmid under the control of the araC/pBAD promoter.

[0170] To produce these constructs, use is made of the Gibson Assembly method (Gibson et al., 2009). The reaction product is transformed in the *E. coli* BL21(DE3) strain (Novagen). The transformed cells are plated out on agar medium with an antibiotic for selection. The colonies obtained are subsequently verified using PCR and the plasmids are sequenced in order to validate them.

[0171] The expression plasmid is subsequently transformed in the donor *E. coli* strain S17.1 in order to be transferred into the *Pseudomonas putida* strains of interest by conjugation.

Results

[0172] The activity of the wild-type *Pseudomonas putida* AroF-I enzyme is indeed inhibited by tyrosine (FIG. **1**). Introduction of the P160L mutation makes it possible to make the enzyme partially resistant to this inhibition by tyrosine (FIG. **2**), and the effect can be slightly improved by introducing a P160L/Q164A double mutation (FIG. **3**). However, the P160L/G191K and P160L/S190 double mutations have a deleterious effect on the AroF-I enzyme, which no longer has any activity, whether in the presence or absence of tyrosine (FIG. **3**).

[0173] In contrast, and particularly beneficially and surprisingly, the introduction of a P160L/S193A double mutation restored virtually all the enzymatic activity and makes the AroF-I enzyme virtually completely resistant to tyrosine inhibition (FIG. **3**).

[0174] These results are confirmed by FIG. **4**, which shows the synergistic effect of this double mutation on the feedback inhibition caused by tyrosine compared to a single P160L or S193A mutation.

Example 2: Genetically Modified Strains of *Pseudomonas putida* for The Synthesis of Phenylpropanoid Compounds

[0175] In order to enable the biosynthesis of phenylpropanoid compounds, a plurality of genetic modifications are made within the *Pseudomonas putida* strain. To this end, additional recombinant genes encoding enzymes for the synthesis of phenylpropanoid compounds such as coumaric acid or frambinone are integrated in the *Pseudomonas putida* chromosome according to the following protocol.

Protocol for Genetic Mutagenesis of *Pseudomonas putida:*

[0176] Genetic mutagenesis in *Pseudomonas putida* is carried out by means of suicide plasmids which integrate into the chromosome and then leave it again, leaving behind the desired gene deletions or insertions. The suicide plasmid used is pK18mobsacB (Schäfer A, Tauch A, Jäger W, Kalinowski J, Thierbach G, Pühler A. Small mobilizable multi-purpose cloning vectors derived from the *Escherichia coli* plasmids pK18 and pK19: selection of defined deletions in the chromosome of *Corynebacterium glutamicum*. Gene 1994 Jul. 22; 145(1): 69-73). This plasmid carries a kanamycin antibiotic resistance cassette and the sacB counter-selection cassette. It also comprises an origin of replication which only works in the bacterium *E. coli*, and an origin of transfer, oriT, which enables it to be transferred by conjugation from *E. coli* to another bacterial strain such as *Pseudomonas putida.*

[0177] The genes to be inserted are cloned in this plasmid, as are the homologous regions at the chosen insertion site in the bacterial chromosome. These homologous regions are cloned on either side of the gene to be inserted, and should have a minimum size of 800 base pairs.

[0178] The clonings are carried out in an *Escherichia coli* strain that is able to conjugate (in general, the strain S17.1, Simon, R., Priefer, U. and A. Pulher, A broad host range mobilization system for in vivo genetic engineering: transposon mutagenesis in gram negative bacteria. Nature BioTechnology volume 1, pages 784-791 (1983)).

[0179] The conjugation protocol is as follows:

[0180] A culture droplet of S17.1/suicide plasma (donor strain) is deposited on an LB agar rich medium (Luria-Miller, Roth, reference X968.2, containing 1.5% agar), and a culture droplet of the *Pseudomonas putida* strain (recipient strain) is deposited on the first droplet. The culture dish is incubated overnight at 30° C. The droplet is subsequently diluted in 10 mM MgSO4 and the protocol is carried out as described in FIG. **6**.

[0181] The pK18mobsacB conjugation plasmid is an integrative plasmid, able to integrate into the genome of the recipient bacterial strain in order to produce transconjugants. Because the plasmid is resistant to kanamycin, the selection medium for the transconjugants is LB agar containing 100 mg/ml ampicillin (*Pseudomonas putida* is naturally resistant to this antibiotic), and 50 mg/ml kanamycin (kanamycin, ROTH reference T832.4; ampicillin, EUROMEDEX, reference EUO400-D). This medium thus makes it possible to select the *Pseudomonas putida* in which a plasmid has been integrated.

[0182] The rest of the protocol comprises the following steps:

- **[0183]** Excision of the suicide plasmid: sample 8 Kn+AmpR clones and culture them in 10 ml of 100 mg/ml LB+Amp medium (dilution 1/2000) at 30° C. with agitation. Incubate for between 12 h and 24 h.
- **[0184]** Counter-selection sacB on 25% sucrose: streak the culture of the pool of clones on YT agar and YT+sucrose 25% agar. Incubate at 30° C. until the following morning. Re-sample 20-25 clones of sucrose+YT agar, YT+sucrose 25% agar, and YT agar+Kn 50 mg/ml. Incubate at 30° C. until the following morning. YT: Yeast extract tryptone
- **[0185]** Verification of gene insertion using PCR.

[0186] Primers that hybridize on either side of the chromosomal insertion region make it possible to verify the mutant clones by carrying out colony PCR for the KnS and Sucrose+clones.

Genes to be Inserted into *Pseudomonas putida*

[0187] One or more of the genes listed below are inserted into *Pseudomonas putida* in order to enable the biosynthesis of phenylpropanoid compounds, for instance coumaric acid or frambinone.

[0188] These genes have been specifically identified and selected in known microorganisms. The genes are synthesized and the codons are optimized for maximum expression in *Pseudomonas putida.*

[0189] Some genes are endogenous to *Pseudomonas putida*, but it is recommended to test the activity of the proteins they code for and, if these enzymes are not active enough or are insufficiently expressed, these activities can be optimized.

Genes Encoding a Mutated 3-Deoxy-D-Arabino-Heptulosonate-7-Phosphate (DAHP) Synthase Enzyme:

[0190] SEQ ID NO:10: sequence of the 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase gene with P160L mutation (*Pseudomonas putida* KT2440), encoding DAHP synthase of sequence SEQ ID NO:2.

[0191] SEQ ID NO:11: sequence of the 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase gene with P160L/Q164A double mutation (*Pseudomonas putida* KT2440), encoding DAHP synthase of sequence SEQ ID NO:3.

[0192] SEQ ID NO:12: sequence of the 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase gene with P160L/S193A double mutation (*Pseudomonas putida* KT2440), encoding DAHP synthase of sequence SEQ ID NO:4.

Additional Genes Enabling the Synthesis of Phenylpropanoid Compounds

[0193] SEQ ID NO:13: gene (*Pseudomonas putida* KT2440) for phenylalanine hydroxylase (phhA), encoding the phhA of sequence SEQ ID NO:5.

[0194] SEQ ID NO:14: gene (*Pseudomonas putida* KT2440) for tetrahydrobiopterin dehydratase (phhB), encoding the phhB of sequence SEQ ID NO:6.

[0195] SEQ ID NO:15: gene (*Rhodotorula glutinis*) for tyrosine ammonia lyase, encoding the beta-xylosidase of sequence SEQ ID NO:7.

[0196] SEQ ID NO:16: gene (*Pseudomonas putida* KT2440 fcs gene) for 4-coumarate-CoA ligase (4-CL), encoding the 4-CL of sequence SEQ ID NO:8.

[0197] SEQ ID NO:17: gene (Rheum *palmatum*) for benzalacetone synthase (BAS), encoding the BAS of sequence SEQ ID NO: 9.

Example 3: In Vivo Activity of a Genetically Modified Strain of *Pseudomonas putida* According to the Invention

Preparation of an AroF-I-Fbr P160L/S193A Mutant

[0198] A *Pseudomonas putida* KT2440 strain was genetically modified as described in example 1 so as to express the AroF-I gene encoding DAHP synthase comprising the P160L/S193A double mutation defined by the amino acid sequence SEQ ID NO: 4 and to express the additional recombinant gene encoding a heterologous tyrosine ammonia lyase (TAL_RG_OPT) of sequence SEQ ID NO: 7.

[0199] This strain is referred to as “AroF-I-fbr P160L/S193A mutant”.

[0200] The activity of the AroF-I-fbr P160L/S193A mutant on the production of coumaric acid (PCA) or cinnamic acid (CA) was determined by measuring the quantities produced of these acids compared to a *Pseudomonas putida* KT2440 strain genetically modified to express the wild-type AroF-I enzyme, the latter strain serving as control (AroF-I-WT mutant).

Results

[0201] Cinnamic acid production increases significantly in a *P. putida* strain expressing the AroF-I enzyme having the P160L/S193A double mutation (FIG. **6**).

Optimization of the AroF-I-Fbr P160L/S193A Mutant for the Production Of Coumaric Acid.

[0202] The AroF-I-fbr P160L/S193A mutant was modified in order to express the additional recombinant genes encoding a phenylalanine hydroxylase (phhA) of sequence SEQ ID NO: 5, and a tetrahydrobiopterin dehydratase (phhB) of sequence SEQ ID NO: 6. The genes encoding these enzymes were cloned in a plasmid and expressed under the control of the araC/pBADopt inducible promoter, induction being carried out with 0.5% arabinose: plasmid pC2F387.

[0203] This strain is referred to as “optimized AroF-I-fbr mutant”.

[0204] The production of coumaric acid (PCA) or cinnamic acid (CA) by the optimized AroF-I-fbr mutant was measured then compared to that obtained with the AroF-I-fbr P160L/S193A mutant and that obtained with two control strains, namely:

- **[0205]** a *Pseudomonas putida* KT2440 strain expressing the wild-type AroF-I enzyme (AroF-I-WT mutant), and
- **[0206]** a *Pseudomonas putida* KT2440 strain expressing the wild-type AroF-I and the two phhA and phhB enzymes of sequences SEQ ID NO:5 and SEQ ID NO:6, respectively (AroF-I-WT phhA/B mutant).

Results

[0207] The results presented in FIG. **6** show that the overexpression of the phhA and phhB genes in the AroF-I-WT mutant strain leads to greater production of phenolic acids (PCA+CA).

[0208] Particularly interestingly, the optimized AroF-I-fbr mutant makes it possible to obtain production of total phenolic acids that is much higher than the AroF-I-fbr P160L/S193A mutant and the AroF-I-WT phhA/B mutant. Proportionally, coumaric acid represents virtually all of the total phenolics produced, and this is advantageous since cinnamic acid is not an industrially useful compound for the production of phenylpropanoid compounds.

[0209] These results clearly demonstrate that the optimized AroF-I-fbr mutant is particularly suited to the production of phenylpropanoid compounds, and particularly of coumaric acid.

Free Sequence Listing Text

[0210] In the present application, reference is made to sequence listings, the identifiers (“SEQ ID NO”) of which are listed in the table below. Regardless of the form in which the listings are provided, they form part of the present application.

TABLE 4

SEQ ID NO: 1 AA Wild-type AroF-I	LAAGTRDLTSNTMADLPIDDLNVASNETLITPDQLKKE IPLSAKALQTVTAGREWVRNILDGKDHRLFVVVGPCSI HDIKAAHEYAERLKVLAEEVSDTLYLVMRVYFEKPRTT VGWKGLINDPYLDDSFKIQDGLHIGRKLLLDLAEMGLP TATEALDPISPQYLQDLISWSAIGARTTESQTHREMAS GLSSAVGFKNGTDGGLTVAINALQSVSKPHRFLGINQE GGVSIVTTKGNPYGHWVLRGGNGKPNYDSVSVALCEQD LAKAKIKANIMVDCSHANSNKDPALQPLVMENVANQIL EGNQSIIGLMVESHLNWGCQSIPKNLDDLQYGVSITDA CIDWSATEKTLRSMHAKLKDVLPQRKRG
SEQ ID NO: 2 AA AroF-I_P160L	LAAGTRDLTSNTMADLPIDDLNVASNETLITPDQLKKE IPLSAKALQTVTAGREWVRNILDGKDHRLFVVVGPCSI HDIKAAHEYAERLKVLAEEVSDTLYLVMRVYFEKPRTT VGWKGLINDPYLDDSFKIQDGLHIGRKLLLDLAEMGLP TATEALDLISPQYLQDLISWSAIGARTTESQTHREMAS GLSSAVGFKNGTDGGLTVAINALQSVSKPHRFLGINQE GGVSIVTTKGNPYGHVVLRGGNGKPNYDSVSVALCEQD LAKAKIKANIMVDCSHANSNKDPALQPLVMENVANQIL EGNQSIIGLMVESHLNWGCQSIPKNLDDLQYGVSITDA CIDWSATEKTLRSMHAKLKDVLPQRKRG
SEQ ID NO: 3 AA AroF-I_P160L/ Q164A	LAAGTRDLTSNTMADLPIDDLNVASNETLITPDQLKKE IPLSAKALQTVTAGREWVRNILDGKDHRLFVVVGPCSI HDIKAAHEYAERLKVLAEEVSDTLYLVMRVYFEKPRTT VGWKGLINDPYLDDSFKIQDGLHIGRKLLLDLAEMGLP TATEALDLISPAYLQDLISWSAIGARTTESQTHREMAS GLSSAVGFKNGTDGGLTVAINALQSVSKPHRFLGINQE GGVSIVTTKGNPYGHVVLRGGNGKPNYDSVSVALCEQD LAKAKIKANIMVDCSHANSNKDPALQPLVMENVANQIL EGNQSIIGLMVESHLNWGCQSIPKNLDDLQYGVSITDA CIDWSATEKTLRSMHAKLKDVLPQRKRG
SEQ ID NO: 4 AA AroF-I_P160L/ S193A	LAAGTRDLTSNTMADLPIDDLNVASNETLITPDQLKKE IPLSAKALQTVTAGREWVRNILDGKDHRLFVVVGPCSI HDIKAAHEYAERLKVLAEEVSDTLYLVMRVYFEKPRTT VGWKGLINDPYLDDSFKIQDGLHIGRKLLLDLAEMGLP TATEALDLISPQYLQDLISWSAIGARTTESQTHREMAS GLASAVGFKNGTDGGLTVAINALQSVSKPHRFLGINQE GGVSIVTTKGNPYGHVVLRGGNGKPNYDSVSVALCEQD LAKAKIKANIMVDCSHANSNKDPALQPLVMENVANQIL EGNQSIIGLMVESHLNWGCQSIPKNLDDLQYGVSITDA CIDWSATEKTLRSMHAKLKDVLPQRKRG
SEQ ID NO: 5 AA Phenylalanine hydroxylase (phhA)	MKQTQYVAREPDAHGFIDYPQQEHAVWNTLITRQLKVI EGRACQEYLDGIDQLKLPHDRIPQLGEINKVLGATTGW QVARVPALIPFQTFFELLASKRFPVATFIRTPEELDYL QEPDIFHEIFGHCPLLTNPWFAEFTHTYGKLGLAATKE QRVYLARLYWMTIEFGLMETAQGRKIYGGGILSSPKET VYSLSDEPEHQAFDPIEAMRTPYRIDILQPVYFVLPNM KRLFDLAHEDIMGMVHKAMQLGLHAPKFPPKVAA
SEQ ID NO: 6 AA Tetrahydrobiopterin dehydratase (phhB)	MNALNQAHCEACRADAPKVTDEELAELIREIPDWNIEV RDGHMELERVFLFKNFKHALAFTNAVGEIAEAEGHHPG LLTEWGKVTVTWWSHSIKGLHRNDFIMCARTDKVAETA EGRK
SEQ ID NO: 7 AA Tyrosine ammonia lyase (TAL_RG_OPT)	MAPRPTSQNQTRTCPTTQVTQVDIVEKMLAAPTDSTLE LDGYSLNLGDVVSAARKGRPVRVKDSDEIRSKIDKSVE FLRSQLSMSVYGVTTGFGGSADTRTEDAISLQKALLEH QLCGVLPSSFDSFRLGRGLENSLPLEVVRGAMTIRVNS LTRGHSAVRLVVLEALTNFLNHGITPIVPLRGTISASG DLSPLSYIAAAISGHPDSKVHVVHEGKEKILYAREAMA LFNLEPVVLGPKEGLGLVNGTAVSASMATLALHDAHML SLLSQSLTAMTVEAMVGHAGSFHPFLHDVTRPHPTQIE VAGNIRKLLEGSRFAVHHEEEVKVKDDEGILRQDRYPL RTSPQWLGPLVSDLIHAHAVLTIEAGQSTTDNPLIDVE NKTSHHGGNFQAAAVANTMEKTRLGLAQIGKLNFTQLT EMLNAGMNRGLPSCLAAEDPSLSYHCKGLDIAAAAYTS ELGHLANPVTTHVQPAEMANQAVNSLALISARRTTESN DVLSLLLATHLYCVLQAIDLRAIVFEFKKQFGPAIVSL IDQHFGSAMTGSNLRDELVEKVNKTLAKRLEQTNSYDL VPRWHDAFSFAAGTVVEVLSSTSLSLAAVNAWKVAAAE SAISLTRQVRETFWSAASTSSPALSYLSPRTQILYAFV REELGVKARRGDVFLGKQEVTIGSNVSKIYEAIKSGRI NNVLLKMLA

TABLE 4-continued

SEQ ID NO: 8 AA 4-coumarate-CoA ligase (4-CL)	MNNEARSGSTDPGQRPRYRQVAIGHPQVQVSHVDDVLR MQPVEPLAPLPARLLERLVHWAQVRPDTTFIAARQADG AWRSISYVQMLADVRTIAANLLGLGLSAERPLALLSGN DIEHLQIALGAMYAGIAYCPVSPAYALLSQDFAKLRHV CEVLTPGVVFVSDSQPFQRAFEAVLDDSVGVISVRGQV AGRPHISFDSLLQPGDLAAADAAFAATGPDTIAKFLFT SGSTKLPKAVITTQRMLCANQQMLLQTFPTFAEEPPVL VDWLPWNHTFGGSHNLGIVLYNGGSFYLDAGKPTPQGF AETLRNLREISPTAYLTVPKGWEELVKALEQDPALREV FFARIKLFFFAAAGLSQSVWDRLDRIAEQHCGERIRMM AGLGMTEASPSCTFTTGPLSMAGYVGLPAPGCEVKLVP VGDKLEARFRGPHIMPGYWRSPQQTAEAFDEEGFYCSG DALKLADARQPELGLMFDGRIAEDFKLSSGVFVSVGPL RNRAVLEGSPYVQDIVVTAPDRECLGLLVFPRLPECRR LAGLAEDASDARVLANDTVRSWFADWLERLNRDAQGNA SRIEWLSLLAEPPSIDAGEITDKGSINQRAVLQRRAAQ VEALYRGEDPDALHAKVRP
SEQ ID NO: 9 AA benzalacetone synthase (BAS)	MATEEMKKLATVMAIGTANPPNCYYQADFPDFYFRVTN SDHLINLKQKFKRLCENSRIEKRYLHVTEEILKENPNI AAYEATSLNVRHKMQVKGVAELGKEAALKAIKEWGQPK SKITHLIVCCLAGVDMPGADYQLTKLLDLDPSVKRFMF YHLGCYAGGTVLRLAKDIAENNKGARVLIVCSEMTTTC FRGPSETHLDSMIGQAILGDGAAAVIVGADPDLTVERP IFELVSTAQTIVPESHGAIEGHLLESGLSFHLYKTVPT LISNNIKTCLSDAFTPLNISDWNSLFWIAHPGGPAILD QVTAKVGLEKEKLKVTRQVLKDYGNMSSATVFFIMDEM RKKSLENGQATTGEGLEWGVLFGFGPGITVETWLRSVP VIS
SEQ ID NO: 10 Nt AroF-I_P160L	5'-TTGGCGGCCGGCACCCGTGACCTGACGAGTAACAC GATGGCTGATTTACCGATCGATGACTTGAACGTTGCCT CCAACGAGACCCTGATCACCCCTGATCAGCTCAAGAAG GAAATCCCCCTCAGCGCCAAGGCCCTGCAGACCGTGAC TGCCGGCCGTGAAGTGGTGCGCAATATTCTCGACGGCA AGGACCATCGCCTGTTCGTCGTGGTCGGCCCTTGCTCC ATCCACGACATCAAGGCAGCCCACGAATACGCCGAGCG CCTGAAAGTGCTGGCCGAAGAAGTGTCCGATACGCTGT ACCTGGTCATGCGCGTGTACTTCGAAAAGCCGCGCACC ACCGTCGGCTGGAAAGGCCTGATCAACGATCCGTACCT GGATGACTCGTTCAAGATCCAGGACGGCCTGCACATCG GCCGCAAGTTGCTGCTGGACCTGGCCGAAATGGGCCTG CCGACCGCCACCGAAGCGCTCGACCTGATTTCGCCGCA GTACCTGCAAGACCTGATCAGCTGGTCGGCCATCGGTG CCCGCACCACCGAATCGCAAACACACCGCGAGATGGCC TCGGGCCTGTCCTCGGCGGTGGGTTTCAAGAACGGTAC CGATGGCGGCCTGACCGTTGCCATCAATGCCCTGCAGT CGGTGTCCAAGCCGCACCGCTTCCTGGGCATCAACCAG GAAGGCGGCGTGTCGATCGTCACCACCAAGGGCAACCC ATACGGCCACGTGGTACTGCGCGGCGGCAATGGCAAGC CGAACTACGACTCGGTCAGCGTCGCCCTGTGCGAACAG GACCTGGCCAAGGCCAAGATCAAGGCCAACATCATGGT CGACTGCAGCCATGCCAACTCCAACAAGGACCCGGCCC TGCAACCGCTGGTGATGGAGAACGTCGCCAACCAGATT CTCGAAGGCAACCAGTCGATCATCGGCCTGATGGTCGA AAGCCACCTGAACTGGGGCTGTCAGTCCATTCCGAAAA ACCTGGACGATTTGCAGTATGGCGTGTCGATCACGGAC GCCTGCATCGACTGGTCGGCTACCGAGAAAACCCTGCG CAGCATGCATGCCAAGCTCAAGGATGTGCTGCCGCAGC GTAAGCGCGGCTGA-3'
SEQ ID NO: 11 Nt AroF-I_P16L/Q164A	5'-TTGGCGGCCGGCACCCGTGACCTGACGAGTAACAC GATGGCTGATTTACCGATCGATGACTTGAACGTTGCCT CCAACGAGACCCTGATCACCCCTGATCAGCTCAAGAAG GAAATCCCCCTCAGCGCCAAGGCCCTGCAGACCGTGAC TGCCGGCCGTGAAGTGGTGCGCAATATTCTCGACGGCA AGGACCATCGCCTGTTCGTCGTGGTCGGCCCTTGCTCC ATCCACGACATCAAGGCAGCCCACGAATACGCCGAGCG CCTGAAAGTGCTGGCCGAAGAAGTGTCCGATACGCTGT ACCTGGTCATGCGCGTGTACTTCGAAAAGCCGCGCACC ACCGTCGGCTGGAAAGGCCTGATCAACGATCCGTACCT GGATGACTCGTTCAAGATCCAGGACGGCCTGCACATCG GCCGCAAGTTGCTGCTGGACCTGGCCGAAATGGGCCTG CCGACCGCCACCGAAGCGCTCGACCTGATTTCGCCGGC CTACCTGCAAGACCTGATCAGCTGGTCGGCCATCGGTG CCCGCACCACCGAATCGCAAACACACCGCGAGATGGCC TCGGGCCTGTCCTCGGCGGTGGGTTTCAAGAACGGTAC

TABLE 4-continued

	CGATGGCGGCCTGACCGTTGCCATCAATGCCCTGCAGT
	CGGTGTCCAAGCCGCACCGCTTCCTGGGCATCAACCAG
	GAAGGCGGCGTGTCGATCGTCACCACCAAGGGCAACCC
	ATACGGCCACGTGGTACTGCGCGGCGGCAATGGCAAGC
	CGAACTACGACTCGGTCAGCGTCGCCCTGTGCGAACAG
	GACCTGGCCAAGGCCAAGATCAAGGCCAACATCATGGT
	CGACTGCAGCCATGCCAACTCCAACAAGGACCCGGCCC
	TGCAACCGCTGGTGATGGAGAACGTCGCCAACCAGATT
	CTCGAAGGCAACCAGTCGATCATCGGCCTGATGGTCGA
	AAGCCACCTGAACTGGGGCTGTCAGTCCATTCCGAAAA
	ACCTGGACGATTTGCAGTATGGCGTGTCGATCACGGAC
	GCCTGCATCGACTGGTCGGCTACCGAGAAAACCCTGCG
	CAGCATGCATGCCAAGCTCAAGGATGTGCTGCCGCAGC
	GTAAGCGCGGCTGA-3'
SEQ ID NO: 12	5'-TTGGCGGCCGGCACCCGTGACCTGACGAGTAACAC
Nt	GATGGCTGATTTACCGATCGATGACTTGAACGTTGCCT
AroF-I_P160L/	CCAACGAGACCCTGATCACCCCTGATCAGCTCAAGAAG
S193A	GAAATCCCCCTCAGCGCCAAGGCCCTGCAGACCGTGAC
	TGCCGGCCGTGAAGTGGTGCGCAATATTCTCGACGGCA
	AGGACCATCGCCTGTTCGTCGTGGTCGGCCCTTGCTCC
	ATCCACGACATCAAGGCAGCCCACGAATACGCCGAGCG
	CCTGAAAGTGCTGGCCGAAGAAGTGTCCGATACGCTGT
	ACCTGGTCATGCGCGTGTACTTCGAAAAGCCGCGCACC
	ACCGTCGGCTGGAAAGGCCTGATCAACGATCCGTACCT
	GGATGACTCGTTCAAGATCCAGGACGGCCTGCACATCG
	GCCGCAAGTTGCTGCTGGACCTGGCCGAAATGGGCCTG
	CCGACCGCCACCGAAGCGCTCGACCTGATTTCGCCGCA
	GTACCTGCAAGACCTGATCAGCTGGTCGGCCATCGGTG
	CCCGCACCACCGAATCGCAAACACACCGCGAGATGGCC
	TCGGGTTTGGCATCGGCGGTGGGTTTCAAGAACGGTAC
	CGATGGCGGCCTGACCGTTGCCATCAATGCCCTGCAGT
	CGGTGTCCAAGCCGCACCGCTTCCTGGGCATCAACCAG
	GAAGGCGGCGTGTCGATCGTCACCACCAAGGGCAACCC
	ATACGGCCACGTGGTACTGCGCGGCGGCAATGGCAAGC
	CGAACTACGACTCGGTCAGCGTCGCCCTGTGCGAACAG
	GACCTGGCCAAGGCCAAGATCAAGGCCAACATCATGGT
	CGACTGCAGCCATGCCAACTCCAACAAGGACCCGGCCC
	TGCAACCGCTGGTGATGGAGAACGTCGCCAACCAGATT
	CTCGAAGGCAACCAGTCGATCATCGGCCTGATGGTCGA
	AAGCCACCTGAACTGGGGCTGTCAGTCCATTCCGAAAA
	ACCTGGACGATTTGCAGTATGGCGTGTCGATCACGGAC
	GCCTGCATCGACTGGTCGGCTACCGAGAAAACCCTGCG
	CAGCATGCATGCCAAGCTCAAGGATGTGCTGCCGCAGC
	GTAAGCGCGGCTGA-3'
SEQ ID NO: 13	5'-ATGAAACAGACGCAATACGTGGCACGCGAGCCCGA
Nt	TGCGCATGGTTTTATCGATTACCCGCAGCAAGAGCATG
Phenylalanine	CGGTGTGGAACACCCTGATCACCCGCCAGCTGAAAGTG
hydroxylase (phhA)	ATCGAAGGCCGTGCGTGCCAGGAATACCTGGACGGCAT
	CGACCAGCTGAAATTGCCGCATGACCGCATTCCGCAAC
	TGGGCGAGATCAACAAGGTGCTGGGTGCCACCACCGGC
	TGGCAGGTTGCCCGGGTTCCGGCGCTGATCCCCTTCCA
	GACCTTCTTCGAATTGCTGGCCAGCAAGCGCTTTCCGG
	TCGCCACCTTCATCCGCACCCCGGAAGAGCTGGACTAC
	CTGCAAGAGCCGGATATCTTCCACGAGATCTTCGGCCA
	CTGCCCGCTGCTGACCAATCCCTGGTTCGCCGAATTCA
	CCCACACCTACGGCAAGCTCGGCCTGGCCGCGACCAAG
	GAACAACGTGTGTACCTGGCACGCTTGTACTGGATGAC
	CATCGAGTTTGGCCTGATGGAAACCGCGCAAGGCCGCA
	AAATCTATGGTGGTGGCATCCTCTCGTCGCCGAAAGAG
	ACCGTCTACAGTCTGTCTGACGAGCCTGAGCACCAGGC
	CTTCGACCCGATCGAGGCCATGCGTACACCCTACCGCA
	TCGACATTCTGCAACCGGTGTATTTCGTACTGCCGAAC
	ATGAAGCGCCTGTTCGACCTGGCCCACGAGGACATCAT
	GGGCATGGTCCATAAAGCCATGCAGCTGGGTCTGCATG
	CACCGAAGTTTCCACCCAAGGTCGCTGCCTGA-3'
SEQ ID NO: 14	5'-ATGAATGCCTTGAACCAAGCCCATTGCGAAGCCTG
Nt	CCGCGCCGACGCACCGAAAGTCACCGACGAAGAGCTGG
Tetrahydrobiopterin	CCGAGCTGATCCGCGAAATCCCGGACTGGAACATCGAA
dehydratase (phhB)	GTACGTGACGGCCACATGGAGCTGGAGCGCGTGTTCCT
	GTTCAAGAACTTCAAGCACGCCCTGGCGTTCACCAATG
	CCGTGGGCGAAATTGCCGAAGCCGAAGGCCACCACCCA
	GGGCTGCTGACTGAATGGGGCAAGGTCACCGTGACCTG

TABLE 4-continued

	GTGGAGCCACTCGATCAAAGGCCTGCACCGCAACGACT TCATCATGTGCGCACGCACTGACAAGGTGGCGGAAACG GCTGAAGGCCGGAAGTAA-3'
SEQ ID NO: 15 Nt Tyrosine ammonia lyase (TAL_RG_OPT)	5'-ATGGCCCCTCGCCCTACCTCACAGAACCAAACCCG CACATGCCCGACGACGCAGGTTACTCAAGTTGATATAG TTGAGAAGATGCTCGCTGCACCAACTGACAGCACCCTA GAGCTCGACGGGTATTCACTAAATCTTGGGGACGTCGT TTCAGCTGCAAGGAAAGGAAGACCTGTAAGAGTAAAAG ATAGTGATGAAATTCGGAGTAAAATAGATAAGTCCGTA GAGTTTTTAAGGTCACAACTTAGCATGTCCGTATACGG GGTCACTACCGGGTTCGGCGGTTCCGCCGACACCCGCA CCGAGGACGCTATATCATTGCAGAAAGCTCTTCTAGAG CATCAGCTCTGCGGCGTTCTTCCAAGTTCCTTCGATTC GTTTAGGCTGGGGCGCGGGCTTGAGAACTCTCTGCCCC TAGAAGTGGTAAGGGGCGCTATGACAATACGGGTGAAC AGTCTAACAAGAGGTCACAGCGCGGTTAGACTAGTTGT ACTTGAAGCTCTGACTAACTTCTTAAACCACGGGATTA CCCCGATTGTCCCACTCCGGGGAACCATCAGTGCGTCC GGTGACCTATCGCCCCTCTCATATATTGCGGCAGCTAT ATCAGGACATCCAGATTCAAAGGTTCATGTAGTACATG AAGGAAAAGAGAAAATACTTTACGCACGCGAGGCCATG GCCCTTTTTAACCTCGAGCCCGTGGTACTTGGTCCGAA AGAGGGCCTCGGACTAGTTAACGGTACTGCCGTCAGTG CCTCAATGGCTACGCTTGCACTCCACGATGCGCACATG CTGAGCCTGCTAAGTCAAAGTCTCACAGCGATGACCGT GGAGGCCATGGTGGGGCATGCGGGGTCATTTCATCCAT TTTTGCATGATGTCACTCGTCCGCATCCTACGCAGATT GAGGTAGCAGGCAACATTCGCAAGCTTCTCGAGGGAAG TCGTTTCGCCGTCCATCATGAGGAAGAAGTAAAAGTAA AGGATGACGAAGGAATATTAAGGCAAGACCGATACCCG CTCCGCACGTCACCGCAATGGTTGGGTCCACTGGTTTC AGACCTCATCCACGCACACGCCGTCTTAACTATTGAAG CAGGGCAATCGACGACAGACAATCCTCTCATCGACGTA GAGAATAAGACCTCGCATCATGGAGGAAATTTTCAAGC TGCAGCTGTCGCGAACACAATGGAAAAGACACGTCTCG GCCTGGCGCAAATAGGGAAACTGAATTTCACCCAGCTC ACGGAAATGCTGAACGCCGGCATGAACCGCGGCCTGCC GTCTTGTCTCGCCGCGGAAGATCCTTCTTTATCATATC ACTGTAAGGGTTTAGATATCGCGGCAGCTGCATATACG TCCGAACTAGGTCATCTGGCTAACCCTGTCACGACCCA CGTACAACCGGCGGAGATGGCTAATCAAGCAGTTAACT CCCTTGCACTAATTTCCGCCCGCCGGACAACAGAGAGT AACGACGTGTTATCACTGCTGCTCGCTACCCACTTATA CTGCGTCTTGCAGGCTATCGACTTACGCGCAATCGTGT TCGAATTTAAGAAGCAATTCGGGCCAGCTATTGTGTCC CTAATTGATCAGCACTTCGGAAGCGCCATGACTGGGTC TAATCTTCGAGACGAGCTAGTCGAAAAAGTAAATAAGA CACTCGCAAAGAGGCTGGAACAGACTAACAGCTACGAC CTAGTTCCACGGTGGCACGACGCCTTTAGTTTTGCAGC GGGAACGGTAGTAGAGGTATTGTCATCGACTTCGTTGT CGTTGGCTGCTGTCAACGCGTGGAAAGTTGCAGCTGCA GAGTCAGCAATTTCGCTGACGCGGCAAGTACGCGAAAC ATTTTGGAGCGCTGCTTCGACAAGCTCGCCAGCCCTTT CTTACCTGTCCCCACGTACGCAGATCTTGTACGCATTC GTAAGAGAGGAGTTAGGAGTCAAAGCCCGAAGGGGTGA CGTATTCCTTGGAAAGCAAGAAGTTACAATTGGATCCA ACGTTTCAAAGATCTATGAGGCCATTAAGAGTGGGCGC ATAAATAACGTCCTGTTGAAGATGCTGGCCTGA-3'
SEQ ID NO: 16 Nt 4-coumarate-CoA ligase (4-CL)	5'-GTGAATAACGAAGCCCGCTCAGGGTCGACCGACCC TGGCCAACGTCCGCGCTACCGCCAGGTGGCCATCGGGC ATCCCCAGGTGCAGGTCAGTCACGTCGACGACGTGCTG CGCATGCAACCTGTCGAGCCACTGGCGCCGCTGCCGGC GCGCCTGCTCGAGCGCCTGGTGCATTGGGCCCAGGTGC GCCCGGACACCACTTTCATCGCGGCACGCCAGGCAGAC GGTGCCTGGCGTTCGATCAGCTACGTGCAGATGCTCGC CGATGTGCGCACCATCGCCGCCAACTTGCTAGGACTGG GCCTCAGTGCCGAGCGCCCGCTGGCGCTGCTTTCCGGC AACGACATCGAACACCTGCAAATCGCCCTCGGCGCCAT GTATGCCGGTATTGCCTATTGCCCGGTGTCGCCGGCCT ACGCGCTGTTGTCGCAAGACTTCGCCAAGTTGCGCCAT GTCTGCGAGGTGCTCACCCCCGGAGTGGTCTTCGTCAG CGACAGCCAGCCGTTCCAGCGCGCCTTCGAGGCGGTGC TGGACGATTCGGTCGGCGTGATCAGCGTGCGTGGCCAG GTCGCAGGTCGCCCCCATATAAGCTTCGACAGCCTGTT GCAACCGGGTGACCTGGCGGCGGCCGATGCGGCTTTCG

TABLE 4-continued

	CCGCCACCGGGCCGGACACCATCGCCAAATTCCTCTTC ACCTCGGGCTCGACCAAGCTGCCCAAGGCGGTGATCAC CACCCAGCGCATGCTGTGCGCCAATCAGCAGATGCTTC TGCAGACTTTTCCGACGTTCGCCGAGGAGCCGCCGGTG CTGGTGGACTGGCTGCCGTGGAACCACACGTTCGGCGG TAGCCACAACCTCGGCATCGTGCTTTACAACGGGGGCA GTTTCTACCTGGACGCCGGCAAGCCGACCCCGCAAGGC TTCGCCGAGACCTTGCGCAATCTGCGCGAGATTTCCCC CACGGCCTACCTCACCGTACCCAAGGGCTGGGAGGAAC TGGTCAAGGCACTGGAGCAGGACCCCGCGCTACGCGAG GTGTTCTTTGCCCGCATCAAGCTGTTCTTCTTTGCCGC CGCAGGCCTGTCGCAAAGCGTCTGGGACCGGCTGGACC GCATTGCCGAGCAACACTGTGGCGAACGCATCCGCATG ATGGCCGGCCTTGGCATGACCGAAGCCTCGCCATCGTG CACCTTCACCACCGGGCCTTTGTCGATGGCCGGCTATG TCGGGCTGCCGGCACCTGGCTGCGAAGTGAAGCTGGTG CCGGTGGGCGACAAGCTCGAGGCGCGCTTCCGTGGCCC GCATATCATGCCGGGCTACTGGCGCTCGCCGCAGCAGA CCGCCGAGGCGTTCGACGAGGAGGGCTTCTACTGTTCG GGCGACGCGTTGAAGCTGGCCGATGCCAGGCAGCCCGA GCTTGGCCTGATGTTCGATGGCCGTATCGCTGAGGACT TCAAACTTTCGTCCGGGGTATTCGTCAGTGTCGGGCCG CTGCGCAACCGCGCAGTGCTGGAGGGCTCGCCTTACGT ACAGGACATCGTGGTCACCGCGCCGGACCGTGAATGCC TGGGCCTGCTGGTGTTCCCGCGTCTGCCCGAGTGTCGG CGCCTGGCCGGGCTGGCAGAGGATGCCAGCGATGCGCG GGTGCTGGCCAACGACACCGTGCGCAGTTGGTTCGCTG ACTGGCTGGAGCGCTTGAACCGCGATGCCCAAGGCAAC GCCAGCCGTATCGAATGGCTGTCGCTGCTGGCCGAGCC GCCGTCGATCGACGCCGGTGAAATCACCGACAAGGGCT CGATCAATCAGCGCGCCGTGCTGCAGCGGCGCGCCGCT CAGGTCGAGGCGCTGTACCGTGGCGAAGACCCCGACGC ATTGCACGCCAAGGTGCGGCCTTGA-3'
SEQ ID NO: 17 Nt Benzalacetone synthase (BAS)	5'-ATGGCAACTGAGGAGATGAAGAAATTGGCCACCGT GATGGCCATTGGCACGGCCAACCCTCCGAACTGCTACT ACCAGGCCGACTTTCCCGACTTCTACTTCCGCGTCACC AACAGCGACCACCTCATCAACCTCAAGCAAAAGTTCAA GCGCCTTTGTGAAAACTCAAGGATTGAGAAGCGTTACC TTCATGTGACCGAAGAGATTCTCAAGGAAAACCCAAAC ATTGCTGCCTACGAGGCAACCTCGTTGAATGTAAGACA CAAAATGCAAGTGAAAGGAGTTGCAGAGCTTGGGAAAG AGGCTGCCCTCAAGGCCATCAAAGAATGGGGCCAACCC AAGTCCAAGATCACACATCTCATCGTGTGTTGCCTAGC CGGCGTTGACATGCCCGGCGCGGATTATCAACTCACTA AGCTTCTTGACCTTGACCCTTCCGTCAAGCGTTTTATG TTTTACCACCTAGGATGCTACGCTGGTGGCACTGTCCT TCGCCTTGCAAAGGACATAGCGGAGAACAACAAGGGAG CTCGTGTTCTCATCGTTTGCTCAGAGATGACAACAACT TGTTTTCGTGGGCCATCTGAAACCCATCTGGACTCCAT GATAGGCCAAGCAATATTAGGCGATGGGGCTGCAGCTG TCATAGTTGGCGCAGATCCAGACCTAACCGTTGAGAGG CCCATATTCGAGTTGGTTTCCACAGCCCAGACTATTGT ACCCGAATCCCATGGTGCAATTGAGGGCCACTTGCTTG AATCTGGACTCAGTTTCCATTTGTACAAGACCGTTCCT ACACTAATCTCTAACAACATTAAAACTTGCCTTTCTGA TGCTTTCACTCCTCTAAACATTAGCGATTGGAACTCTC TTTTCTGGATCGCACACCCTGGTGGTCCTGCCATCCTA GACCAAGTTACTGCTAAGGTTGGTCTTGAAAAGGAGAA ACTCAAGGTAACTAGACAAGTGTTGAAGGACTATGGAA ACATGTCGAGTGCTACGGTGTTTTTCATCATGGATGAG ATGAGGAAGAAGTCACTCGAAAACGGTCAAGCAACCAC TGGAGAAGGGCTCGAGTGGGGTGTTTTGTTTGGGTTCG GGCCTGGAATCACCGTTGAAACTGTAGTGCTACGCAGT GTGCCCGTAATTAGCTAG-3'

LIST OF CITED DOCUMENTS

Patent Documents

[0211] The following patent document(s) is (are) cited, in case they may be of use:

[0212] patcit1: FR1234567 (filing number);

[0213] patcit2: US2004230550 (publication number); and

[0214] patcit3: FR2795457 (publication number).

Non-Patent Literature

[0215] The following non-patent items are cited, in case they may be of use:

[0216] nplcit1: Klesk et al., 2004, J. Agric. Food Chem. 52, 5155-61;

[0217] nplcit2: Larsen et al., 1991, Acta Agric. Scand. 41, 447-54).

[0218] nplcit3: Kikuchi Y et al. "Mutational analysis of the feedback sites of phenylalanine-sensitive 3-deoxy-d-arabino-heptulosonate-7-phosphate synthase of *Escherichia coli*." Appl Environ Microbiol 63:761-762 (1997).): P160L, Q164A, S190A, G191K, S193A, 1225P.

[0219] nplcit4: Cui, Di et al., "Molecular basis for feedback inhibition of tyrosine-regulated 3-deoxy-d-arabino-heptulosonate-7-phosphate synthase from *Escherichia coli*", J Struct Biol. 2019 Jun. 1; 206(3):322-334.

[0220] nplcit5: Lütke-Eversloh and Stephanopoulos, 2005

[0221] nplcit6: Lütke-Eversloh and Stephanopoulos, 2007. "L-Tyrosine production by deregulated strains of *Escherichia coli"*.

[0222] nplcit7: Kang et al., 2012 "Artificial biosynthesis of phenylpropanoic acids in a tyrosine overproducing *Escherichia coli* strain"

[0223] nplcit8: Santos et al., 2011. "Optimization of a heterologous pathway for the production of flavonoids from glucose",

[0224] nplcit9: Juminaga et al., 2011. "Modular engineering of L-tyrosine production in *E. coli"*.

[0225] nplcit10: Rodriguez et al., 2015, "Establishment of a yeast platform strain for production of p-coumaric acid through metabolic engineering of aromatic amino acid biosynthesis."

[0226] nplcit11: Calero et al., 2016. "Broad-host range ProUSER vectors enable fast characterization of inducible promoters and optimization of p-coumaric acid production in *Pseudomonas putida* KT2440".

[0227] nplcit12: Calero et al., 2017. "Genome-wide identification of tolerance mechanisms toward p-coumaric acid in *Pseudomonas putida"*.

[0228] nplcit13: Koeduka, T., et al. (2011). "Characterization of raspberry ketone/zingerone synthase, catalyzing the alpha, beta-hydrogenation of phenylbutenones in raspberry fruits". Biochemical and Biophysical Research Communications 412, 104-108.

[0229] nplcit14: Beekwilder et al., 2007. "Microbial production of natural raspberry ketone".

[0230] nplcit15: Lee et al., 2017. "Heterologous production of raspberry ketone in the wine yeast *Saccharomyces cerevisiae* via pathway engineering and synthetic enzyme fusion".

[0231] nplcit16: Waterhouse, A. et al., "SWISS-MODEL: homology modelling of protein structures and complexes". Nucleic Acids Res. 46(W1), W296-W303 (2018)

[0232] nplcit17: O. Trott et al., "AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization and multithreading", Journal of computational chemistry, 2010).

[0233] nplcit18: Ding, R. et al. "Introduction of two mutations into AroG increases phenylalanine production in *Escherichia coli*". Biotechnol Lett 36, 2103-2108 (2014).

[0234] nplcit19: (Gibson et al., 2009).

[0235] nplcit20: SchsferA, et al., "Small mobilizable multi-purpose cloning vectors derived from the *Escherichia coli* plasmids pK18 and pK19: selection of defined deletions in the chromosome of *Corynebacterium glutamicum*". Gene 1994 Jul. 22, 145(1): 69-73)

[0236] nplcit21: Simon, R., et al. "A broad host range mobilization system for in vivo genetic engineering: transposon mutagenesis in gram negative bacteria". Nature BioTechnology volume 1, pages784-791 (1983)).

[0237] nplcit22: O. Trott et al., 2010. "AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization and multithreading", Journal of computational chemistry.

[0238] nplcit23: Zhou et al., "Characterization of mutants of a tyrosine ammonia-lyase from *Rhodotorula glutinis*". Appl. Microbiol. Biotechnol, 2015.

[0239] nplcit23: Prior et al., "Broad-host-range vectors for protein expression across gram negative hosts". Biotechnol Bioeng, 2010 Jun. 1; 106(2): 326-32.

SEQUENCE LISTING

```
<160> NUMBER OF SEQ ID NOS: 17

<210> SEQ ID NO 1
<211> LENGTH: 370
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 1

Leu Ala Ala Gly Thr Arg Asp Leu Thr Ser Asn Thr Met Ala Asp Leu
1               5                   10                  15

Pro Ile Asp Asp Leu Asn Val Ala Ser Asn Glu Thr Leu Ile Thr Pro
            20                  25                  30

Asp Gln Leu Lys Lys Glu Ile Pro Leu Ser Ala Lys Ala Leu Gln Thr
        35                  40                  45
```

-continued

```
Val Thr Ala Gly Arg Glu Val Val Arg Asn Ile Leu Asp Gly Lys Asp
    50                  55                  60

His Arg Leu Phe Val Val Val Gly Pro Cys Ser Ile His Asp Ile Lys
65                  70                  75                  80

Ala Ala His Glu Tyr Ala Glu Arg Leu Lys Val Leu Ala Glu Glu Val
                85                  90                  95

Ser Asp Thr Leu Tyr Leu Val Met Arg Val Tyr Phe Glu Lys Pro Arg
            100                 105                 110

Thr Thr Val Gly Trp Lys Gly Leu Ile Asn Asp Pro Tyr Leu Asp Asp
        115                 120                 125

Ser Phe Lys Ile Gln Asp Gly Leu His Ile Gly Arg Lys Leu Leu Leu
    130                 135                 140

Asp Leu Ala Glu Met Gly Leu Pro Thr Ala Thr Glu Ala Leu Asp Pro
145                 150                 155                 160

Ile Ser Pro Gln Tyr Leu Gln Asp Leu Ile Ser Trp Ser Ala Ile Gly
                165                 170                 175

Ala Arg Thr Thr Glu Ser Gln Thr His Arg Glu Met Ala Ser Gly Leu
            180                 185                 190

Ser Ser Ala Val Gly Phe Lys Asn Gly Thr Asp Gly Gly Leu Thr Val
        195                 200                 205

Ala Ile Asn Ala Leu Gln Ser Val Ser Lys Pro His Arg Phe Leu Gly
    210                 215                 220

Ile Asn Gln Glu Gly Gly Val Ser Ile Val Thr Thr Lys Gly Asn Pro
225                 230                 235                 240

Tyr Gly His Val Val Leu Arg Gly Gly Asn Gly Lys Pro Asn Tyr Asp
                245                 250                 255

Ser Val Ser Val Ala Leu Cys Glu Gln Asp Leu Ala Lys Ala Lys Ile
            260                 265                 270

Lys Ala Asn Ile Met Val Asp Cys Ser His Ala Asn Ser Asn Lys Asp
        275                 280                 285

Pro Ala Leu Gln Pro Leu Val Met Glu Asn Val Ala Asn Gln Ile Leu
    290                 295                 300

Glu Gly Asn Gln Ser Ile Ile Gly Leu Met Val Glu Ser His Leu Asn
305                 310                 315                 320

Trp Gly Cys Gln Ser Ile Pro Lys Asn Leu Asp Asp Leu Gln Tyr Gly
                325                 330                 335

Val Ser Ile Thr Asp Ala Cys Ile Asp Trp Ser Ala Thr Glu Lys Thr
            340                 345                 350

Leu Arg Ser Met His Ala Lys Leu Lys Asp Val Leu Pro Gln Arg Lys
        355                 360                 365

Arg Gly
    370


<210> SEQ ID NO 2
<211> LENGTH: 370
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 2

Leu Ala Ala Gly Thr Arg Asp Leu Thr Ser Asn Thr Met Ala Asp Leu
1               5                   10                  15

Pro Ile Asp Asp Leu Asn Val Ala Ser Asn Glu Thr Leu Ile Thr Pro
```

-continued

```
            20                  25                  30

Asp Gln Leu Lys Lys Glu Ile Pro Leu Ser Ala Lys Ala Leu Gln Thr
        35                  40                  45

Val Thr Ala Gly Arg Glu Val Val Arg Asn Ile Leu Asp Gly Lys Asp
    50                  55                  60

His Arg Leu Phe Val Val Val Gly Pro Cys Ser Ile His Asp Ile Lys
65                  70                  75                  80

Ala Ala His Glu Tyr Ala Glu Arg Leu Lys Val Leu Ala Glu Glu Val
                85                  90                  95

Ser Asp Thr Leu Tyr Leu Val Met Arg Val Tyr Phe Glu Lys Pro Arg
            100                 105                 110

Thr Thr Val Gly Trp Lys Gly Leu Ile Asn Asp Pro Tyr Leu Asp Asp
        115                 120                 125

Ser Phe Lys Ile Gln Asp Gly Leu His Ile Gly Arg Lys Leu Leu Leu
    130                 135                 140

Asp Leu Ala Glu Met Gly Leu Pro Thr Ala Thr Glu Ala Leu Asp Leu
145                 150                 155                 160

Ile Ser Pro Gln Tyr Leu Gln Asp Leu Ile Ser Trp Ser Ala Ile Gly
                165                 170                 175

Ala Arg Thr Thr Glu Ser Gln Thr His Arg Glu Met Ala Ser Gly Leu
            180                 185                 190

Ser Ser Ala Val Gly Phe Lys Asn Gly Thr Asp Gly Gly Leu Thr Val
        195                 200                 205

Ala Ile Asn Ala Leu Gln Ser Val Ser Lys Pro His Arg Phe Leu Gly
    210                 215                 220

Ile Asn Gln Glu Gly Gly Val Ser Ile Val Thr Thr Lys Gly Asn Pro
225                 230                 235                 240

Tyr Gly His Val Val Leu Arg Gly Gly Asn Gly Lys Pro Asn Tyr Asp
                245                 250                 255

Ser Val Ser Val Ala Leu Cys Glu Gln Asp Leu Ala Lys Ala Lys Ile
            260                 265                 270

Lys Ala Asn Ile Met Val Asp Cys Ser His Ala Asn Ser Asn Lys Asp
        275                 280                 285

Pro Ala Leu Gln Pro Leu Val Met Glu Asn Val Ala Asn Gln Ile Leu
    290                 295                 300

Glu Gly Asn Gln Ser Ile Ile Gly Leu Met Val Glu Ser His Leu Asn
305                 310                 315                 320

Trp Gly Cys Gln Ser Ile Pro Lys Asn Leu Asp Asp Leu Gln Tyr Gly
                325                 330                 335

Val Ser Ile Thr Asp Ala Cys Ile Asp Trp Ser Ala Thr Glu Lys Thr
            340                 345                 350

Leu Arg Ser Met His Ala Lys Leu Lys Asp Val Leu Pro Gln Arg Lys
        355                 360                 365

Arg Gly
    370


<210> SEQ ID NO 3
<211> LENGTH: 740
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 3
```

-continued

```
Leu Ala Ala Gly Thr Arg Asp Leu Thr Ser Asn Thr Met Ala Asp Leu
1               5                   10                  15

Pro Ile Asp Asp Leu Asn Val Ala Ser Asn Glu Thr Leu Ile Thr Pro
            20                  25                  30

Asp Gln Leu Lys Lys Glu Ile Pro Leu Ser Ala Lys Ala Leu Gln Thr
        35                  40                  45

Val Thr Ala Gly Arg Glu Val Val Arg Asn Ile Leu Asp Gly Lys Asp
    50                  55                  60

His Arg Leu Phe Val Val Val Gly Pro Cys Ser Ile His Asp Ile Lys
65                  70                  75                  80

Ala Ala His Glu Tyr Ala Glu Arg Leu Lys Val Leu Ala Glu Glu Val
                85                  90                  95

Ser Asp Thr Leu Tyr Leu Val Met Arg Val Tyr Phe Glu Lys Pro Arg
            100                 105                 110

Thr Thr Val Gly Trp Lys Gly Leu Ile Asn Asp Pro Tyr Leu Asp Asp
        115                 120                 125

Ser Phe Lys Ile Gln Asp Gly Leu His Ile Gly Arg Lys Leu Leu Leu
    130                 135                 140

Asp Leu Ala Glu Met Gly Leu Pro Thr Ala Thr Glu Ala Leu Asp Leu
145                 150                 155                 160

Ile Ser Pro Ala Tyr Leu Gln Asp Leu Ile Ser Trp Ser Ala Ile Gly
                165                 170                 175

Ala Arg Thr Thr Glu Ser Gln Thr His Arg Glu Met Ala Ser Gly Leu
            180                 185                 190

Ser Ser Ala Val Gly Phe Lys Asn Gly Thr Asp Gly Gly Leu Thr Val
        195                 200                 205

Ala Ile Asn Ala Leu Gln Ser Val Ser Lys Pro His Arg Phe Leu Gly
    210                 215                 220

Ile Asn Gln Glu Gly Gly Val Ser Ile Val Thr Thr Lys Gly Asn Pro
225                 230                 235                 240

Tyr Gly His Val Val Leu Arg Gly Gly Asn Gly Lys Pro Asn Tyr Asp
                245                 250                 255

Ser Val Ser Val Ala Leu Cys Glu Gln Asp Leu Ala Lys Ala Lys Ile
            260                 265                 270

Lys Ala Asn Ile Met Val Asp Cys Ser His Ala Asn Ser Asn Lys Asp
        275                 280                 285

Pro Ala Leu Gln Pro Leu Val Met Glu Asn Val Ala Asn Gln Ile Leu
    290                 295                 300

Glu Gly Asn Gln Ser Ile Ile Gly Leu Met Val Glu Ser His Leu Asn
305                 310                 315                 320

Trp Gly Cys Gln Ser Ile Pro Lys Asn Leu Asp Asp Leu Gln Tyr Gly
                325                 330                 335

Val Ser Ile Thr Asp Ala Cys Ile Asp Trp Ser Ala Thr Glu Lys Thr
            340                 345                 350

Leu Arg Ser Met His Ala Lys Leu Lys Asp Val Leu Pro Gln Arg Lys
        355                 360                 365

Arg Gly Leu Ala Ala Gly Thr Arg Asp Leu Thr Ser Asn Thr Met Ala
    370                 375                 380

Asp Leu Pro Ile Asp Asp Leu Asn Val Ala Ser Asn Glu Thr Leu Ile
385                 390                 395                 400

Thr Pro Asp Gln Leu Lys Lys Glu Ile Pro Leu Ser Ala Lys Ala Leu
```

-continued

```
                405                 410                 415

Gln Thr Val Thr Ala Gly Arg Glu Val Val Arg Asn Ile Leu Asp Gly
            420                 425                 430

Lys Asp His Arg Leu Phe Val Val Val Gly Pro Cys Ser Ile His Asp
        435                 440                 445

Ile Lys Ala Ala His Glu Tyr Ala Glu Arg Leu Lys Val Leu Ala Glu
    450                 455                 460

Glu Val Ser Asp Thr Leu Tyr Leu Val Met Arg Val Tyr Phe Glu Lys
465                 470                 475                 480

Pro Arg Thr Thr Val Gly Trp Lys Gly Leu Ile Asn Asp Pro Tyr Leu
                485                 490                 495

Asp Asp Ser Phe Lys Ile Gln Asp Gly Leu His Ile Gly Arg Lys Leu
            500                 505                 510

Leu Leu Asp Leu Ala Glu Met Gly Leu Pro Thr Ala Thr Glu Ala Leu
        515                 520                 525

Asp Leu Ile Ser Pro Ala Tyr Leu Gln Asp Leu Ile Ser Trp Ser Ala
    530                 535                 540

Ile Gly Ala Arg Thr Thr Glu Ser Gln Thr His Arg Glu Met Ala Ser
545                 550                 555                 560

Gly Leu Ser Ser Ala Val Gly Phe Lys Asn Gly Thr Asp Gly Gly Leu
                565                 570                 575

Thr Val Ala Ile Asn Ala Leu Gln Ser Val Ser Lys Pro His Arg Phe
            580                 585                 590

Leu Gly Ile Asn Gln Glu Gly Gly Val Ser Ile Val Thr Thr Lys Gly
        595                 600                 605

Asn Pro Tyr Gly His Val Val Leu Arg Gly Gly Asn Gly Lys Pro Asn
    610                 615                 620

Tyr Asp Ser Val Ser Val Ala Leu Cys Glu Gln Asp Leu Ala Lys Ala
625                 630                 635                 640

Lys Ile Lys Ala Asn Ile Met Val Asp Cys Ser His Ala Asn Ser Asn
                645                 650                 655

Lys Asp Pro Ala Leu Gln Pro Leu Val Met Glu Asn Val Ala Asn Gln
            660                 665                 670

Ile Leu Glu Gly Asn Gln Ser Ile Ile Gly Leu Met Val Glu Ser His
        675                 680                 685

Leu Asn Trp Gly Cys Gln Ser Ile Pro Lys Asn Leu Asp Asp Leu Gln
    690                 695                 700

Tyr Gly Val Ser Ile Thr Asp Ala Cys Ile Asp Trp Ser Ala Thr Glu
705                 710                 715                 720

Lys Thr Leu Arg Ser Met His Ala Lys Leu Lys Asp Val Leu Pro Gln
                725                 730                 735

Arg Lys Arg Gly
            740


<210> SEQ ID NO 4
<211> LENGTH: 370
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 4

Leu Ala Ala Gly Thr Arg Asp Leu Thr Ser Asn Thr Met Ala Asp Leu
1               5                   10                  15
```

-continued

```
Pro Ile Asp Asp Leu Asn Val Ala Ser Asn Glu Thr Leu Ile Thr Pro
        20                  25                  30

Asp Gln Leu Lys Lys Glu Ile Pro Leu Ser Ala Lys Ala Leu Gln Thr
    35                  40                  45

Val Thr Ala Gly Arg Glu Val Val Arg Asn Ile Leu Asp Gly Lys Asp
    50                  55                  60

His Arg Leu Phe Val Val Val Gly Pro Cys Ser Ile His Asp Ile Lys
65                  70                  75                  80

Ala Ala His Glu Tyr Ala Glu Arg Leu Lys Val Leu Ala Glu Glu Val
                85                  90                  95

Ser Asp Thr Leu Tyr Leu Val Met Arg Val Tyr Phe Glu Lys Pro Arg
            100                 105                 110

Thr Thr Val Gly Trp Lys Gly Leu Ile Asn Asp Pro Tyr Leu Asp Asp
        115                 120                 125

Ser Phe Lys Ile Gln Asp Gly Leu His Ile Gly Arg Lys Leu Leu Leu
    130                 135                 140

Asp Leu Ala Glu Met Gly Leu Pro Thr Ala Thr Glu Ala Leu Asp Leu
145                 150                 155                 160

Ile Ser Pro Gln Tyr Leu Gln Asp Leu Ile Ser Trp Ser Ala Ile Gly
                165                 170                 175

Ala Arg Thr Thr Glu Ser Gln Thr His Arg Glu Met Ala Ser Gly Leu
            180                 185                 190

Ala Ser Ala Val Gly Phe Lys Asn Gly Thr Asp Gly Gly Leu Thr Val
        195                 200                 205

Ala Ile Asn Ala Leu Gln Ser Val Ser Lys Pro His Arg Phe Leu Gly
    210                 215                 220

Ile Asn Gln Glu Gly Gly Val Ser Ile Val Thr Thr Lys Gly Asn Pro
225                 230                 235                 240

Tyr Gly His Val Val Leu Arg Gly Gly Asn Gly Lys Pro Asn Tyr Asp
                245                 250                 255

Ser Val Ser Val Ala Leu Cys Glu Gln Asp Leu Ala Lys Ala Lys Ile
            260                 265                 270

Lys Ala Asn Ile Met Val Asp Cys Ser His Ala Asn Ser Asn Lys Asp
        275                 280                 285

Pro Ala Leu Gln Pro Leu Val Met Glu Asn Val Ala Asn Gln Ile Leu
    290                 295                 300

Glu Gly Asn Gln Ser Ile Ile Gly Leu Met Val Glu Ser His Leu Asn
305                 310                 315                 320

Trp Gly Cys Gln Ser Ile Pro Lys Asn Leu Asp Asp Leu Gln Tyr Gly
                325                 330                 335

Val Ser Ile Thr Asp Ala Cys Ile Asp Trp Ser Ala Thr Glu Lys Thr
            340                 345                 350

Leu Arg Ser Met His Ala Lys Leu Lys Asp Val Leu Pro Gln Arg Lys
        355                 360                 365

Arg Gly
    370
```

<210> SEQ ID NO 5
<211> LENGTH: 262
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 5

-continued

```
Met Lys Gln Thr Gln Tyr Val Ala Arg Glu Pro Asp Ala His Gly Phe
1               5                   10                  15

Ile Asp Tyr Pro Gln Gln Glu His Ala Val Trp Asn Thr Leu Ile Thr
            20                  25                  30

Arg Gln Leu Lys Val Ile Glu Gly Arg Ala Cys Gln Glu Tyr Leu Asp
        35                  40                  45

Gly Ile Asp Gln Leu Lys Leu Pro His Asp Arg Ile Pro Gln Leu Gly
    50                  55                  60

Glu Ile Asn Lys Val Leu Gly Ala Thr Thr Gly Trp Gln Val Ala Arg
65                  70                  75                  80

Val Pro Ala Leu Ile Pro Phe Gln Thr Phe Phe Glu Leu Leu Ala Ser
                85                  90                  95

Lys Arg Phe Pro Val Ala Thr Phe Ile Arg Thr Pro Glu Glu Leu Asp
            100                 105                 110

Tyr Leu Gln Glu Pro Asp Ile Phe His Glu Ile Phe Gly His Cys Pro
        115                 120                 125

Leu Leu Thr Asn Pro Trp Phe Ala Glu Phe Thr His Thr Tyr Gly Lys
    130                 135                 140

Leu Gly Leu Ala Ala Thr Lys Glu Gln Arg Val Tyr Leu Ala Arg Leu
145                 150                 155                 160

Tyr Trp Met Thr Ile Glu Phe Gly Leu Met Glu Thr Ala Gln Gly Arg
                165                 170                 175

Lys Ile Tyr Gly Gly Gly Ile Leu Ser Ser Pro Lys Glu Thr Val Tyr
            180                 185                 190

Ser Leu Ser Asp Glu Pro Glu His Gln Ala Phe Asp Pro Ile Glu Ala
        195                 200                 205

Met Arg Thr Pro Tyr Arg Ile Asp Ile Leu Gln Pro Val Tyr Phe Val
    210                 215                 220

Leu Pro Asn Met Lys Arg Leu Phe Asp Leu Ala His Glu Asp Ile Met
225                 230                 235                 240

Gly Met Val His Lys Ala Met Gln Leu Gly Leu His Ala Pro Lys Phe
                245                 250                 255

Pro Pro Lys Val Ala Ala
            260


<210> SEQ ID NO 6
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 6

Met Asn Ala Leu Asn Gln Ala His Cys Glu Ala Cys Arg Ala Asp Ala
1               5                   10                  15

Pro Lys Val Thr Asp Glu Glu Leu Ala Glu Leu Ile Arg Glu Ile Pro
            20                  25                  30

Asp Trp Asn Ile Glu Val Arg Asp Gly His Met Glu Leu Glu Arg Val
        35                  40                  45

Phe Leu Phe Lys Asn Phe Lys His Ala Leu Ala Phe Thr Asn Ala Val
    50                  55                  60

Gly Glu Ile Ala Glu Ala Glu Gly His His Pro Gly Leu Leu Thr Glu
65                  70                  75                  80

Trp Gly Lys Val Thr Val Thr Trp Trp Ser His Ser Ile Lys Gly Leu
```

-continued

```
                85                  90                  95

His Arg Asn Asp Phe Ile Met Cys Ala Arg Thr Asp Lys Val Ala Glu
            100                 105                 110

Thr Ala Glu Gly Arg Lys
        115


<210> SEQ ID NO 7
<211> LENGTH: 693
<212> TYPE: PRT
<213> ORGANISM: Rhodotorula glutinis

<400> SEQUENCE: 7

Met Ala Pro Arg Pro Thr Ser Gln Asn Gln Thr Arg Thr Cys Pro Thr
1               5                   10                  15

Thr Gln Val Thr Gln Val Asp Ile Val Glu Lys Met Leu Ala Ala Pro
            20                  25                  30

Thr Asp Ser Thr Leu Glu Leu Asp Gly Tyr Ser Leu Asn Leu Gly Asp
        35                  40                  45

Val Val Ser Ala Ala Arg Lys Gly Arg Pro Val Arg Val Lys Asp Ser
    50                  55                  60

Asp Glu Ile Arg Ser Lys Ile Asp Lys Ser Val Glu Phe Leu Arg Ser
65                  70                  75                  80

Gln Leu Ser Met Ser Val Tyr Gly Val Thr Thr Gly Phe Gly Gly Ser
                85                  90                  95

Ala Asp Thr Arg Thr Glu Asp Ala Ile Ser Leu Gln Lys Ala Leu Leu
            100                 105                 110

Glu His Gln Leu Cys Gly Val Leu Pro Ser Ser Phe Asp Ser Phe Arg
        115                 120                 125

Leu Gly Arg Gly Leu Glu Asn Ser Leu Pro Leu Glu Val Val Arg Gly
    130                 135                 140

Ala Met Thr Ile Arg Val Asn Ser Leu Thr Arg Gly His Ser Ala Val
145                 150                 155                 160

Arg Leu Val Val Leu Glu Ala Leu Thr Asn Phe Leu Asn His Gly Ile
                165                 170                 175

Thr Pro Ile Val Pro Leu Arg Gly Thr Ile Ser Ala Ser Gly Asp Leu
            180                 185                 190

Ser Pro Leu Ser Tyr Ile Ala Ala Ala Ile Ser Gly His Pro Asp Ser
        195                 200                 205

Lys Val His Val Val His Glu Gly Lys Glu Lys Ile Leu Tyr Ala Arg
    210                 215                 220

Glu Ala Met Ala Leu Phe Asn Leu Glu Pro Val Val Leu Gly Pro Lys
225                 230                 235                 240

Glu Gly Leu Gly Leu Val Asn Gly Thr Ala Val Ser Ala Ser Met Ala
                245                 250                 255

Thr Leu Ala Leu His Asp Ala His Met Leu Ser Leu Leu Ser Gln Ser
            260                 265                 270

Leu Thr Ala Met Thr Val Glu Ala Met Val Gly His Ala Gly Ser Phe
        275                 280                 285

His Pro Phe Leu His Asp Val Thr Arg Pro His Pro Thr Gln Ile Glu
    290                 295                 300

Val Ala Gly Asn Ile Arg Lys Leu Leu Glu Gly Ser Arg Phe Ala Val
305                 310                 315                 320
```

-continued

```
His His Glu Glu Glu Val Lys Val Lys Asp Asp Glu Gly Ile Leu Arg
                325                 330                 335

Gln Asp Arg Tyr Pro Leu Arg Thr Ser Pro Gln Trp Leu Gly Pro Leu
            340                 345                 350

Val Ser Asp Leu Ile His Ala His Ala Val Leu Thr Ile Glu Ala Gly
        355                 360                 365

Gln Ser Thr Thr Asp Asn Pro Leu Ile Asp Val Glu Asn Lys Thr Ser
    370                 375                 380

His His Gly Gly Asn Phe Gln Ala Ala Ala Val Ala Asn Thr Met Glu
385                 390                 395                 400

Lys Thr Arg Leu Gly Leu Ala Gln Ile Gly Lys Leu Asn Phe Thr Gln
                405                 410                 415

Leu Thr Glu Met Leu Asn Ala Gly Met Asn Arg Gly Leu Pro Ser Cys
            420                 425                 430

Leu Ala Ala Glu Asp Pro Ser Leu Ser Tyr His Cys Lys Gly Leu Asp
        435                 440                 445

Ile Ala Ala Ala Ala Tyr Thr Ser Glu Leu Gly His Leu Ala Asn Pro
    450                 455                 460

Val Thr Thr His Val Gln Pro Ala Glu Met Ala Asn Gln Ala Val Asn
465                 470                 475                 480

Ser Leu Ala Leu Ile Ser Ala Arg Arg Thr Thr Glu Ser Asn Asp Val
                485                 490                 495

Leu Ser Leu Leu Leu Ala Thr His Leu Tyr Cys Val Leu Gln Ala Ile
            500                 505                 510

Asp Leu Arg Ala Ile Val Phe Glu Phe Lys Lys Gln Phe Gly Pro Ala
        515                 520                 525

Ile Val Ser Leu Ile Asp Gln His Phe Gly Ser Ala Met Thr Gly Ser
    530                 535                 540

Asn Leu Arg Asp Glu Leu Val Glu Lys Val Asn Lys Thr Leu Ala Lys
545                 550                 555                 560

Arg Leu Glu Gln Thr Asn Ser Tyr Asp Leu Val Pro Arg Trp His Asp
                565                 570                 575

Ala Phe Ser Phe Ala Ala Gly Thr Val Val Glu Val Leu Ser Ser Thr
            580                 585                 590

Ser Leu Ser Leu Ala Ala Val Asn Ala Trp Lys Val Ala Ala Ala Glu
        595                 600                 605

Ser Ala Ile Ser Leu Thr Arg Gln Val Arg Glu Thr Phe Trp Ser Ala
    610                 615                 620

Ala Ser Thr Ser Ser Pro Ala Leu Ser Tyr Leu Ser Pro Arg Thr Gln
625                 630                 635                 640

Ile Leu Tyr Ala Phe Val Arg Glu Glu Leu Gly Val Lys Ala Arg Arg
                645                 650                 655

Gly Asp Val Phe Leu Gly Lys Gln Glu Val Thr Ile Gly Ser Asn Val
            660                 665                 670

Ser Lys Ile Tyr Glu Ala Ile Lys Ser Gly Arg Ile Asn Asn Val Leu
        675                 680                 685

Leu Lys Met Leu Ala
    690
```

<210> SEQ ID NO 8
<211> LENGTH: 627
<212> TYPE: PRT

-continued

<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 8

```
Met Asn Asn Glu Ala Arg Ser Gly Ser Thr Asp Pro Gly Gln Arg Pro
1               5                   10                  15

Arg Tyr Arg Gln Val Ala Ile Gly His Pro Gln Val Gln Val Ser His
            20                  25                  30

Val Asp Asp Val Leu Arg Met Gln Pro Val Glu Pro Leu Ala Pro Leu
        35                  40                  45

Pro Ala Arg Leu Leu Glu Arg Leu Val His Trp Ala Gln Val Arg Pro
    50                  55                  60

Asp Thr Thr Phe Ile Ala Ala Arg Gln Ala Asp Gly Ala Trp Arg Ser
65                  70                  75                  80

Ile Ser Tyr Val Gln Met Leu Ala Asp Val Arg Thr Ile Ala Ala Asn
                85                  90                  95

Leu Leu Gly Leu Gly Leu Ser Ala Glu Arg Pro Leu Ala Leu Leu Ser
            100                 105                 110

Gly Asn Asp Ile Glu His Leu Gln Ile Ala Leu Gly Ala Met Tyr Ala
        115                 120                 125

Gly Ile Ala Tyr Cys Pro Val Ser Pro Ala Tyr Ala Leu Leu Ser Gln
    130                 135                 140

Asp Phe Ala Lys Leu Arg His Val Cys Glu Val Leu Thr Pro Gly Val
145                 150                 155                 160

Val Phe Val Ser Asp Ser Gln Pro Phe Gln Arg Ala Phe Glu Ala Val
                165                 170                 175

Leu Asp Asp Ser Val Gly Val Ile Ser Val Arg Gly Gln Val Ala Gly
            180                 185                 190

Arg Pro His Ile Ser Phe Asp Ser Leu Leu Gln Pro Gly Asp Leu Ala
        195                 200                 205

Ala Ala Asp Ala Ala Phe Ala Ala Thr Gly Pro Asp Thr Ile Ala Lys
    210                 215                 220

Phe Leu Phe Thr Ser Gly Ser Thr Lys Leu Pro Lys Ala Val Ile Thr
225                 230                 235                 240

Thr Gln Arg Met Leu Cys Ala Asn Gln Gln Met Leu Leu Gln Thr Phe
                245                 250                 255

Pro Thr Phe Ala Glu Glu Pro Pro Val Leu Val Asp Trp Leu Pro Trp
            260                 265                 270

Asn His Thr Phe Gly Gly Ser His Asn Leu Gly Ile Val Leu Tyr Asn
        275                 280                 285

Gly Gly Ser Phe Tyr Leu Asp Ala Gly Lys Pro Thr Pro Gln Gly Phe
    290                 295                 300

Ala Glu Thr Leu Arg Asn Leu Arg Glu Ile Ser Pro Thr Ala Tyr Leu
305                 310                 315                 320

Thr Val Pro Lys Gly Trp Glu Glu Leu Val Lys Ala Leu Glu Gln Asp
                325                 330                 335

Pro Ala Leu Arg Glu Val Phe Phe Ala Arg Ile Lys Leu Phe Phe Phe
            340                 345                 350

Ala Ala Ala Gly Leu Ser Gln Ser Val Trp Asp Arg Leu Asp Arg Ile
        355                 360                 365

Ala Glu Gln His Cys Gly Glu Arg Ile Arg Met Met Ala Gly Leu Gly
    370                 375                 380
```

-continued

Met Thr Glu Ala Ser Pro Ser Cys Thr Phe Thr Thr Gly Pro Leu Ser
385 390 395 400

Met Ala Gly Tyr Val Gly Leu Pro Ala Pro Gly Cys Glu Val Lys Leu
405 410 415

Val Pro Val Gly Asp Lys Leu Glu Ala Arg Phe Arg Gly Pro His Ile
420 425 430

Met Pro Gly Tyr Trp Arg Ser Pro Gln Gln Thr Ala Glu Ala Phe Asp
435 440 445

Glu Glu Gly Phe Tyr Cys Ser Gly Asp Ala Leu Lys Leu Ala Asp Ala
450 455 460

Arg Gln Pro Glu Leu Gly Leu Met Phe Asp Gly Arg Ile Ala Glu Asp
465 470 475 480

Phe Lys Leu Ser Ser Gly Val Phe Val Ser Val Gly Pro Leu Arg Asn
485 490 495

Arg Ala Val Leu Glu Gly Ser Pro Tyr Val Gln Asp Ile Val Val Thr
500 505 510

Ala Pro Asp Arg Glu Cys Leu Gly Leu Leu Val Phe Pro Arg Leu Pro
515 520 525

Glu Cys Arg Arg Leu Ala Gly Leu Ala Glu Asp Ala Ser Asp Ala Arg
530 535 540

Val Leu Ala Asn Asp Thr Val Arg Ser Trp Phe Ala Asp Trp Leu Glu
545 550 555 560

Arg Leu Asn Arg Asp Ala Gln Gly Asn Ala Ser Arg Ile Glu Trp Leu
565 570 575

Ser Leu Leu Ala Glu Pro Pro Ser Ile Asp Ala Gly Glu Ile Thr Asp
580 585 590

Lys Gly Ser Ile Asn Gln Arg Ala Val Leu Gln Arg Arg Ala Ala Gln
595 600 605

Val Glu Ala Leu Tyr Arg Gly Glu Asp Pro Asp Ala Leu His Ala Lys
610 615 620

Val Arg Pro
625

<210> SEQ ID NO 9
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Rheum palmatum

<400> SEQUENCE: 9

Met Ala Thr Glu Glu Met Lys Lys Leu Ala Thr Val Met Ala Ile Gly
1 5 10 15

Thr Ala Asn Pro Pro Asn Cys Tyr Tyr Gln Ala Asp Phe Pro Asp Phe
20 25 30

Tyr Phe Arg Val Thr Asn Ser Asp His Leu Ile Asn Leu Lys Gln Lys
35 40 45

Phe Lys Arg Leu Cys Glu Asn Ser Arg Ile Glu Lys Arg Tyr Leu His
50 55 60

Val Thr Glu Glu Ile Leu Lys Glu Asn Pro Asn Ile Ala Ala Tyr Glu
65 70 75 80

Ala Thr Ser Leu Asn Val Arg His Lys Met Gln Val Lys Gly Val Ala
85 90 95

Glu Leu Gly Lys Glu Ala Ala Leu Lys Ala Ile Lys Glu Trp Gly Gln
100 105 110

-continued

```
Pro Lys Ser Lys Ile Thr His Leu Ile Val Cys Cys Leu Ala Gly Val
        115                 120                 125

Asp Met Pro Gly Ala Asp Tyr Gln Leu Thr Lys Leu Leu Asp Leu Asp
    130                 135                 140

Pro Ser Val Lys Arg Phe Met Phe Tyr His Leu Gly Cys Tyr Ala Gly
145                 150                 155                 160

Gly Thr Val Leu Arg Leu Ala Lys Asp Ile Ala Glu Asn Asn Lys Gly
                165                 170                 175

Ala Arg Val Leu Ile Val Cys Ser Glu Met Thr Thr Thr Cys Phe Arg
            180                 185                 190

Gly Pro Ser Glu Thr His Leu Asp Ser Met Ile Gly Gln Ala Ile Leu
        195                 200                 205

Gly Asp Gly Ala Ala Ala Val Ile Val Gly Ala Asp Pro Asp Leu Thr
    210                 215                 220

Val Glu Arg Pro Ile Phe Glu Leu Val Ser Thr Ala Gln Thr Ile Val
225                 230                 235                 240

Pro Glu Ser His Gly Ala Ile Glu Gly His Leu Leu Glu Ser Gly Leu
                245                 250                 255

Ser Phe His Leu Tyr Lys Thr Val Pro Thr Leu Ile Ser Asn Asn Ile
            260                 265                 270

Lys Thr Cys Leu Ser Asp Ala Phe Thr Pro Leu Asn Ile Ser Asp Trp
        275                 280                 285

Asn Ser Leu Phe Trp Ile Ala His Pro Gly Gly Pro Ala Ile Leu Asp
    290                 295                 300

Gln Val Thr Ala Lys Val Gly Leu Glu Lys Glu Lys Leu Lys Val Thr
305                 310                 315                 320

Arg Gln Val Leu Lys Asp Tyr Gly Asn Met Ser Ser Ala Thr Val Phe
                325                 330                 335

Phe Ile Met Asp Glu Met Arg Lys Lys Ser Leu Glu Asn Gly Gln Ala
            340                 345                 350

Thr Thr Gly Glu Gly Leu Glu Trp Gly Val Leu Phe Gly Phe Gly Pro
        355                 360                 365

Gly Ile Thr Val Glu Thr Val Val Leu Arg Ser Val Pro Val Ile Ser
    370                 375                 380


<210> SEQ ID NO 10
<211> LENGTH: 1113
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 10

Thr Thr Gly Gly Cys Gly Gly Cys Cys Gly Gly Cys Ala Cys Cys Cys
1               5                   10                  15

Gly Thr Gly Ala Cys Cys Thr Gly Ala Cys Gly Ala Gly Thr Ala Ala
            20                  25                  30

Cys Ala Cys Gly Ala Thr Gly Gly Cys Thr Gly Ala Thr Thr Thr Ala
        35                  40                  45

Cys Cys Gly Ala Thr Cys Gly Ala Thr Gly Ala Cys Thr Thr Gly Ala
    50                  55                  60

Ala Cys Gly Thr Thr Gly Cys Cys Thr Cys Cys Ala Ala Cys Gly Ala
65                  70                  75                  80

Gly Ala Cys Cys Cys Thr Gly Ala Thr Cys Ala Cys Cys Cys Cys Thr
```

-continued

```
                85                  90                  95

Gly Ala Thr Cys Ala Gly Cys Thr Cys Ala Ala Gly Ala Ala Gly Gly
            100                 105                 110

Ala Ala Ala Thr Cys Cys Cys Cys Cys Thr Cys Ala Gly Cys Gly Cys
        115                 120                 125

Cys Ala Ala Gly Gly Cys Cys Cys Thr Gly Cys Ala Gly Ala Cys Cys
    130                 135                 140

Gly Thr Gly Ala Cys Thr Gly Cys Cys Gly Gly Cys Cys Gly Thr Gly
145                 150                 155                 160

Ala Ala Gly Thr Gly Gly Thr Gly Cys Gly Cys Ala Ala Thr Ala Thr
                165                 170                 175

Thr Cys Thr Cys Gly Ala Cys Gly Gly Cys Ala Ala Gly Gly Ala Cys
            180                 185                 190

Cys Ala Thr Cys Gly Cys Cys Thr Gly Thr Thr Cys Gly Thr Cys Gly
        195                 200                 205

Thr Gly Gly Thr Cys Gly Gly Cys Cys Cys Thr Thr Gly Cys Thr Cys
    210                 215                 220

Cys Ala Thr Cys Cys Ala Cys Gly Ala Cys Ala Thr Cys Ala Ala Gly
225                 230                 235                 240

Gly Cys Ala Gly Cys Cys Cys Ala Cys Gly Ala Ala Thr Ala Cys Gly
                245                 250                 255

Cys Cys Gly Ala Gly Cys Gly Cys Cys Thr Gly Ala Ala Ala Gly Thr
            260                 265                 270

Gly Cys Thr Gly Gly Cys Cys Gly Ala Ala Gly Ala Ala Gly Thr Gly
        275                 280                 285

Thr Cys Cys Gly Ala Thr Ala Cys Gly Cys Thr Gly Thr Ala Cys Cys
    290                 295                 300

Thr Gly Gly Thr Cys Ala Thr Gly Cys Gly Cys Gly Thr Gly Thr Ala
305                 310                 315                 320

Cys Thr Thr Cys Gly Ala Ala Ala Ala Gly Cys Cys Gly Cys Gly Cys
                325                 330                 335

Ala Cys Cys Ala Cys Cys Gly Thr Cys Gly Gly Cys Thr Gly Gly Ala
            340                 345                 350

Ala Ala Gly Gly Cys Cys Thr Gly Ala Thr Cys Ala Ala Cys Gly Ala
        355                 360                 365

Thr Cys Cys Gly Thr Ala Cys Cys Thr Gly Gly Ala Thr Gly Ala Cys
    370                 375                 380

Thr Cys Gly Thr Thr Cys Ala Ala Gly Ala Thr Cys Cys Ala Gly Gly
385                 390                 395                 400

Ala Cys Gly Gly Cys Cys Thr Gly Cys Ala Cys Ala Thr Cys Gly Gly
                405                 410                 415

Cys Cys Gly Cys Ala Ala Gly Thr Thr Gly Cys Thr Gly Cys Thr Gly
            420                 425                 430

Gly Ala Cys Cys Thr Gly Gly Cys Cys Gly Ala Ala Ala Thr Gly Gly
        435                 440                 445

Gly Cys Cys Thr Gly Cys Cys Gly Ala Cys Cys Gly Cys Cys Ala Cys
    450                 455                 460

Cys Gly Ala Ala Gly Cys Gly Cys Thr Cys Gly Ala Cys Cys Thr Gly
465                 470                 475                 480

Ala Thr Thr Thr Cys Gly Cys Cys Gly Cys Ala Gly Thr Ala Cys Cys
                485                 490                 495
```

-continued

```
Thr Gly Cys Ala Ala Gly Ala Cys Cys Thr Gly Ala Thr Cys Ala Gly
            500                 505                 510

Cys Thr Gly Gly Thr Cys Gly Gly Cys Cys Ala Thr Cys Gly Gly Thr
        515                 520                 525

Gly Cys Cys Cys Gly Cys Ala Cys Cys Ala Cys Cys Gly Ala Ala Thr
    530                 535                 540

Cys Gly Cys Ala Ala Ala Cys Ala Cys Ala Cys Cys Gly Cys Gly Ala
545                 550                 555                 560

Gly Ala Thr Gly Gly Cys Cys Thr Cys Gly Gly Gly Cys Cys Thr Gly
                565                 570                 575

Thr Cys Cys Thr Cys Gly Gly Cys Gly Gly Thr Gly Gly Gly Thr Thr
            580                 585                 590

Thr Cys Ala Ala Gly Ala Ala Cys Gly Gly Thr Ala Cys Cys Gly Ala
        595                 600                 605

Thr Gly Gly Cys Gly Gly Cys Cys Thr Gly Ala Cys Cys Gly Thr Thr
    610                 615                 620

Gly Cys Cys Ala Thr Cys Ala Ala Thr Gly Cys Cys Cys Thr Gly Cys
625                 630                 635                 640

Ala Gly Thr Cys Gly Gly Thr Gly Thr Cys Cys Ala Ala Gly Cys Cys
                645                 650                 655

Gly Cys Ala Cys Cys Gly Cys Thr Thr Cys Cys Thr Gly Gly Gly Cys
            660                 665                 670

Ala Thr Cys Ala Ala Cys Cys Ala Gly Gly Ala Ala Gly Gly Cys Gly
        675                 680                 685

Gly Cys Gly Thr Gly Thr Cys Gly Ala Thr Cys Gly Thr Cys Ala Cys
    690                 695                 700

Cys Ala Cys Cys Ala Ala Gly Gly Gly Cys Ala Ala Cys Cys Cys Ala
705                 710                 715                 720

Thr Ala Cys Gly Gly Cys Cys Ala Cys Gly Thr Gly Gly Thr Ala Cys
                725                 730                 735

Thr Gly Cys Gly Cys Gly Gly Cys Gly Gly Cys Ala Ala Thr Gly Gly
            740                 745                 750

Cys Ala Ala Gly Cys Cys Gly Ala Ala Cys Thr Ala Cys Gly Ala Cys
        755                 760                 765

Thr Cys Gly Gly Thr Cys Ala Gly Cys Gly Thr Cys Gly Cys Cys Cys
    770                 775                 780

Thr Gly Thr Gly Cys Gly Ala Ala Cys Ala Gly Gly Ala Cys Cys Thr
785                 790                 795                 800

Gly Gly Cys Cys Ala Ala Gly Gly Cys Cys Ala Ala Gly Ala Thr Cys
                805                 810                 815

Ala Ala Gly Gly Cys Cys Ala Ala Cys Ala Thr Cys Ala Thr Gly Gly
            820                 825                 830

Thr Cys Gly Ala Cys Thr Gly Cys Ala Gly Cys Cys Ala Thr Gly Cys
        835                 840                 845

Cys Ala Ala Cys Thr Cys Cys Ala Ala Cys Ala Ala Gly Gly Ala Cys
    850                 855                 860

Cys Cys Gly Gly Cys Cys Cys Thr Gly Cys Ala Ala Cys Cys Gly Cys
865                 870                 875                 880

Thr Gly Gly Thr Gly Ala Thr Gly Gly Ala Gly Ala Ala Cys Gly Thr
                885                 890                 895
```

-continued

```
Cys Gly Cys Cys Ala Ala Cys Cys Ala Gly Ala Thr Thr Cys Thr Cys
            900                 905                 910

Gly Ala Ala Gly Gly Cys Ala Ala Cys Cys Ala Gly Thr Cys Gly Ala
        915                 920                 925

Thr Cys Ala Thr Cys Gly Gly Cys Cys Thr Gly Ala Thr Gly Gly Thr
    930                 935                 940

Cys Gly Ala Ala Ala Gly Cys Cys Ala Cys Cys Thr Gly Ala Ala Cys
945                 950                 955                 960

Thr Gly Gly Gly Gly Cys Thr Gly Thr Cys Ala Gly Thr Cys Cys Ala
                965                 970                 975

Thr Thr Cys Cys Gly Ala Ala Ala Ala Ala Cys Cys Thr Gly Gly Ala
            980                 985                 990

Cys Gly Ala Thr Thr Thr Gly Cys  Ala Gly Thr Ala Thr  Gly Gly Cys
        995                 1000                1005

Gly Thr  Gly Thr Cys Gly Ala  Thr Cys Ala Cys Gly  Gly Ala Cys
    1010                1015                1020

Gly Cys  Cys Thr Gly Cys Ala  Thr Cys Gly Ala Cys  Thr Gly Gly
    1025                1030                1035

Thr Cys  Gly Gly Cys Thr Ala  Cys Cys Gly Ala Gly  Ala Ala Ala
    1040                1045                1050

Ala Cys  Cys Cys Thr Gly Cys  Gly Cys Ala Gly Cys  Ala Thr Gly
    1055                1060                1065

Cys Ala  Thr Gly Cys Cys Ala  Ala Gly Cys Thr Cys  Ala Ala Gly
    1070                1075                1080

Gly Ala  Thr Gly Thr Gly Cys  Thr Gly Cys Cys Gly  Cys Ala Gly
    1085                1090                1095

Cys Gly  Thr Ala Ala Gly Cys  Gly Cys Gly Gly Cys  Thr Gly Ala
    1100                1105                1110
```

<210> SEQ ID NO 11
<211> LENGTH: 1113
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 11

```
Thr Thr Gly Gly Cys Gly Gly Cys Cys Gly Gly Cys Ala Cys Cys Cys
1               5                   10                  15

Gly Thr Gly Ala Cys Cys Thr Gly Ala Cys Gly Ala Gly Thr Ala Ala
            20                  25                  30

Cys Ala Cys Gly Ala Thr Gly Gly Cys Thr Gly Ala Thr Thr Thr Ala
        35                  40                  45

Cys Cys Gly Ala Thr Cys Gly Ala Thr Gly Ala Cys Thr Thr Gly Ala
    50                  55                  60

Ala Cys Gly Thr Thr Gly Cys Cys Thr Cys Cys Ala Ala Cys Gly Ala
65                  70                  75                  80

Gly Ala Cys Cys Cys Thr Gly Ala Thr Cys Ala Cys Cys Cys Cys Thr
                85                  90                  95

Gly Ala Thr Cys Ala Gly Cys Thr Cys Ala Ala Gly Ala Ala Gly Gly
            100                 105                 110

Ala Ala Ala Thr Cys Cys Cys Cys Cys Thr Cys Ala Gly Cys Gly Cys
        115                 120                 125

Cys Ala Ala Gly Gly Cys Cys Cys Thr Gly Cys Ala Gly Ala Cys Cys
    130                 135                 140
```

-continued

```
Gly Thr Gly Ala Cys Thr Gly Cys Cys Gly Gly Cys Cys Gly Thr Gly
145                 150                 155                 160

Ala Ala Gly Thr Gly Gly Thr Gly Cys Gly Cys Ala Ala Thr Ala Thr
                165                 170                 175

Thr Cys Thr Cys Gly Ala Cys Gly Gly Cys Ala Ala Gly Gly Ala Cys
            180                 185                 190

Cys Ala Thr Cys Gly Cys Cys Thr Gly Thr Thr Cys Gly Thr Cys Gly
        195                 200                 205

Thr Gly Gly Thr Cys Gly Gly Cys Cys Cys Thr Thr Gly Cys Thr Cys
    210                 215                 220

Cys Ala Thr Cys Cys Ala Cys Gly Ala Cys Ala Thr Cys Ala Ala Gly
225                 230                 235                 240

Gly Cys Ala Gly Cys Cys Cys Ala Cys Gly Ala Ala Thr Ala Cys Gly
                245                 250                 255

Cys Cys Gly Ala Gly Cys Gly Cys Cys Thr Gly Ala Ala Ala Gly Thr
            260                 265                 270

Gly Cys Thr Gly Gly Cys Cys Gly Ala Ala Gly Ala Ala Gly Thr Gly
        275                 280                 285

Thr Cys Cys Gly Ala Thr Ala Cys Gly Cys Thr Gly Thr Ala Cys Cys
    290                 295                 300

Thr Gly Gly Thr Cys Ala Thr Gly Cys Gly Cys Gly Thr Gly Thr Ala
305                 310                 315                 320

Cys Thr Thr Cys Gly Ala Ala Ala Ala Gly Cys Cys Gly Cys Gly Cys
                325                 330                 335

Ala Cys Cys Ala Cys Cys Gly Thr Cys Gly Gly Cys Thr Gly Gly Ala
            340                 345                 350

Ala Ala Gly Gly Cys Cys Thr Gly Ala Thr Cys Ala Ala Cys Gly Ala
        355                 360                 365

Thr Cys Cys Gly Thr Ala Cys Cys Thr Gly Gly Ala Thr Gly Ala Cys
    370                 375                 380

Thr Cys Gly Thr Thr Cys Ala Ala Gly Ala Thr Cys Cys Ala Gly Gly
385                 390                 395                 400

Ala Cys Gly Gly Cys Cys Thr Gly Cys Ala Cys Ala Thr Cys Gly Gly
                405                 410                 415

Cys Cys Gly Cys Ala Ala Gly Thr Thr Gly Cys Thr Gly Cys Thr Gly
            420                 425                 430

Gly Ala Cys Cys Thr Gly Gly Cys Cys Gly Ala Ala Ala Thr Gly Gly
        435                 440                 445

Gly Cys Cys Thr Gly Cys Cys Gly Ala Cys Cys Gly Cys Cys Ala Cys
    450                 455                 460

Cys Gly Ala Ala Gly Cys Gly Cys Thr Cys Gly Ala Cys Cys Thr Gly
465                 470                 475                 480

Ala Thr Thr Thr Cys Gly Cys Cys Gly Gly Cys Cys Thr Ala Cys Cys
                485                 490                 495

Thr Gly Cys Ala Ala Gly Ala Cys Cys Thr Gly Ala Thr Cys Ala Gly
            500                 505                 510

Cys Thr Gly Gly Thr Cys Gly Gly Cys Cys Ala Thr Cys Gly Gly Thr
        515                 520                 525

Gly Cys Cys Cys Gly Cys Ala Cys Cys Ala Cys Cys Gly Ala Ala Thr
    530                 535                 540
```

-continued

Cys Gly Cys Ala Ala Ala Cys Ala Cys Ala Cys Cys Gly Cys Gly Ala
545 550 555 560

Gly Ala Thr Gly Gly Cys Cys Thr Cys Gly Gly Gly Cys Cys Thr Gly
565 570 575

Thr Cys Cys Thr Cys Gly Gly Cys Gly Gly Thr Gly Gly Gly Thr Thr
580 585 590

Thr Cys Ala Ala Gly Ala Ala Cys Gly Gly Thr Ala Cys Cys Gly Ala
595 600 605

Thr Gly Gly Cys Gly Gly Cys Cys Thr Gly Ala Cys Cys Gly Thr Thr
610 615 620

Gly Cys Cys Ala Thr Cys Ala Ala Thr Gly Cys Cys Cys Thr Gly Cys
625 630 635 640

Ala Gly Thr Cys Gly Gly Thr Gly Thr Cys Cys Ala Ala Gly Cys Cys
645 650 655

Gly Cys Ala Cys Cys Gly Cys Thr Thr Cys Cys Thr Gly Gly Gly Cys
660 665 670

Ala Thr Cys Ala Ala Cys Cys Ala Gly Gly Ala Ala Gly Gly Cys Gly
675 680 685

Gly Cys Gly Thr Gly Thr Cys Gly Ala Thr Cys Gly Thr Cys Ala Cys
690 695 700

Cys Ala Cys Cys Ala Ala Gly Gly Gly Cys Ala Ala Cys Cys Cys Ala
705 710 715 720

Thr Ala Cys Gly Gly Cys Cys Ala Cys Gly Thr Gly Gly Thr Ala Cys
725 730 735

Thr Gly Cys Gly Cys Gly Gly Cys Gly Gly Cys Ala Ala Thr Gly Gly
740 745 750

Cys Ala Ala Gly Cys Cys Gly Ala Ala Cys Thr Ala Cys Gly Ala Cys
755 760 765

Thr Cys Gly Gly Thr Cys Ala Gly Cys Gly Thr Cys Gly Cys Cys Cys
770 775 780

Thr Gly Thr Gly Cys Gly Ala Ala Cys Ala Gly Gly Ala Cys Cys Thr
785 790 795 800

Gly Gly Cys Cys Ala Ala Gly Gly Cys Cys Ala Ala Gly Ala Thr Cys
805 810 815

Ala Ala Gly Gly Cys Cys Ala Ala Cys Ala Thr Cys Ala Thr Gly Gly
820 825 830

Thr Cys Gly Ala Cys Thr Gly Cys Ala Gly Cys Cys Ala Thr Gly Cys
835 840 845

Cys Ala Ala Cys Thr Cys Cys Ala Ala Cys Ala Ala Gly Gly Ala Cys
850 855 860

Cys Cys Gly Gly Cys Cys Cys Thr Gly Cys Ala Ala Cys Cys Gly Cys
865 870 875 880

Thr Gly Gly Thr Gly Ala Thr Gly Gly Ala Gly Ala Ala Cys Gly Thr
885 890 895

Cys Gly Cys Cys Ala Ala Cys Cys Ala Gly Ala Thr Thr Cys Thr Cys
900 905 910

Gly Ala Ala Gly Gly Cys Ala Ala Cys Cys Ala Gly Thr Cys Gly Ala
915 920 925

Thr Cys Ala Thr Cys Gly Gly Cys Cys Thr Gly Ala Thr Gly Gly Thr
930 935 940

Cys Gly Ala Ala Ala Gly Cys Cys Ala Cys Cys Thr Gly Ala Ala Cys

-continued

945 950 955 960

Thr Gly Gly Gly Gly Cys Thr Gly Thr Cys Ala Gly Thr Cys Cys Ala
965 970 975

Thr Thr Cys Cys Gly Ala Ala Ala Ala Ala Cys Cys Thr Gly Gly Ala
980 985 990

Cys Gly Ala Thr Thr Thr Gly Cys Ala Gly Thr Ala Thr Gly Gly Cys
995 1000 1005

Gly Thr Gly Thr Cys Gly Ala Thr Cys Ala Cys Gly Gly Ala Cys
1010 1015 1020

Gly Cys Cys Thr Gly Cys Ala Thr Cys Gly Ala Cys Thr Gly Gly
1025 1030 1035

Thr Cys Gly Gly Cys Thr Ala Cys Cys Gly Ala Gly Ala Ala Ala
1040 1045 1050

Ala Cys Cys Cys Thr Gly Cys Gly Cys Ala Gly Cys Ala Thr Gly
1055 1060 1065

Cys Ala Thr Gly Cys Cys Ala Ala Gly Cys Thr Cys Ala Ala Gly
1070 1075 1080

Gly Ala Thr Gly Thr Gly Cys Thr Gly Cys Cys Gly Cys Ala Gly
1085 1090 1095

Cys Gly Thr Ala Ala Gly Cys Gly Cys Gly Gly Cys Thr Gly Ala
1100 1105 1110

<210> SEQ ID NO 12
<211> LENGTH: 1113
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 12

Thr Thr Gly Gly Cys Gly Gly Cys Cys Gly Gly Cys Ala Cys Cys Cys
1 5 10 15

Gly Thr Gly Ala Cys Cys Thr Gly Ala Cys Gly Ala Gly Thr Ala Ala
20 25 30

Cys Ala Cys Gly Ala Thr Gly Gly Cys Thr Gly Ala Thr Thr Thr Ala
35 40 45

Cys Cys Gly Ala Thr Cys Gly Ala Thr Gly Ala Cys Thr Thr Gly Ala
50 55 60

Ala Cys Gly Thr Thr Gly Cys Cys Thr Cys Cys Ala Ala Cys Gly Ala
65 70 75 80

Gly Ala Cys Cys Cys Thr Gly Ala Thr Cys Ala Cys Cys Cys Cys Thr
85 90 95

Gly Ala Thr Cys Ala Gly Cys Thr Cys Ala Ala Gly Ala Ala Gly Gly
100 105 110

Ala Ala Ala Thr Cys Cys Cys Cys Cys Thr Cys Ala Gly Cys Gly Cys
115 120 125

Cys Ala Ala Gly Gly Cys Cys Cys Thr Gly Cys Ala Gly Ala Cys Cys
130 135 140

Gly Thr Gly Ala Cys Thr Gly Cys Cys Gly Gly Cys Cys Gly Thr Gly
145 150 155 160

Ala Ala Gly Thr Gly Gly Thr Gly Cys Gly Cys Ala Ala Thr Ala Thr
165 170 175

Thr Cys Thr Cys Gly Ala Cys Gly Gly Cys Ala Ala Gly Gly Ala Cys
180 185 190

-continued

Cys Ala Thr Cys Gly Cys Cys Thr Gly Thr Thr Cys Gly Thr Cys Gly
195 200 205

Thr Gly Gly Thr Cys Gly Gly Cys Cys Cys Thr Thr Gly Cys Thr Cys
210 215 220

Cys Ala Thr Cys Cys Ala Cys Gly Ala Cys Ala Thr Cys Ala Ala Gly
225 230 235 240

Gly Cys Ala Gly Cys Cys Cys Ala Cys Gly Ala Ala Thr Ala Cys Gly
245 250 255

Cys Cys Gly Ala Gly Cys Gly Cys Cys Thr Gly Ala Ala Ala Gly Thr
260 265 270

Gly Cys Thr Gly Gly Cys Cys Gly Ala Ala Gly Ala Ala Gly Thr Gly
275 280 285

Thr Cys Cys Gly Ala Thr Ala Cys Gly Cys Thr Gly Thr Ala Cys Cys
290 295 300

Thr Gly Gly Thr Cys Ala Thr Gly Cys Gly Cys Gly Thr Gly Thr Ala
305 310 315 320

Cys Thr Thr Cys Gly Ala Ala Ala Ala Gly Cys Cys Gly Cys Gly Cys
325 330 335

Ala Cys Cys Ala Cys Cys Gly Thr Cys Gly Gly Cys Thr Gly Gly Ala
340 345 350

Ala Ala Gly Gly Cys Cys Thr Gly Ala Thr Cys Ala Ala Cys Gly Ala
355 360 365

Thr Cys Cys Gly Thr Ala Cys Cys Thr Gly Gly Ala Thr Gly Ala Cys
370 375 380

Thr Cys Gly Thr Thr Cys Ala Ala Gly Ala Thr Cys Cys Ala Gly Gly
385 390 395 400

Ala Cys Gly Gly Cys Cys Thr Gly Cys Ala Cys Ala Thr Cys Gly Gly
405 410 415

Cys Cys Gly Cys Ala Ala Gly Thr Thr Gly Cys Thr Gly Cys Thr Gly
420 425 430

Gly Ala Cys Cys Thr Gly Gly Cys Cys Gly Ala Ala Ala Thr Gly Gly
435 440 445

Gly Cys Cys Thr Gly Cys Cys Gly Ala Cys Cys Gly Cys Cys Ala Cys
450 455 460

Cys Gly Ala Ala Gly Cys Gly Cys Thr Cys Gly Ala Cys Cys Thr Gly
465 470 475 480

Ala Thr Thr Thr Cys Gly Cys Cys Gly Cys Ala Gly Thr Ala Cys Cys
485 490 495

Thr Gly Cys Ala Ala Gly Ala Cys Cys Thr Gly Ala Thr Cys Ala Gly
500 505 510

Cys Thr Gly Gly Thr Cys Gly Gly Cys Cys Ala Thr Cys Gly Gly Thr
515 520 525

Gly Cys Cys Cys Gly Cys Ala Cys Cys Ala Cys Cys Gly Ala Ala Thr
530 535 540

Cys Gly Cys Ala Ala Ala Cys Ala Cys Ala Cys Cys Gly Cys Gly Ala
545 550 555 560

Gly Ala Thr Gly Gly Cys Cys Thr Cys Gly Gly Gly Thr Thr Thr Gly
565 570 575

Gly Cys Ala Thr Cys Gly Gly Cys Gly Gly Thr Gly Gly Gly Thr Thr
580 585 590

Thr Cys Ala Ala Gly Ala Ala Cys Gly Gly Thr Ala Cys Cys Gly Ala

-continued

```
        595                 600                 605

Thr Gly Gly Cys Gly Gly Cys Cys Thr Gly Ala Cys Cys Gly Thr Thr
    610                 615                 620

Gly Cys Cys Ala Thr Cys Ala Ala Thr Gly Cys Cys Cys Thr Gly Cys
625                 630                 635                 640

Ala Gly Thr Cys Gly Gly Thr Gly Thr Cys Cys Ala Ala Gly Cys Cys
                645                 650                 655

Gly Cys Ala Cys Cys Gly Cys Thr Thr Cys Cys Thr Gly Gly Gly Cys
            660                 665                 670

Ala Thr Cys Ala Ala Cys Cys Ala Gly Gly Ala Ala Gly Gly Cys Gly
        675                 680                 685

Gly Cys Gly Thr Gly Thr Cys Gly Ala Thr Cys Gly Thr Cys Ala Cys
    690                 695                 700

Cys Ala Cys Cys Ala Ala Gly Gly Gly Cys Ala Ala Cys Cys Cys Ala
705                 710                 715                 720

Thr Ala Cys Gly Gly Cys Cys Ala Cys Gly Thr Gly Gly Thr Ala Cys
                725                 730                 735

Thr Gly Cys Gly Cys Gly Gly Cys Gly Gly Cys Ala Ala Thr Gly Gly
            740                 745                 750

Cys Ala Ala Gly Cys Cys Gly Ala Ala Cys Thr Ala Cys Gly Ala Cys
        755                 760                 765

Thr Cys Gly Gly Thr Cys Ala Gly Cys Gly Thr Cys Gly Cys Cys Cys
    770                 775                 780

Thr Gly Thr Gly Cys Gly Ala Ala Cys Ala Gly Gly Ala Cys Cys Thr
785                 790                 795                 800

Gly Gly Cys Cys Ala Ala Gly Gly Cys Cys Ala Ala Gly Ala Thr Cys
                805                 810                 815

Ala Ala Gly Gly Cys Cys Ala Ala Cys Ala Thr Cys Ala Thr Gly Gly
            820                 825                 830

Thr Cys Gly Ala Cys Thr Gly Cys Ala Gly Cys Cys Ala Thr Gly Cys
        835                 840                 845

Cys Ala Ala Cys Thr Cys Cys Ala Ala Cys Ala Ala Gly Gly Ala Cys
    850                 855                 860

Cys Cys Gly Gly Cys Cys Cys Thr Gly Cys Ala Ala Cys Cys Gly Cys
865                 870                 875                 880

Thr Gly Gly Thr Gly Ala Thr Gly Gly Ala Gly Ala Ala Cys Gly Thr
                885                 890                 895

Cys Gly Cys Cys Ala Ala Cys Cys Ala Gly Ala Thr Thr Cys Thr Cys
            900                 905                 910

Gly Ala Ala Gly Gly Cys Ala Ala Cys Cys Ala Gly Thr Cys Gly Ala
        915                 920                 925

Thr Cys Ala Thr Cys Gly Gly Cys Cys Thr Gly Ala Thr Gly Gly Thr
    930                 935                 940

Cys Gly Ala Ala Ala Gly Cys Cys Ala Cys Cys Thr Gly Ala Ala Cys
945                 950                 955                 960

Thr Gly Gly Gly Gly Cys Thr Gly Thr Cys Ala Gly Thr Cys Cys Ala
                965                 970                 975

Thr Thr Cys Cys Gly Ala Ala Ala Ala Ala Cys Cys Thr Gly Gly Ala
            980                 985                 990

Cys Gly Ala Thr Thr Thr Gly Cys  Ala Gly Thr Ala Thr  Gly Gly Cys
        995                 1000                1005
```

-continued

```
Gly Thr  Gly Thr Cys Gly Ala  Thr Cys Ala Cys Gly  Gly Ala Cys
    1010                 1015                 1020

Gly Cys  Cys Thr Gly Cys Ala  Thr Cys Gly Ala Cys  Thr Gly Gly
    1025                 1030                 1035

Thr Cys  Gly Gly Cys Thr Ala  Cys Cys Gly Ala Gly  Ala Ala Ala
    1040                 1045                 1050

Ala Cys  Cys Cys Thr Gly Cys  Gly Cys Ala Gly Cys  Ala Thr Gly
    1055                 1060                 1065

Cys Ala  Thr Gly Cys Cys Ala  Ala Gly Cys Thr Cys  Ala Ala Gly
    1070                 1075                 1080

Gly Ala  Thr Gly Thr Gly Cys  Thr Gly Cys Cys Gly  Cys Ala Gly
    1085                 1090                 1095

Cys Gly  Thr Ala Ala Gly Cys  Gly Cys Gly Gly Cys  Thr Gly Ala
    1100                 1105                 1110


<210> SEQ ID NO 13
<211> LENGTH: 789
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 13

Ala Thr Gly Ala Ala Ala Cys Ala Gly Ala Cys Gly Cys Ala Ala Thr
1               5                   10                  15

Ala Cys Gly Thr Gly Gly Cys Ala Cys Gly Cys Gly Ala Gly Cys Cys
            20                  25                  30

Cys Gly Ala Thr Gly Cys Gly Cys Ala Thr Gly Gly Thr Thr Thr Thr
        35                  40                  45

Ala Thr Cys Gly Ala Thr Thr Ala Cys Cys Cys Gly Cys Ala Gly Cys
    50                  55                  60

Ala Ala Gly Ala Gly Cys Ala Thr Gly Cys Gly Gly Thr Gly Thr Gly
65                  70                  75                  80

Gly Ala Ala Cys Ala Cys Cys Cys Thr Gly Ala Thr Cys Ala Cys Cys
                85                  90                  95

Cys Gly Cys Cys Ala Gly Cys Thr Gly Ala Ala Ala Gly Thr Gly Ala
            100                 105                 110

Thr Cys Gly Ala Ala Gly Gly Cys Cys Gly Thr Gly Cys Gly Thr Gly
        115                 120                 125

Cys Cys Ala Gly Gly Ala Ala Thr Ala Cys Cys Thr Gly Gly Ala Cys
    130                 135                 140

Gly Gly Cys Ala Thr Cys Gly Ala Cys Cys Ala Gly Cys Thr Gly Ala
145                 150                 155                 160

Ala Ala Thr Thr Gly Cys Cys Gly Cys Ala Thr Gly Ala Cys Cys Gly
                165                 170                 175

Cys Ala Thr Thr Cys Cys Gly Cys Ala Ala Cys Thr Gly Gly Gly Cys
            180                 185                 190

Gly Ala Gly Ala Thr Cys Ala Ala Cys Ala Ala Gly Gly Thr Gly Cys
        195                 200                 205

Thr Gly Gly Gly Thr Gly Cys Cys Ala Cys Cys Ala Cys Cys Gly Gly
    210                 215                 220

Cys Thr Gly Gly Cys Ala Gly Gly Thr Thr Gly Cys Cys Cys Gly Gly
225                 230                 235                 240

Gly Thr Thr Cys Cys Gly Gly Cys Gly Cys Thr Gly Ala Thr Cys Cys
```

-continued

```
                245                 250                 255

Cys Cys Thr Thr Cys Cys Ala Gly Ala Cys Cys Thr Thr Cys Thr Thr
            260                 265                 270

Cys Gly Ala Ala Thr Thr Gly Cys Thr Gly Gly Cys Cys Ala Gly Cys
        275                 280                 285

Ala Ala Gly Cys Gly Cys Thr Thr Thr Cys Cys Gly Gly Thr Cys Gly
    290                 295                 300

Cys Cys Ala Cys Cys Thr Thr Cys Ala Thr Cys Cys Gly Cys Ala Cys
305                 310                 315                 320

Cys Cys Cys Gly Gly Ala Ala Gly Ala Gly Cys Thr Gly Gly Ala Cys
                325                 330                 335

Thr Ala Cys Cys Thr Gly Cys Ala Ala Gly Ala Gly Cys Cys Gly Gly
            340                 345                 350

Ala Thr Ala Thr Cys Thr Thr Cys Cys Ala Cys Gly Ala Gly Ala Thr
        355                 360                 365

Cys Thr Thr Cys Gly Gly Cys Cys Ala Cys Thr Gly Cys Cys Cys Gly
    370                 375                 380

Cys Thr Gly Cys Thr Gly Ala Cys Cys Ala Ala Thr Cys Cys Cys Thr
385                 390                 395                 400

Gly Gly Thr Thr Cys Gly Cys Cys Gly Ala Ala Thr Thr Cys Ala Cys
                405                 410                 415

Cys Cys Ala Cys Ala Cys Cys Thr Ala Cys Gly Gly Cys Ala Ala Gly
            420                 425                 430

Cys Thr Cys Gly Gly Cys Cys Thr Gly Gly Cys Cys Gly Cys Gly Ala
        435                 440                 445

Cys Cys Ala Ala Gly Gly Ala Ala Cys Ala Ala Cys Gly Thr Gly Thr
    450                 455                 460

Gly Thr Ala Cys Cys Thr Gly Gly Cys Ala Cys Gly Cys Thr Thr Gly
465                 470                 475                 480

Thr Ala Cys Thr Gly Gly Ala Thr Gly Ala Cys Cys Ala Thr Cys Gly
                485                 490                 495

Ala Gly Thr Thr Thr Gly Gly Cys Cys Thr Gly Ala Thr Gly Gly Ala
            500                 505                 510

Ala Ala Cys Cys Gly Cys Gly Cys Ala Ala Gly Gly Cys Cys Gly Cys
        515                 520                 525

Ala Ala Ala Ala Thr Cys Thr Ala Thr Gly Gly Thr Gly Gly Thr Gly
    530                 535                 540

Gly Cys Ala Thr Cys Cys Thr Cys Thr Cys Gly Thr Cys Gly Cys Cys
545                 550                 555                 560

Gly Ala Ala Ala Gly Ala Gly Ala Cys Cys Gly Thr Cys Thr Ala Cys
                565                 570                 575

Ala Gly Thr Cys Thr Gly Thr Cys Thr Gly Ala Cys Gly Ala Gly Cys
            580                 585                 590

Cys Thr Gly Ala Gly Cys Ala Cys Cys Ala Gly Gly Cys Cys Thr Thr
        595                 600                 605

Cys Gly Ala Cys Cys Cys Gly Ala Thr Cys Gly Ala Gly Gly Cys Cys
    610                 615                 620

Ala Thr Gly Cys Gly Thr Ala Cys Ala Cys Cys Cys Thr Ala Cys Cys
625                 630                 635                 640

Gly Cys Ala Thr Cys Gly Ala Cys Ala Thr Thr Cys Thr Gly Cys Ala
                645                 650                 655
```

-continued

Ala Cys Cys Gly Gly Thr Gly Thr Ala Thr Thr Thr Cys Gly Thr Ala
660 665 670

Cys Thr Gly Cys Cys Gly Ala Ala Cys Ala Thr Gly Ala Ala Gly Cys
675 680 685

Gly Cys Cys Thr Gly Thr Thr Cys Gly Ala Cys Cys Thr Gly Gly Cys
690 695 700

Cys Cys Ala Cys Gly Ala Gly Gly Ala Cys Ala Thr Cys Ala Thr Gly
705 710 715 720

Gly Gly Cys Ala Thr Gly Gly Thr Cys Cys Ala Thr Ala Ala Ala Gly
725 730 735

Cys Cys Ala Thr Gly Cys Ala Gly Cys Thr Gly Gly Gly Thr Cys Thr
740 745 750

Gly Cys Ala Thr Gly Cys Ala Cys Cys Gly Ala Ala Gly Thr Thr Thr
755 760 765

Cys Cys Ala Cys Cys Cys Ala Ala Gly Gly Thr Cys Gly Cys Thr Gly
770 775 780

Cys Cys Thr Gly Ala
785

<210> SEQ ID NO 14
<211> LENGTH: 357
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 14

Ala Thr Gly Ala Ala Thr Gly Cys Cys Thr Thr Gly Ala Ala Cys Cys
1 5 10 15

Ala Ala Gly Cys Cys Cys Ala Thr Thr Gly Cys Gly Ala Ala Gly Cys
20 25 30

Cys Thr Gly Cys Cys Gly Cys Gly Cys Cys Gly Ala Cys Gly Cys Ala
35 40 45

Cys Cys Gly Ala Ala Ala Gly Thr Cys Ala Cys Cys Gly Ala Cys Gly
50 55 60

Ala Ala Gly Ala Gly Cys Thr Gly Gly Cys Cys Gly Ala Gly Cys Thr
65 70 75 80

Gly Ala Thr Cys Cys Gly Cys Gly Ala Ala Ala Thr Cys Cys Cys Gly
85 90 95

Gly Ala Cys Thr Gly Gly Ala Ala Cys Ala Thr Cys Gly Ala Ala Gly
100 105 110

Thr Ala Cys Gly Thr Gly Ala Cys Gly Gly Cys Cys Ala Cys Ala Thr
115 120 125

Gly Gly Ala Gly Cys Thr Gly Gly Ala Gly Cys Gly Cys Gly Thr Gly
130 135 140

Thr Thr Cys Cys Thr Gly Thr Thr Cys Ala Ala Gly Ala Ala Cys Thr
145 150 155 160

Thr Cys Ala Ala Gly Cys Ala Cys Gly Cys Cys Cys Thr Gly Gly Cys
165 170 175

Gly Thr Thr Cys Ala Cys Cys Ala Ala Thr Gly Cys Cys Gly Thr Gly
180 185 190

Gly Gly Cys Gly Ala Ala Ala Thr Thr Gly Cys Cys Gly Ala Ala Gly
195 200 205

Cys Cys Gly Ala Ala Gly Gly Cys Cys Ala Cys Cys Ala Cys Cys Cys

-continued

```
    210                 215                 220

Ala Gly Gly Gly Cys Thr Gly Cys Thr Gly Ala Cys Thr Gly Ala Ala
225                 230                 235                 240

Thr Gly Gly Gly Gly Cys Ala Ala Gly Gly Thr Cys Ala Cys Cys Gly
                245                 250                 255

Thr Gly Ala Cys Cys Thr Gly Gly Thr Gly Gly Ala Gly Cys Cys Ala
            260                 265                 270

Cys Thr Cys Gly Ala Thr Cys Ala Ala Ala Gly Gly Cys Cys Thr Gly
        275                 280                 285

Cys Ala Cys Cys Gly Cys Ala Ala Cys Gly Ala Cys Thr Thr Cys Ala
    290                 295                 300

Thr Cys Ala Thr Gly Thr Gly Cys Gly Cys Ala Cys Gly Cys Ala Cys
305                 310                 315                 320

Thr Gly Ala Cys Ala Ala Gly Gly Thr Gly Gly Cys Gly Gly Ala Ala
                325                 330                 335

Ala Cys Gly Gly Cys Thr Gly Ala Ala Gly Gly Cys Cys Gly Gly Ala
            340                 345                 350

Ala Gly Thr Ala Ala
        355


<210> SEQ ID NO 15
<211> LENGTH: 2082
<212> TYPE: PRT
<213> ORGANISM: Rhodotorula glutinis

<400> SEQUENCE: 15

Ala Thr Gly Gly Cys Cys Cys Cys Thr Cys Gly Cys Cys Cys Thr Ala
1               5                   10                  15

Cys Cys Thr Cys Ala Cys Ala Gly Ala Ala Cys Cys Ala Ala Ala Cys
            20                  25                  30

Cys Cys Gly Cys Ala Cys Ala Thr Gly Cys Cys Cys Gly Ala Cys Gly
        35                  40                  45

Ala Cys Gly Cys Ala Gly Gly Thr Thr Ala Cys Thr Cys Ala Ala Gly
    50                  55                  60

Thr Thr Gly Ala Thr Ala Thr Ala Gly Thr Thr Gly Ala Gly Ala Ala
65                  70                  75                  80

Gly Ala Thr Gly Cys Thr Cys Gly Cys Thr Gly Cys Ala Cys Cys Ala
                85                  90                  95

Ala Cys Thr Gly Ala Cys Ala Gly Cys Ala Cys Cys Cys Thr Ala Gly
            100                 105                 110

Ala Gly Cys Thr Cys Gly Ala Cys Gly Gly Gly Thr Ala Thr Thr Cys
        115                 120                 125

Ala Cys Thr Ala Ala Ala Thr Cys Thr Thr Gly Gly Gly Gly Ala Cys
    130                 135                 140

Gly Thr Cys Gly Thr Thr Thr Cys Ala Gly Cys Thr Gly Cys Ala Ala
145                 150                 155                 160

Gly Gly Ala Ala Ala Gly Gly Ala Ala Gly Ala Cys Cys Thr Gly Thr
                165                 170                 175

Ala Ala Gly Ala Gly Thr Ala Ala Ala Ala Gly Ala Thr Ala Gly Thr
            180                 185                 190

Gly Ala Thr Gly Ala Ala Ala Thr Thr Cys Gly Gly Ala Gly Thr Ala
        195                 200                 205
```

-continued

```
Ala Ala Ala Thr Ala Gly Ala Thr Ala Ala Gly Thr Cys Cys Gly Thr
    210                 215                 220

Ala Gly Ala Gly Thr Thr Thr Thr Thr Ala Ala Gly Gly Thr Cys Ala
225                 230                 235                 240

Cys Ala Ala Cys Thr Thr Ala Gly Cys Ala Thr Gly Thr Cys Cys Gly
                245                 250                 255

Thr Ala Thr Ala Cys Gly Gly Gly Gly Thr Cys Ala Cys Thr Ala Cys
            260                 265                 270

Cys Gly Gly Gly Thr Thr Cys Gly Gly Cys Gly Gly Thr Thr Cys Cys
        275                 280                 285

Gly Cys Cys Gly Ala Cys Ala Cys Cys Cys Gly Cys Ala Cys Cys Gly
    290                 295                 300

Ala Gly Gly Ala Cys Gly Cys Thr Ala Thr Ala Thr Cys Ala Thr Thr
305                 310                 315                 320

Gly Cys Ala Gly Ala Ala Ala Gly Cys Thr Cys Thr Thr Cys Thr Ala
                325                 330                 335

Gly Ala Gly Cys Ala Thr Cys Ala Gly Cys Thr Cys Thr Gly Cys Gly
            340                 345                 350

Gly Cys Gly Thr Thr Cys Thr Thr Cys Cys Ala Ala Gly Thr Thr Cys
        355                 360                 365

Cys Thr Thr Cys Gly Ala Thr Thr Cys Gly Thr Thr Thr Ala Gly Gly
    370                 375                 380

Cys Thr Gly Gly Gly Gly Cys Gly Cys Gly Gly Gly Cys Thr Thr Gly
385                 390                 395                 400

Ala Gly Ala Ala Cys Thr Cys Thr Cys Thr Gly Cys Cys Cys Cys Thr
                405                 410                 415

Ala Gly Ala Ala Gly Thr Gly Gly Thr Ala Ala Gly Gly Gly Gly Cys
            420                 425                 430

Gly Cys Thr Ala Thr Gly Ala Cys Ala Ala Thr Ala Cys Gly Gly Gly
        435                 440                 445

Thr Gly Ala Ala Cys Ala Gly Thr Cys Thr Ala Ala Cys Ala Ala Gly
    450                 455                 460

Ala Gly Gly Thr Cys Ala Cys Ala Gly Cys Gly Cys Gly Gly Thr Thr
465                 470                 475                 480

Ala Gly Ala Cys Thr Ala Gly Thr Thr Gly Thr Ala Cys Thr Thr Gly
                485                 490                 495

Ala Ala Gly Cys Thr Cys Thr Gly Ala Cys Thr Ala Ala Cys Thr Thr
            500                 505                 510

Cys Thr Thr Ala Ala Ala Cys Cys Ala Cys Gly Gly Gly Ala Thr Thr
        515                 520                 525

Ala Cys Cys Cys Cys Gly Ala Thr Thr Gly Thr Cys Cys Cys Ala Cys
    530                 535                 540

Thr Cys Cys Gly Gly Gly Gly Ala Ala Cys Cys Ala Thr Cys Ala Gly
545                 550                 555                 560

Thr Gly Cys Gly Thr Cys Cys Gly Gly Thr Gly Ala Cys Cys Thr Ala
                565                 570                 575

Thr Cys Gly Cys Cys Cys Cys Thr Cys Thr Cys Ala Thr Ala Thr Ala
            580                 585                 590

Thr Thr Gly Cys Gly Gly Cys Ala Gly Cys Thr Ala Thr Ala Thr Cys
        595                 600                 605

Ala Gly Gly Ala Cys Ala Thr Cys Cys Ala Gly Ala Thr Thr Cys Ala
```

-continued

```
    610                 615                 620

Ala Ala Gly Gly Thr Thr Cys Ala Thr Gly Thr Ala Gly Thr Ala Cys
625                 630                 635                 640

Ala Thr Gly Ala Ala Gly Gly Ala Ala Ala Ala Gly Ala Gly Ala Ala
                645                 650                 655

Ala Ala Thr Ala Cys Thr Thr Thr Ala Cys Gly Cys Ala Cys Gly Cys
            660                 665                 670

Gly Ala Gly Gly Cys Cys Ala Thr Gly Gly Cys Cys Cys Thr Thr Thr
        675                 680                 685

Thr Thr Ala Ala Cys Cys Thr Cys Gly Ala Gly Cys Cys Cys Gly Thr
    690                 695                 700

Gly Gly Thr Ala Cys Thr Thr Gly Gly Thr Cys Cys Gly Ala Ala Ala
705                 710                 715                 720

Gly Ala Gly Gly Gly Cys Cys Thr Cys Gly Gly Ala Cys Thr Ala Gly
                725                 730                 735

Thr Thr Ala Ala Cys Gly Gly Thr Ala Cys Thr Gly Cys Cys Gly Thr
            740                 745                 750

Cys Ala Gly Thr Gly Cys Cys Thr Cys Ala Ala Thr Gly Gly Cys Thr
        755                 760                 765

Ala Cys Gly Cys Thr Thr Gly Cys Ala Cys Thr Cys Cys Ala Cys Gly
    770                 775                 780

Ala Thr Gly Cys Gly Cys Ala Cys Ala Thr Gly Cys Thr Gly Ala Gly
785                 790                 795                 800

Cys Cys Thr Gly Cys Thr Ala Ala Gly Thr Cys Ala Ala Ala Gly Thr
                805                 810                 815

Cys Thr Cys Ala Cys Ala Gly Cys Gly Ala Thr Gly Ala Cys Cys Gly
            820                 825                 830

Thr Gly Gly Ala Gly Gly Cys Cys Ala Thr Gly Gly Thr Gly Gly Gly
        835                 840                 845

Gly Cys Ala Thr Gly Cys Gly Gly Gly Gly Thr Cys Ala Thr Thr Thr
    850                 855                 860

Cys Ala Thr Cys Cys Ala Thr Thr Thr Thr Thr Gly Cys Ala Thr Gly
865                 870                 875                 880

Ala Thr Gly Thr Cys Ala Cys Thr Cys Gly Thr Cys Cys Gly Cys Ala
                885                 890                 895

Thr Cys Cys Thr Ala Cys Gly Cys Ala Gly Ala Thr Thr Gly Ala Gly
            900                 905                 910

Gly Thr Ala Gly Cys Ala Gly Gly Cys Ala Ala Cys Ala Thr Thr Cys
        915                 920                 925

Gly Cys Ala Ala Gly Cys Thr Thr Cys Thr Cys Gly Ala Gly Gly Gly
    930                 935                 940

Ala Ala Gly Thr Cys Gly Thr Thr Thr Cys Gly Cys Cys Gly Thr Cys
945                 950                 955                 960

Cys Ala Thr Cys Ala Thr Gly Ala Gly Gly Ala Ala Gly Ala Ala Gly
                965                 970                 975

Thr Ala Ala Ala Ala Gly Thr Ala Ala Ala Gly Gly Ala Thr Gly Ala
            980                 985                 990

Cys Gly Ala Ala Gly Gly Ala Ala  Thr Ala Thr Thr Ala  Ala Gly Gly
        995                 1000                 1005

Cys Ala  Ala Gly Ala Cys Cys  Gly Ala Thr Ala Cys  Cys Cys Gly
    1010                 1015                 1020
```

-continued

Cys Thr Cys Cys Gly Cys Ala Cys Gly Thr Cys Ala Cys Cys Gly
1025 1030 1035

Cys Ala Ala Thr Gly Gly Thr Thr Gly Gly Gly Thr Cys Cys Ala
1040 1045 1050

Cys Thr Gly Gly Thr Thr Thr Cys Ala Gly Ala Cys Cys Thr Cys
1055 1060 1065

Ala Thr Cys Cys Ala Cys Gly Cys Ala Cys Ala Cys Gly Cys Cys
1070 1075 1080

Gly Thr Cys Thr Thr Ala Ala Cys Thr Ala Thr Thr Gly Ala Ala
1085 1090 1095

Gly Cys Ala Gly Gly Gly Cys Ala Ala Thr Cys Gly Ala Cys Gly
1100 1105 1110

Ala Cys Ala Gly Ala Cys Ala Ala Thr Cys Cys Thr Cys Thr Cys
1115 1120 1125

Ala Thr Cys Gly Ala Cys Gly Thr Ala Gly Ala Gly Ala Ala Thr
1130 1135 1140

Ala Ala Gly Ala Cys Cys Thr Cys Gly Cys Ala Thr Cys Ala Thr
1145 1150 1155

Gly Gly Ala Gly Gly Ala Ala Ala Thr Thr Thr Thr Cys Ala Ala
1160 1165 1170

Gly Cys Thr Gly Cys Ala Gly Cys Thr Gly Thr Cys Gly Cys Gly
1175 1180 1185

Ala Ala Cys Ala Cys Ala Ala Thr Gly Gly Ala Ala Ala Ala Gly
1190 1195 1200

Ala Cys Ala Cys Gly Thr Cys Thr Cys Gly Gly Cys Cys Thr Gly
1205 1210 1215

Gly Cys Gly Cys Ala Ala Ala Thr Ala Gly Gly Gly Ala Ala Ala
1220 1225 1230

Cys Thr Gly Ala Ala Thr Thr Thr Cys Ala Cys Cys Cys Ala Gly
1235 1240 1245

Cys Thr Cys Ala Cys Gly Gly Ala Ala Ala Thr Gly Cys Thr Gly
1250 1255 1260

Ala Ala Cys Gly Cys Cys Gly Gly Cys Ala Thr Gly Ala Ala Cys
1265 1270 1275

Cys Gly Cys Gly Gly Cys Cys Thr Gly Cys Cys Gly Thr Cys Thr
1280 1285 1290

Thr Gly Thr Cys Thr Cys Gly Cys Cys Gly Cys Gly Gly Ala Ala
1295 1300 1305

Gly Ala Thr Cys Cys Thr Thr Cys Thr Thr Thr Ala Thr Cys Ala
1310 1315 1320

Thr Ala Thr Cys Ala Cys Thr Gly Thr Ala Ala Gly Gly Gly Thr
1325 1330 1335

Thr Thr Ala Gly Ala Thr Ala Thr Cys Gly Cys Gly Gly Cys Ala
1340 1345 1350

Gly Cys Thr Gly Cys Ala Thr Ala Thr Ala Cys Gly Thr Cys Cys
1355 1360 1365

Gly Ala Ala Cys Thr Ala Gly Gly Thr Cys Ala Thr Cys Thr Gly
1370 1375 1380

Gly Cys Thr Ala Ala Cys Cys Cys Thr Gly Thr Cys Ala Cys Gly
1385 1390 1395

-continued

```
Ala Cys  Cys Cys Ala Cys Gly  Thr Ala Cys Ala Ala  Cys Cys Gly
    1400                 1405                 1410

Gly Cys  Gly Gly Ala Gly Ala  Thr Gly Gly Cys Thr  Ala Ala Thr
    1415                 1420                 1425

Cys Ala  Ala Gly Cys Ala Gly  Thr Thr Ala Ala Cys  Thr Cys Cys
    1430                 1435                 1440

Cys Thr  Thr Gly Cys Ala Cys  Thr Ala Ala Thr Thr  Thr Cys Cys
    1445                 1450                 1455

Gly Cys  Cys Cys Gly Cys Cys  Gly Gly Ala Cys Ala  Ala Cys Ala
    1460                 1465                 1470

Gly Ala  Gly Ala Gly Thr Ala  Ala Cys Gly Ala Cys  Gly Thr Gly
    1475                 1480                 1485

Thr Thr  Ala Thr Cys Ala Cys  Thr Gly Cys Thr Gly  Cys Thr Cys
    1490                 1495                 1500

Gly Cys  Thr Ala Cys Cys Cys  Ala Cys Thr Thr Ala  Thr Ala Cys
    1505                 1510                 1515

Thr Gly  Cys Gly Thr Cys Thr  Thr Gly Cys Ala Gly  Gly Cys Thr
    1520                 1525                 1530

Ala Thr  Cys Gly Ala Cys Thr  Thr Ala Cys Gly Cys  Gly Cys Ala
    1535                 1540                 1545

Ala Thr  Cys Gly Thr Gly Thr  Thr Cys Gly Ala Ala  Thr Thr Thr
    1550                 1555                 1560

Ala Ala  Gly Ala Ala Gly Cys  Ala Ala Thr Thr Cys  Gly Gly Gly
    1565                 1570                 1575

Cys Cys  Ala Gly Cys Thr Ala  Thr Thr Gly Thr Gly  Thr Cys Cys
    1580                 1585                 1590

Cys Thr  Ala Ala Thr Thr Gly  Ala Thr Cys Ala Gly  Cys Ala Cys
    1595                 1600                 1605

Thr Thr  Cys Gly Gly Ala Ala  Gly Cys Gly Cys Cys  Ala Thr Gly
    1610                 1615                 1620

Ala Cys  Thr Gly Gly Gly Thr  Cys Thr Ala Ala Thr  Cys Thr Thr
    1625                 1630                 1635

Cys Gly  Ala Gly Ala Cys Gly  Ala Gly Cys Thr Ala  Gly Thr Cys
    1640                 1645                 1650

Gly Ala  Ala Ala Ala Ala Gly  Thr Ala Ala Ala Thr  Ala Ala Gly
    1655                 1660                 1665

Ala Cys  Ala Cys Thr Cys Gly  Cys Ala Ala Ala Gly  Ala Gly Gly
    1670                 1675                 1680

Cys Thr  Gly Gly Ala Ala Cys  Ala Gly Ala Cys Thr  Ala Ala Cys
    1685                 1690                 1695

Ala Gly  Cys Thr Ala Cys Gly  Ala Cys Cys Thr Ala  Gly Thr Thr
    1700                 1705                 1710

Cys Cys  Ala Cys Gly Gly Thr  Gly Gly Cys Ala Cys  Gly Ala Cys
    1715                 1720                 1725

Gly Cys  Cys Thr Thr Thr Ala  Gly Thr Thr Thr Thr  Gly Cys Ala
    1730                 1735                 1740

Gly Cys  Gly Gly Gly Ala Ala  Cys Gly Gly Thr Ala  Gly Thr Ala
    1745                 1750                 1755

Gly Ala  Gly Gly Thr Ala Thr  Thr Gly Thr Cys Ala  Thr Cys Gly
    1760                 1765                 1770

Ala Cys  Thr Thr Cys Gly Thr  Thr Gly Thr Cys Gly  Thr Thr Gly
```

-continued

1775 1780 1785

Gly Cys Thr Gly Cys Thr Gly Thr Cys Ala Ala Cys Gly Cys Gly
1790 1795 1800

Thr Gly Gly Ala Ala Ala Gly Thr Thr Gly Cys Ala Gly Cys Thr
1805 1810 1815

Gly Cys Ala Gly Ala Gly Thr Cys Ala Gly Cys Ala Ala Thr Thr
1820 1825 1830

Thr Cys Gly Cys Thr Gly Ala Cys Gly Cys Gly Gly Cys Ala Ala
1835 1840 1845

Gly Thr Ala Cys Gly Cys Gly Ala Ala Ala Cys Ala Thr Thr Thr
1850 1855 1860

Thr Gly Gly Ala Gly Cys Gly Cys Thr Gly Cys Thr Thr Cys Gly
1865 1870 1875

Ala Cys Ala Ala Gly Cys Thr Cys Gly Cys Cys Ala Gly Cys Cys
1880 1885 1890

Cys Thr Thr Thr Cys Thr Thr Ala Cys Cys Thr Gly Thr Cys Cys
1895 1900 1905

Cys Cys Ala Cys Gly Thr Ala Cys Gly Cys Ala Gly Ala Thr Cys
1910 1915 1920

Thr Thr Gly Thr Ala Cys Gly Cys Ala Thr Thr Cys Gly Thr Ala
1925 1930 1935

Ala Gly Ala Gly Ala Gly Gly Ala Gly Thr Thr Ala Gly Gly Ala
1940 1945 1950

Gly Thr Cys Ala Ala Ala Gly Cys Cys Cys Gly Ala Ala Gly Gly
1955 1960 1965

Gly Gly Thr Gly Ala Cys Gly Thr Ala Thr Thr Cys Cys Thr Thr
1970 1975 1980

Gly Gly Ala Ala Ala Gly Cys Ala Ala Gly Ala Ala Gly Thr Thr
1985 1990 1995

Ala Cys Ala Ala Thr Thr Gly Gly Ala Thr Cys Cys Ala Ala Cys
2000 2005 2010

Gly Thr Thr Thr Cys Ala Ala Ala Gly Ala Thr Cys Thr Ala Thr
2015 2020 2025

Gly Ala Gly Gly Cys Cys Ala Thr Thr Ala Ala Gly Ala Gly Thr
2030 2035 2040

Gly Gly Gly Cys Gly Cys Ala Thr Ala Ala Ala Thr Ala Ala Cys
2045 2050 2055

Gly Thr Cys Cys Thr Gly Thr Thr Gly Ala Ala Gly Ala Thr Gly
2060 2065 2070

Cys Thr Gly Gly Cys Cys Thr Gly Ala
2075 2080

<210> SEQ ID NO 16
<211> LENGTH: 1884
<212> TYPE: PRT
<213> ORGANISM: Pseudomonas putida

<400> SEQUENCE: 16

Gly Thr Gly Ala Ala Thr Ala Ala Cys Gly Ala Ala Gly Cys Cys Cys
1 5 10 15

Gly Cys Thr Cys Ala Gly Gly Gly Thr Cys Gly Ala Cys Cys Gly Ala
20 25 30

-continued

```
Cys Cys Cys Thr Gly Gly Cys Cys Ala Ala Cys Gly Thr Cys Cys Gly
        35                  40                  45

Cys Gly Cys Thr Ala Cys Cys Gly Cys Cys Ala Gly Gly Thr Gly Gly
    50                  55                  60

Cys Cys Ala Thr Cys Gly Gly Gly Cys Ala Thr Cys Cys Cys Cys Ala
65                  70                  75                  80

Gly Gly Thr Gly Cys Ala Gly Gly Thr Cys Ala Gly Thr Cys Ala Cys
                85                  90                  95

Gly Thr Cys Gly Ala Cys Gly Ala Cys Gly Thr Gly Cys Thr Gly Cys
            100                 105                 110

Gly Cys Ala Thr Gly Cys Ala Ala Cys Cys Thr Gly Thr Cys Gly Ala
        115                 120                 125

Gly Cys Cys Ala Cys Thr Gly Gly Cys Gly Cys Cys Gly Cys Thr Gly
    130                 135                 140

Cys Cys Gly Gly Cys Gly Cys Gly Cys Cys Thr Gly Cys Thr Cys Gly
145                 150                 155                 160

Ala Gly Cys Gly Cys Cys Thr Gly Gly Thr Gly Cys Ala Thr Thr Gly
                165                 170                 175

Gly Gly Cys Cys Cys Ala Gly Gly Thr Gly Cys Gly Cys Cys Cys Gly
            180                 185                 190

Gly Ala Cys Ala Cys Cys Ala Cys Thr Thr Thr Cys Ala Thr Cys Gly
        195                 200                 205

Cys Gly Gly Cys Ala Cys Gly Cys Cys Ala Gly Gly Cys Ala Gly Ala
    210                 215                 220

Cys Gly Gly Thr Gly Cys Cys Thr Gly Gly Cys Gly Thr Thr Cys Gly
225                 230                 235                 240

Ala Thr Cys Ala Gly Cys Thr Ala Cys Gly Thr Gly Cys Ala Gly Ala
                245                 250                 255

Thr Gly Cys Thr Cys Gly Cys Cys Gly Ala Thr Gly Thr Gly Cys Gly
            260                 265                 270

Cys Ala Cys Cys Ala Thr Cys Gly Cys Cys Gly Cys Cys Ala Ala Cys
        275                 280                 285

Thr Thr Gly Cys Thr Ala Gly Gly Ala Cys Thr Gly Gly Gly Cys Cys
    290                 295                 300

Thr Cys Ala Gly Thr Gly Cys Cys Gly Ala Gly Cys Gly Cys Cys Cys
305                 310                 315                 320

Gly Cys Thr Gly Gly Cys Gly Cys Thr Gly Cys Thr Thr Thr Cys Cys
                325                 330                 335

Gly Gly Cys Ala Ala Cys Gly Ala Cys Ala Thr Cys Gly Ala Ala Cys
            340                 345                 350

Ala Cys Cys Thr Gly Cys Ala Ala Ala Thr Cys Gly Cys Cys Cys Thr
        355                 360                 365

Cys Gly Gly Cys Gly Cys Cys Ala Thr Gly Thr Ala Thr Gly Cys Cys
    370                 375                 380

Gly Gly Thr Ala Thr Thr Gly Cys Cys Thr Ala Thr Thr Gly Cys Cys
385                 390                 395                 400

Cys Gly Gly Thr Gly Thr Cys Gly Cys Cys Gly Gly Cys Cys Thr Ala
                405                 410                 415

Cys Gly Cys Gly Cys Thr Gly Thr Thr Gly Thr Cys Gly Cys Ala Ala
            420                 425                 430

Gly Ala Cys Thr Thr Cys Gly Cys Cys Ala Ala Gly Thr Thr Gly Cys
```

-continued

```
        435                 440                 445

Gly Cys Cys Ala Thr Gly Thr Cys Thr Gly Cys Gly Ala Gly Gly Thr
    450                 455                 460

Gly Cys Thr Cys Ala Cys Cys Cys Cys Cys Gly Gly Ala Gly Thr Gly
465                 470                 475                 480

Gly Thr Cys Thr Thr Cys Gly Thr Cys Ala Gly Cys Gly Ala Cys Ala
                485                 490                 495

Gly Cys Cys Ala Gly Cys Cys Gly Thr Thr Cys Cys Ala Gly Cys Gly
            500                 505                 510

Cys Gly Cys Cys Thr Thr Cys Gly Ala Gly Gly Cys Gly Gly Thr Gly
        515                 520                 525

Cys Thr Gly Gly Ala Cys Gly Ala Thr Thr Cys Gly Gly Thr Cys Gly
    530                 535                 540

Gly Cys Gly Thr Gly Ala Thr Cys Ala Gly Cys Gly Thr Gly Cys Gly
545                 550                 555                 560

Thr Gly Gly Cys Cys Ala Gly Gly Thr Cys Gly Cys Ala Gly Gly Thr
                565                 570                 575

Cys Gly Cys Cys Cys Cys Cys Ala Thr Ala Thr Ala Ala Gly Cys Thr
            580                 585                 590

Thr Cys Gly Ala Cys Ala Gly Cys Cys Thr Gly Thr Thr Gly Cys Ala
        595                 600                 605

Ala Cys Cys Gly Gly Gly Thr Gly Ala Cys Cys Thr Gly Gly Cys Gly
    610                 615                 620

Gly Cys Gly Gly Cys Cys Gly Ala Thr Gly Cys Gly Gly Cys Thr Thr
625                 630                 635                 640

Thr Cys Gly Cys Cys Gly Cys Cys Ala Cys Cys Gly Gly Gly Cys Cys
                645                 650                 655

Gly Gly Ala Cys Ala Cys Cys Ala Thr Cys Gly Cys Cys Ala Ala Ala
            660                 665                 670

Thr Thr Cys Cys Thr Cys Thr Thr Cys Ala Cys Cys Thr Cys Gly Gly
        675                 680                 685

Gly Cys Thr Cys Gly Ala Cys Cys Ala Ala Gly Cys Thr Gly Cys Cys
    690                 695                 700

Cys Ala Ala Gly Gly Cys Gly Gly Thr Gly Ala Thr Cys Ala Cys Cys
705                 710                 715                 720

Ala Cys Cys Cys Ala Gly Cys Gly Cys Ala Thr Gly Cys Thr Gly Thr
                725                 730                 735

Gly Cys Gly Cys Cys Ala Ala Thr Cys Ala Gly Cys Ala Gly Ala Thr
            740                 745                 750

Gly Cys Thr Thr Cys Thr Gly Cys Ala Gly Ala Cys Thr Thr Thr Thr
        755                 760                 765

Cys Cys Gly Ala Cys Gly Thr Thr Cys Gly Cys Cys Gly Ala Gly Gly
    770                 775                 780

Ala Gly Cys Cys Gly Cys Cys Gly Gly Thr Gly Cys Thr Gly Gly Thr
785                 790                 795                 800

Gly Gly Ala Cys Thr Gly Gly Cys Thr Gly Cys Cys Gly Thr Gly Gly
                805                 810                 815

Ala Ala Cys Cys Ala Cys Ala Cys Gly Thr Thr Cys Gly Gly Cys Gly
            820                 825                 830

Gly Thr Ala Gly Cys Cys Ala Cys Ala Ala Cys Cys Thr Cys Gly Gly
        835                 840                 845
```

-continued

```
Cys Ala Thr Cys Gly Thr Gly Cys Thr Thr Thr Ala Cys Ala Ala Cys
    850                 855                 860

Gly Gly Gly Gly Gly Cys Ala Gly Thr Thr Thr Cys Thr Ala Cys Cys
865                 870                 875                 880

Thr Gly Gly Ala Cys Gly Cys Cys Gly Gly Cys Ala Ala Gly Cys Cys
                885                 890                 895

Gly Ala Cys Cys Cys Cys Gly Cys Ala Ala Gly Gly Cys Thr Thr Cys
            900                 905                 910

Gly Cys Cys Gly Ala Gly Ala Cys Cys Thr Thr Gly Cys Gly Cys Ala
        915                 920                 925

Ala Thr Cys Thr Gly Cys Gly Cys Gly Ala Gly Ala Thr Thr Thr Cys
    930                 935                 940

Cys Cys Cys Cys Ala Cys Gly Gly Cys Cys Thr Ala Cys Cys Thr Cys
945                 950                 955                 960

Ala Cys Cys Gly Thr Ala Cys Cys Cys Ala Ala Gly Gly Gly Cys Thr
                965                 970                 975

Gly Gly Gly Ala Gly Gly Ala Ala Cys Thr Gly Gly Thr Cys Ala Ala
            980                 985                 990

Gly Gly Cys Ala Cys Thr Gly Gly  Ala Gly Cys Ala Gly  Gly Ala Cys
        995                 1000                1005

Cys Cys  Cys Gly Cys Gly Cys  Thr Ala Cys Gly Cys  Gly Ala Gly
    1010                1015                1020

Gly Thr  Gly Thr Thr Cys Thr  Thr Thr Gly Cys Cys  Cys Gly Cys
    1025                1030                1035

Ala Thr  Cys Ala Ala Gly Cys  Thr Gly Thr Thr Cys  Thr Thr Cys
    1040                1045                1050

Thr Thr  Thr Gly Cys Cys Gly  Cys Cys Gly Cys Ala  Gly Gly Cys
    1055                1060                1065

Cys Thr  Gly Thr Cys Gly Cys  Ala Ala Ala Gly Cys  Gly Thr Cys
    1070                1075                1080

Thr Gly  Gly Gly Ala Cys Cys  Gly Gly Cys Thr Gly  Gly Ala Cys
    1085                1090                1095

Cys Gly  Cys Ala Thr Thr Gly  Cys Cys Gly Ala Gly  Cys Ala Ala
    1100                1105                1110

Cys Ala  Cys Thr Gly Thr Gly  Gly Cys Gly Ala Ala  Cys Gly Cys
    1115                1120                1125

Ala Thr  Cys Cys Gly Cys Ala  Thr Gly Ala Thr Gly  Gly Cys Cys
    1130                1135                1140

Gly Gly  Cys Cys Thr Thr Gly  Gly Cys Ala Thr Gly  Ala Cys Cys
    1145                1150                1155

Gly Ala  Ala Gly Cys Cys Thr  Cys Gly Cys Cys Ala  Thr Cys Gly
    1160                1165                1170

Thr Gly  Cys Ala Cys Cys Thr  Thr Cys Ala Cys Cys  Ala Cys Cys
    1175                1180                1185

Gly Gly  Gly Cys Cys Thr Thr  Thr Gly Thr Cys Gly  Ala Thr Gly
    1190                1195                1200

Gly Cys  Cys Gly Gly Cys Thr  Ala Thr Gly Thr Cys  Gly Gly Gly
    1205                1210                1215

Cys Thr  Gly Cys Cys Gly Gly  Cys Ala Cys Cys Thr  Gly Gly Cys
    1220                1225                1230
```

-continued

```
Thr Gly  Cys Gly Ala Ala Gly  Thr Gly Ala Ala Gly  Cys Thr Gly
    1235                 1240                 1245

Gly Thr  Gly Cys Cys Gly Gly  Thr Gly Gly Gly Cys  Gly Ala Cys
    1250                 1255                 1260

Ala Ala  Gly Cys Thr Cys Gly  Ala Gly Gly Cys Gly  Cys Gly Cys
    1265                 1270                 1275

Thr Thr  Cys Cys Gly Thr Gly  Gly Cys Cys Cys Gly  Cys Ala Thr
    1280                 1285                 1290

Ala Thr  Cys Ala Thr Gly Cys  Cys Gly Gly Gly Cys  Thr Ala Cys
    1295                 1300                 1305

Thr Gly  Gly Cys Gly Cys Thr  Cys Gly Cys Cys Gly  Cys Ala Gly
    1310                 1315                 1320

Cys Ala  Gly Ala Cys Cys Gly  Cys Cys Gly Ala Gly  Gly Cys Gly
    1325                 1330                 1335

Thr Thr  Cys Gly Ala Cys Gly  Ala Gly Gly Ala Gly  Gly Gly Cys
    1340                 1345                 1350

Thr Thr  Cys Thr Ala Cys Thr  Gly Thr Thr Cys Gly  Gly Gly Cys
    1355                 1360                 1365

Gly Ala  Cys Gly Cys Gly Thr  Thr Gly Ala Ala Gly  Cys Thr Gly
    1370                 1375                 1380

Gly Cys  Cys Gly Ala Thr Gly  Cys Cys Ala Gly Gly  Cys Ala Gly
    1385                 1390                 1395

Cys Cys  Cys Gly Ala Gly Cys  Thr Thr Gly Gly Cys  Cys Thr Gly
    1400                 1405                 1410

Ala Thr  Gly Thr Thr Cys Gly  Ala Thr Gly Gly Cys  Cys Gly Thr
    1415                 1420                 1425

Ala Thr  Cys Gly Cys Thr Gly  Ala Gly Gly Ala Cys  Thr Thr Cys
    1430                 1435                 1440

Ala Ala  Ala Cys Thr Thr Thr  Cys Gly Thr Cys Cys  Gly Gly Gly
    1445                 1450                 1455

Gly Thr  Ala Thr Thr Cys Gly  Thr Cys Ala Gly Thr  Gly Thr Cys
    1460                 1465                 1470

Gly Gly  Gly Cys Cys Gly Cys  Thr Gly Cys Gly Cys  Ala Ala Cys
    1475                 1480                 1485

Cys Gly  Cys Gly Cys Ala Gly  Thr Gly Cys Thr Gly  Gly Ala Gly
    1490                 1495                 1500

Gly Gly  Cys Thr Cys Gly Cys  Cys Thr Thr Ala Cys  Gly Thr Ala
    1505                 1510                 1515

Cys Ala  Gly Gly Ala Cys Ala  Thr Cys Gly Thr Gly  Gly Thr Cys
    1520                 1525                 1530

Ala Cys  Cys Gly Cys Gly Cys  Cys Gly Gly Ala Cys  Cys Gly Thr
    1535                 1540                 1545

Gly Ala  Ala Thr Gly Cys Cys  Thr Gly Gly Gly Cys  Cys Thr Gly
    1550                 1555                 1560

Cys Thr  Gly Gly Thr Gly Thr  Thr Cys Cys Cys Gly  Cys Gly Thr
    1565                 1570                 1575

Cys Thr  Gly Cys Cys Cys Gly  Ala Gly Thr Gly Thr  Cys Gly Gly
    1580                 1585                 1590

Cys Gly  Cys Cys Thr Gly Gly  Cys Cys Gly Gly Gly  Cys Thr Gly
    1595                 1600                 1605

Gly Cys  Ala Gly Ala Gly Gly  Ala Thr Gly Cys Cys  Ala Gly Cys
```

-continued

```
    1610                1615                1620

Gly Ala  Thr Gly Cys Gly Cys  Gly Gly Gly Thr Gly  Cys Thr Gly
    1625                1630                1635

Gly Cys  Cys Ala Ala Cys Gly  Ala Cys Ala Cys Cys  Gly Thr Gly
    1640                1645                1650

Cys Gly  Cys Ala Gly Thr Thr  Gly Gly Thr Thr Cys  Gly Cys Thr
    1655                1660                1665

Gly Ala  Cys Thr Gly Gly Cys  Thr Gly Gly Ala Gly  Cys Gly Cys
    1670                1675                1680

Thr Thr  Gly Ala Ala Cys Cys  Gly Cys Gly Ala Thr  Gly Cys Cys
    1685                1690                1695

Cys Ala  Ala Gly Gly Cys Ala  Ala Cys Gly Cys Cys  Ala Gly Cys
    1700                1705                1710

Cys Gly  Thr Ala Thr Cys Gly  Ala Ala Thr Gly Gly  Cys Thr Gly
    1715                1720                1725

Thr Cys  Gly Cys Thr Gly Cys  Thr Gly Gly Cys Cys  Gly Ala Gly
    1730                1735                1740

Cys Cys  Gly Cys Cys Gly Thr  Cys Gly Ala Thr Cys  Gly Ala Cys
    1745                1750                1755

Gly Cys  Cys Gly Gly Thr Gly  Ala Ala Ala Thr Cys  Ala Cys Cys
    1760                1765                1770

Gly Ala  Cys Ala Ala Gly Gly  Gly Cys Thr Cys Gly  Ala Thr Cys
    1775                1780                1785

Ala Ala  Thr Cys Ala Gly Cys  Gly Cys Gly Cys Cys  Gly Thr Gly
    1790                1795                1800

Cys Thr  Gly Cys Ala Gly Cys  Gly Gly Cys Gly Cys  Gly Cys Cys
    1805                1810                1815

Gly Cys  Thr Cys Ala Gly Gly  Thr Cys Gly Ala Gly  Gly Cys Gly
    1820                1825                1830

Cys Thr  Gly Thr Ala Cys Cys  Gly Thr Gly Gly Cys  Gly Ala Ala
    1835                1840                1845

Gly Ala  Cys Cys Cys Cys Gly  Ala Cys Gly Cys Ala  Thr Thr Gly
    1850                1855                1860

Cys Ala  Cys Gly Cys Cys Ala  Ala Gly Gly Thr Gly  Cys Gly Gly
    1865                1870                1875

Cys Cys  Thr Thr Gly Ala
    1880


<210> SEQ ID NO 17
<211> LENGTH: 1155
<212> TYPE: PRT
<213> ORGANISM: Rheum palmatum

<400> SEQUENCE: 17

Ala Thr Gly Gly Cys Ala Ala Cys Thr Gly Ala Gly Gly Ala Gly Ala
1               5                   10                  15

Thr Gly Ala Ala Gly Ala Ala Ala Thr Thr Gly Gly Cys Cys Ala Cys
            20                  25                  30

Cys Gly Thr Gly Ala Thr Gly Gly Cys Cys Ala Thr Thr Gly Gly Cys
        35                  40                  45

Ala Cys Gly Gly Cys Cys Ala Ala Cys Cys Cys Thr Cys Cys Gly Ala
    50                  55                  60
```

-continued

```
Ala Cys Thr Gly Cys Thr Ala Cys Thr Ala Cys Cys Ala Gly Gly Cys
65                  70                  75                  80

Cys Gly Ala Cys Thr Thr Thr Cys Cys Cys Gly Ala Cys Thr Thr Cys
                85                  90                  95

Thr Ala Cys Thr Thr Cys Cys Gly Cys Gly Thr Cys Ala Cys Cys Ala
            100                 105                 110

Ala Cys Ala Gly Cys Gly Ala Cys Cys Ala Cys Cys Thr Cys Ala Thr
        115                 120                 125

Cys Ala Ala Cys Cys Thr Cys Ala Ala Gly Cys Ala Ala Ala Ala Gly
    130                 135                 140

Thr Thr Cys Ala Ala Gly Cys Gly Cys Cys Thr Thr Thr Gly Thr Gly
145                 150                 155                 160

Ala Ala Ala Ala Cys Thr Cys Ala Ala Gly Gly Ala Thr Thr Gly Ala
                165                 170                 175

Gly Ala Ala Gly Cys Gly Thr Thr Ala Cys Cys Thr Thr Cys Ala Thr
            180                 185                 190

Gly Thr Gly Ala Cys Cys Gly Ala Ala Gly Ala Gly Ala Thr Thr Cys
        195                 200                 205

Thr Cys Ala Ala Gly Gly Ala Ala Ala Ala Cys Cys Cys Ala Ala Ala
    210                 215                 220

Cys Ala Thr Thr Gly Cys Thr Gly Cys Cys Thr Ala Cys Gly Ala Gly
225                 230                 235                 240

Gly Cys Ala Ala Cys Cys Thr Cys Gly Thr Thr Gly Ala Ala Thr Gly
                245                 250                 255

Thr Ala Ala Gly Ala Cys Ala Cys Ala Ala Ala Ala Thr Gly Cys Ala
            260                 265                 270

Ala Gly Thr Gly Ala Ala Ala Gly Gly Ala Gly Thr Thr Gly Cys Ala
        275                 280                 285

Gly Ala Gly Cys Thr Thr Gly Gly Gly Ala Ala Ala Gly Ala Gly Gly
    290                 295                 300

Cys Thr Gly Cys Cys Cys Thr Cys Ala Ala Gly Gly Cys Cys Ala Thr
305                 310                 315                 320

Cys Ala Ala Ala Gly Ala Ala Thr Gly Gly Gly Gly Cys Cys Ala Ala
                325                 330                 335

Cys Cys Cys Ala Ala Gly Thr Cys Cys Ala Ala Gly Ala Thr Cys Ala
            340                 345                 350

Cys Ala Cys Ala Thr Cys Thr Cys Ala Thr Cys Gly Thr Gly Thr Gly
        355                 360                 365

Thr Thr Gly Cys Cys Thr Ala Gly Cys Cys Gly Gly Cys Gly Thr Thr
    370                 375                 380

Gly Ala Cys Ala Thr Gly Cys Cys Cys Gly Gly Cys Gly Cys Gly Gly
385                 390                 395                 400

Ala Thr Thr Ala Thr Cys Ala Ala Cys Thr Cys Ala Cys Thr Ala Ala
                405                 410                 415

Gly Cys Thr Thr Cys Thr Thr Gly Ala Cys Cys Thr Thr Gly Ala Cys
            420                 425                 430

Cys Cys Thr Thr Cys Cys Gly Thr Cys Ala Ala Gly Cys Gly Thr Thr
        435                 440                 445

Thr Thr Ala Thr Gly Thr Thr Thr Thr Ala Cys Cys Ala Cys Cys Thr
    450                 455                 460

Ala Gly Gly Ala Thr Gly Cys Thr Ala Cys Gly Cys Thr Gly Gly Thr
```

-continued

465 470 475 480

Gly Gly Cys Ala Cys Thr Gly Thr Cys Cys Thr Thr Cys Gly Cys Cys
485 490 495

Thr Thr Gly Cys Ala Ala Ala Gly Gly Ala Cys Ala Thr Ala Gly Cys
500 505 510

Gly Gly Ala Gly Ala Ala Cys Ala Ala Cys Ala Ala Gly Gly Gly Ala
515 520 525

Gly Cys Thr Cys Gly Thr Gly Thr Thr Cys Thr Cys Ala Thr Cys Gly
530 535 540

Thr Thr Thr Gly Cys Thr Cys Ala Gly Ala Gly Ala Thr Gly Ala Cys
545 550 555 560

Ala Ala Cys Ala Ala Cys Thr Thr Gly Thr Thr Thr Thr Cys Gly Thr
565 570 575

Gly Gly Gly Cys Cys Ala Thr Cys Thr Gly Ala Ala Ala Cys Cys Cys
580 585 590

Ala Thr Cys Thr Gly Gly Ala Cys Thr Cys Cys Ala Thr Gly Ala Thr
595 600 605

Ala Gly Gly Cys Cys Ala Ala Gly Cys Ala Ala Thr Ala Thr Thr Ala
610 615 620

Gly Gly Cys Gly Ala Thr Gly Gly Gly Gly Cys Thr Gly Cys Ala Gly
625 630 635 640

Cys Thr Gly Thr Cys Ala Thr Ala Gly Thr Thr Gly Gly Cys Gly Cys
645 650 655

Ala Gly Ala Thr Cys Cys Ala Gly Ala Cys Cys Thr Ala Ala Cys Cys
660 665 670

Gly Thr Thr Gly Ala Gly Ala Gly Gly Cys Cys Cys Ala Thr Ala Thr
675 680 685

Thr Cys Gly Ala Gly Thr Thr Gly Gly Thr Thr Thr Cys Cys Ala Cys
690 695 700

Ala Gly Cys Cys Cys Ala Gly Ala Cys Thr Ala Thr Thr Gly Thr Ala
705 710 715 720

Cys Cys Cys Gly Ala Ala Thr Cys Cys Cys Ala Thr Gly Gly Thr Gly
725 730 735

Cys Ala Ala Thr Thr Gly Ala Gly Gly Gly Cys Cys Ala Cys Thr Thr
740 745 750

Gly Cys Thr Thr Gly Ala Ala Thr Cys Thr Gly Gly Ala Cys Thr Cys
755 760 765

Ala Gly Thr Thr Thr Cys Cys Ala Thr Thr Thr Gly Thr Ala Cys Ala
770 775 780

Ala Gly Ala Cys Cys Gly Thr Thr Cys Cys Thr Ala Cys Ala Cys Thr
785 790 795 800

Ala Ala Thr Cys Thr Cys Thr Ala Ala Cys Ala Ala Cys Ala Thr Thr
805 810 815

Ala Ala Ala Ala Cys Thr Thr Gly Cys Cys Thr Thr Thr Cys Thr Gly
820 825 830

Ala Thr Gly Cys Thr Thr Thr Cys Ala Cys Thr Cys Cys Thr Cys Thr
835 840 845

Ala Ala Ala Cys Ala Thr Thr Ala Gly Cys Gly Ala Thr Thr Gly Gly
850 855 860

Ala Ala Cys Thr Cys Thr Cys Thr Thr Thr Thr Cys Thr Gly Gly Ala
865 870 875 880

-continued

```
Thr Cys Gly Cys Ala Cys Ala Cys Cys Cys Thr Gly Gly Thr Gly Gly
                885                 890                 895

Thr Cys Cys Thr Gly Cys Cys Ala Thr Cys Cys Thr Ala Gly Ala Cys
            900                 905                 910

Cys Ala Ala Gly Thr Thr Ala Cys Thr Gly Cys Thr Ala Ala Gly Gly
        915                 920                 925

Thr Thr Gly Gly Thr Cys Thr Thr Gly Ala Ala Ala Ala Gly Gly Ala
    930                 935                 940

Gly Ala Ala Ala Cys Thr Cys Ala Ala Gly Gly Thr Ala Ala Cys Thr
945                 950                 955                 960

Ala Gly Ala Cys Ala Ala Gly Thr Gly Thr Thr Gly Ala Ala Gly Gly
                965                 970                 975

Ala Cys Thr Ala Thr Gly Gly Ala Ala Ala Cys Ala Thr Gly Thr Cys
            980                 985                 990

Gly Ala Gly Thr Gly Cys Thr Ala  Cys Gly Gly Thr Gly  Thr Thr Thr
        995                 1000                1005

Thr Thr  Cys Ala Thr Cys Ala  Thr Gly Gly Ala Thr  Gly Ala Gly
    1010                 1015                 1020

Ala Thr  Gly Ala Gly Gly Ala  Ala Gly Ala Ala Gly  Thr Cys Ala
    1025                 1030                 1035

Cys Thr  Cys Gly Ala Ala Ala  Ala Cys Gly Gly Thr  Cys Ala Ala
    1040                 1045                 1050

Gly Cys  Ala Ala Cys Cys Ala  Cys Thr Gly Gly Ala  Gly Ala Ala
    1055                 1060                 1065

Gly Gly  Gly Cys Thr Cys Gly  Ala Gly Thr Gly Gly  Gly Gly Thr
    1070                 1075                 1080

Gly Thr  Thr Thr Thr Gly Thr  Thr Thr Gly Gly Gly  Thr Thr Cys
    1085                 1090                 1095

Gly Gly  Gly Cys Cys Thr Gly  Gly Ala Ala Thr Cys  Ala Cys Cys
    1100                 1105                 1110

Gly Thr  Thr Gly Ala Ala Ala  Cys Thr Gly Thr Ala  Gly Thr Gly
    1115                 1120                 1125

Cys Thr  Ala Cys Gly Cys Ala  Gly Thr Gly Thr Gly  Cys Cys Cys
    1130                 1135                 1140

Gly Thr  Ala Ala Thr Thr Ala  Gly Cys Thr Ala Gly
    1145                 1150                 1155
```

1: A genetically modified strain of *Pseudomonas putida*, wherein it comprises a mutated AroF-I gene encoding 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase, the sequence of which has at least 90% identity to the sequence SEQ ID NO:1 and having at least one P160L mutation, a P160L/Q164A double mutation, or a P160L/S193A double mutation, preferably a P160L/S193A double mutation.

2: The genetically modified strain of *Pseudomonas putida* as claimed in claim **1**, wherein it is able to express a recombinant DAHP synthase that is resistant to feedback inhibition by tyrosine.

3: The genetically modified strain as claimed in claim **1**, wherein it comprises an additional recombinant gene encoding a phenylalanine hydroxylase (phhA), the sequence of which has at least 80% identity to the sequence SEQ ID NO: 5, and an additional recombinant gene encoding a tetrahydrobiopterin dehydratase (phhB), the sequence of which has at least 80% identity to the sequence SEQ ID NO: 6; said recombinant genes phhA and phhB preferably being placed under the control of a heterologous promoter, enabling their overexpression.

4: The genetically modified strain as claimed in claim **1**, wherein it expresses a DAHP synthase having an amino acid sequence that has at least 80% identity to the sequence SEQ ID NO: 2, the sequence SEQ ID NO:3, or the sequence SEQ ID NO:4.

5: The genetically modified strain as claimed in claim **4**, wherein it expresses a DAHP synthase having an amino acid sequence defined by the sequence SEQ ID NO:2, SEQ ID NO:3, or SEQ ID NO:4.

6: The genetically modified strain as claimed in claim **1**, wherein it further comprises one or more additional recombinant genes selected from:

the recombinant gene encoding a polypeptide having tyrosine ammonia lyase (TAL) activity, in particular a TAL polypeptide having at least 80% identity to the amino acid sequence SEQ ID NO:7 of the polypeptide TAL_RG_OPT,

the recombinant gene encoding a polypeptide having 4-coumarate-CoA ligase (4-CL) activity, in particular a 4CL polypeptide having at least 80% identity to the amino acid sequence SEQ ID NO:8, and/or

the recombinant gene encoding a polypeptide having benzalacetone synthase (BAS) activity, in particular a BAS polypeptide having at least 80% identity to the sequence SEQ ID NO:9.

7: The genetically modified strain as claimed in claim **1**, wherein it is able to produce a phenylpropanoid compound.

8: A method for synthesizing a phenylpropanoid compound, wherein it comprises a step of growing the genetically modified strain as claimed in claim **7** in a culture medium under conditions that enable the expression of the recombinant genes required for the synthesis of said phenylpropanoid, said phenylpropanoid compound being synthesized by said genetically modified strain.

9: The method as claimed in claim **8**, wherein it also comprises a step of recovering the phenylpropanoid compound from the culture medium.

10. (canceled)

* * * * *